



UNIVERSITÀ DI SIENA 1240

DEPARTMENT OF PHYSICAL, EARTH AND ENVIRONMENTAL SCIENCES

Bachelor's degree in physics and advanced technologies

APPLICATION OF ARTIFICIAL INTELLIGENCE TECHNIQUES TO THE TRACKING OF CHARGED COSMIC RAYS IN THE CALET DETECTOR

Supervisor:
Prof. Paolo Maestro

Thesis by:
Giulio Giamello

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Introduction

Since their discovery in 1912 by the physicist Victor Hess [1] (Nobel prize in 1936), *cosmic rays* (CR), or what was initially defined as ‘invisible and penetrating radiation’, have been and still are the subject of scientific study and research.

In particular, major interests are focused on understanding their origin and composition, as well as the mechanisms of acceleration and propagation in the interstellar medium. The goal of the direct study¹ of cosmic rays is therefore to perform measurements of the energy spectra of individual elements, their abundances, and to identify in space the ‘regions’ of their origin, so that it is possible to investigate their source and acceleration phenomena.

Data collection and the direct study of CR energy spectra are the basis for the development of astrophysical theories that deal with modeling and describing such phenomena. For this purpose, in recent decades, calorimetric instruments have been developed that are capable of detecting CRs up to energies on the order of hundreds of TeV, by measuring the energy deposited in them by particle showers originating from the CRs that pass through them. Thanks to the use of other precision detectors, these calorimeters are capable of making precise measurements also on the charge and the arrival direction in the calorimeter.

The recent development of research and techniques in *artificial intelligence* (AI), particularly *machine* and *deep learning*, has allowed the use of these ‘new’ technologies in data analysis as well: having access to large amounts of data allows the development of models capable of making accurate predictions on certain quantities of scientific interest. For example, in high-energy physics experiments, the use of such techniques allows studying and approaching the CR *tracking* problem from another perspective.

The CALET (*CALorimetric Electron Telescope*) mission is a space calorimetry experiment in orbit on the ISS (*International Space Station*), which also involves a research group from the DSFTA (*Department of Physical Sciences and Advanced Technologies* of the University of Siena).

The study presented in this thesis aimed to develop a convolutional neural network (CNN) model capable of predicting the crossing trajectory followed by helium nuclei in the energy range (20, 2e4) [GeV]. Therefore, starting from Monte Carlo (MC) simulations of the crossing events in the CALET calorimeter, the CNN was *trained* on such data, its resolution was measured, and its performance was compared with other algorithms used in CALET to reconstruct the CR crossing trajectory.

Once the model was trained and its efficiency verified, it was then applied to a

¹Via measuring instruments in orbit in the stratosphere or in space.

dataset of flight data² (FD). To evaluate the correctness of the predictions made, a comparison was performed with the values calculated via a *combinatorial Kalman filter* (KF), an algorithm already implemented in CALET to reconstruct CR trajectories.

Finally, through an estimation of the KF algorithm's resolution on the same dataset used for training the model, it was possible to measure the CNN's resolution on the flight data.

The chapters of this thesis work are briefly introduced below:

- [chapter 1](#) provides an introduction to the operating principles of neural networks, in particular to the type used in this thesis study, namely *convolutional neural networks*;
- [chapter 2](#), after an introduction to particle showers, describes the CALET experiment, its structure, and the main results obtained;
- [chapter 3](#) illustrates the structure chosen for building the CNN model used and explains how its *training* was performed;
- [chapter 4](#) reports the results produced by the CNN on a dataset of CALET data simulated via Monte Carlo;
- [chapter 5](#), the CNN is used to predict the impact points of events relating to real data, and then its efficiency is measured by comparison with the combinatorial Kalman filter algorithm.

²'Real' data of crossing events recorded by CALET in the period between January and March 2019.

Chapter 1

Neural networks and their applications

1.1 Introduction to neural networks

Artificial intelligence (AI) constitutes a field of *computer science* that studies how computers can “learn” from experience and “reason” based on hierarchies, starting from simple concepts and defining more complex ones based on these [2, 3, 4].

The branch of AI that studies how a computer can go beyond what it is taught and learn *on its own* how to solve a problem is called *machine learning* (ML). This research field deals with self-learning: that is, it researches algorithms that allow not having to pass as input to the computer, in addition to the data (*dataset*), also the criteria according to which to process them, but which, starting from data, allow the computer to autonomously understand which criteria to follow to process them. To make the computer able to autonomously learn the criteria and rules, machine learning resorts to algorithms that accept the data as input and, in some types of learning, e.g., in *supervised learning*, also the desired response (the true value, which the network must return as output when the corresponding data is passed as input).

The input information is then processed via algorithms with the aim of producing a statistical model capable of making predictions, that is, of returning an output starting from data that are not contained in the information used for the construction of the model itself.

A generic machine learning model is built using the following elements:

- an input *dataset*;
- the examples of the expected outputs (*targets*), corresponding to each element of the dataset (e.g., in *supervised learning*);
- a *loss function*, that is, a function that measures the distance between the predicted value and the one calculated by the model at a certain iteration of the training.

The value of the loss function constitutes an indication of how to modify the algorithm to minimize the distance between expected and calculated data.

A machine learning model transforms the input data into an output with a certain “meaning”. It then follows that the choice of the *representation* of the input data plays a fundamental role in:

- producing output with useful “meaning”;
- improving the performance of machine learning algorithms (e.g., to build a more precise model or to reduce the required computational resources).

To try to produce “meaningful” outputs, a sub-field of ML is used: *deep learning* (DL).

DL deals with identifying the best representations of the input data to solve a practical problem via *layers* of successive representations, increasingly meaningful. The number of layers is called the *depth* of the model and the name deep learning follows from this. The representations of each layer are learned via models called *neural networks*.

What is of interest is therefore to investigate how the learning of the information contained in the dataset by the machine occurs, that is, how *machine learning* occurs. It is possible to identify three main categories of machine learning algorithms: *Supervised learning*, *Unsupervised learning* and *Self-supervised learning*.

(More specific notation and terminology will be introduced below).

1.1.1 Supervised learning

In supervised learning the *dataset* consists of a collection of *labeled examples*

$$\{(\mathbf{x}_i, y_i)\}_{i=1}^N$$

Where each element $\mathbf{x}_i$ is called a *feature vector* and has dimension D . Each element of a feature vector ($j = 1, \dots, D$) indicates a specific feature of the i -th event we are describing.

For example, if we are describing a particle detection event we can have that the j -th component of each feature vector $x^{(j)}$ describes a specific feature (e.g., the charge of the particle), while the $j+1$ -th $x^{(j+1)}$ describes another feature (e.g., the direction) and so on.

The element y_i is called a *label* and can be an element belonging to: a finite set of *classes* $\{1, 2, \dots, C\}$, the set of real numbers $\mathbb{R}$, or even more complex structures such as matrices, tensors or graphs.

A *class* can be defined as a category to which a certain event belongs.

The goal of a supervised machine learning algorithm is to use the dataset to produce a statistical model that accepts as input a feature vector $\mathbf{x}$ and returns as output information that allows tracing back to the label of that feature vector.

1.1.2 Unsupervised learning

It is unsupervised learning when the dataset is formed by a collection of *unlabeled examples*

$$\{\mathbf{x}_i\}_{i=1}^N$$

with $\mathbf{x}_i$ feature vector. In this case the goal consists in creating a model that accepts as input the feature vector and transforms it into another vector or into a numerical value that can be exploited to solve the initial concrete problem, via *data visualization*, *data compression* or *data denoising* procedures.

Usually unsupervised learning is exploited as an initial step to have a better understanding of the starting dataset, before passing it to try to solve the problem via supervised learning. Some examples of unsupervised learning are: *clustering* and *dimensionality reduction*.

1.1.3 Self-supervised learning

We speak of self-supervised learning when the dataset contains only *unlabeled* examples, as in the unsupervised case, and it is the network itself, without the need to pass the labels, that “extracts” and “learns” them from the input data.

It is called *self-supervised* because it does not require the intervention of an external human control (*supervised*), but it is the network itself that identifies the labels.

1.2 Deep learning: representations and weights

In machine learning a *mapping* is therefore performed between the inputs and the targets starting from the reference examples passed to the computer. In deep learning the mapping occurs via a “deep” succession of simple transformations of the data (*layers*) and of the information acquired from the previous layers.

The transformation that each layer performs on the data depends on the “weights” of the layer, which, in most cases, are constant numerical values. It is said that the transformation implemented by a layer is *parameterized* by its weights.

It is then possible to trace the learning problem back to the problem of choosing the optimal weights: this is done via the *loss function*, which measures the distance between the predictions and the true target values, and via an *optimizer* that intervenes

on the weights to reduce the value of the measured distance. This procedure is called the *Backpropagation algorithm*.

Initially, random values are chosen for the weights, consequently the network begins to implement a series of transformations, also random: subsequently, as the number of processed events increases, the weights are adjusted in the “correct direction”, that is, the one for which the loss function is minimum.

This procedure, called the *training loop*, is repeated a sufficient number of times (typically dozens of iterations on thousands of events) to reach a value of the loss function close to the minimum, so as to have predicted values as “close” as possible to the true target values.

1.2.1 Overfitting and underfitting

Overfitting and *underfitting* constitute two of the main problems to pay attention to when training a model for solving a machine learning problem. It is in fact observed that it is necessary to find a balance between two characteristics of machine learning models: *optimization* and *generalization*. The first refers to the process of modifying the model to obtain better performance; while the second indicates the ability of the model to make predictions on data never seen before. The goal is to obtain a good generalization and the only way to do it is to modify the training data.

At the beginning of the training phase, optimization and generalization are linked: as the error on the training data decreases, the error on the test data also decreases, in this phase we are in *underfit*. The model has not yet completely learned the patterns to understand the input data as required. It then happens that, after a certain number of iterations on the training data, the generalization stops improving and then begins to worsen, the model is in *overfit*: that is, the network is learning patterns that are specific to the training data, but not general and therefore not valid also on the test data, thus causing a deterioration in performance.

The first solution to prevent this phenomenon from occurring is to resort to training datasets with a higher number of data.

[Image of overfitting and underfitting graph]

However, in cases where this is not possible, attention must be paid to interrupting the iterations (or epochs) of the model on the training data before it goes into overfit, so as to obtain the best performance.

1.3 Convolutional neural networks

Convolutional networks [5], also known as *convolutional neural networks* (CNNs) are a specific type of neural networks used to process data with a known “grid-like” structure. Some examples of datasets with this structure can be: series of values as a function of time, which can be thought of as a 1D grid of data, sampled at regular intervals; or images, which can be considered as 2D grids (matrices) of pixels.

The name “convolutional neural networks” indicates that networks of this type perform a mathematical operation called *convolution* on the data. *Convolution* is a particular type of linear operation: convolutional networks are simple neural networks that, instead of matrix multiplication, use the convolution operation in at least one of their layers.

1.3.1 The convolution operation

In its most general definition, *convolution* is an operation on two real-valued functions.

$$(1.1) \quad s(t) = (x * w)(t)$$

$$(1.2) \quad s(t) = \int x(a)w(t-a) da$$

We indicate the first argument of the convolution operation x as *input* and the second w as *kernel*.

For example, if we indicate with $x(t)$ the position of an object moving over time and suppose that the measurement of this position is affected by some noise, to have a better estimate of this position we can resort to convolution. We can in fact calculate the average of multiple measurements, weighting each one with a certain value $w(a)$, where, in this case, a indicates the moment in time in which the measurement was made and it is therefore possible to obtain a function $s(t)$ with which to estimate the position $x(t)$.

More generally, convolution is defined for any function for which the integral given above is defined. In [Figure 1.1](#) there is a graphical representation of the convolution operation.

Usually, when working with certain data via a computer, it uses discrete time intervals in which measurements are made, which are also discrete. Therefore, if we assume instead that x and w are discrete, that is, they are defined only on integer values of t , we can define the *discrete convolution* as

$$(1.3) \quad s(t) = (x * w)(t) = \sum_{a=-\infty}^{\infty} x(a)w(t-a)$$

The output $s(t)$ is usually called a *feature map*.

Note that, although in the definition of the sum the index a runs from $-\infty$ to $+\infty$, since each element of the input and the kernel must be stored separately, in practice it is assumed that these functions are zero everywhere except in the points for which the values are stored. This allows implementing Equation 1.3 as a summation over a finite number of elements.

If, for example, we apply convolution on more than one axis simultaneously, that is, if we use a two-dimensional image I as input, the kernel will also be two-dimensional:

$$(1.4) \quad S(i, j) = (I * K)(i, j) = \sum_m \sum_n I(m, n) K(i - m, j - n)$$

The convolution operation enjoys the commutative property:

$$(1.5) \quad S(i, j) = (K * I)(i, j) = \sum_m \sum_n I(i - m, j - n) K(m, n)$$

The commutative property follows from the fact that we have *flipped* the kernel with respect to the input: as m increases, the index of the input increases and that of the kernel decreases or vice versa.

We can also define the *cross-correlation* function, which performs the same operation as convolution, but without flipping the kernel.

$$(1.6) \quad S(i, j) = (K * I)(i, j) = \sum_m \sum_n I(i + m, j + n) K(m, n)$$

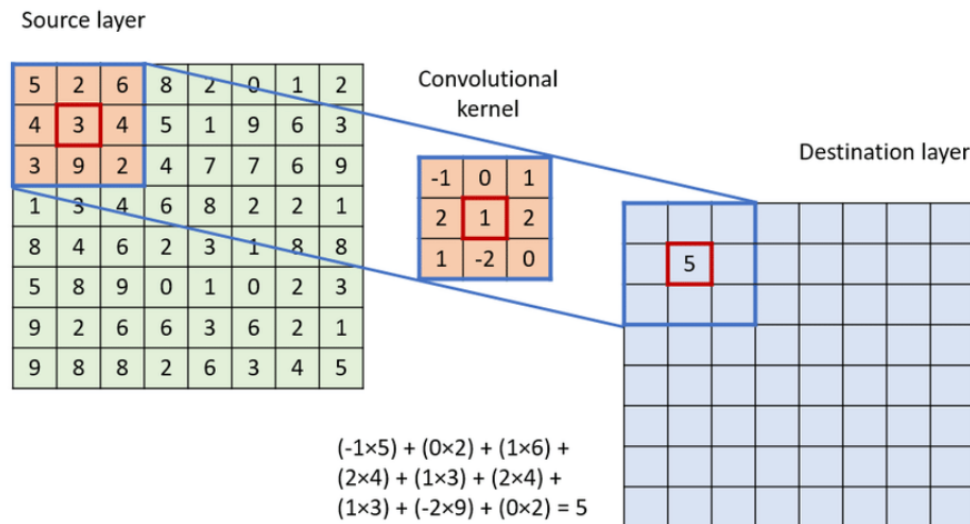


Figure 1.1. the image shows an example of application of the discrete convolution operation between a kernel matrix (*Convolutional kernel*) and an input matrix (*Source layer*).

1.3.2 Advantages and properties of CNNs

The convolution operation is particularly advantageous for machine learning systems because it enjoys the following properties:

- *Sparse interactions*;
- *Parameter sharing*;
- *Equivariance to translation*.

As mentioned previously, the layers of “traditional” neural networks use the matrix multiplication of the parameters, that is, they use a distinct parameter to describe every single interaction between an input and output node. This means that every input node (or neuron) interacts with all output nodes.

Convolutional networks instead resort to *sparse interactions* by going to reduce the size of the kernel with respect to that of the input. For example, in processing an image, the input (image) can have thousands or millions of pixels, but we can detect “small” and “meaningful” characteristics (*features*), such as the edges of the figures, even with a kernel that covers only tens or hundreds of pixels and this translates into the possibility of having to keep in memory a smaller number of parameters, improving the statistical efficiency of the model.

Furthermore, sparse interactions reduce the number of operations required during computation, thus increasing the efficiency of the model: if we have m inputs and n outputs, matrix multiplication requires $m \times n$ parameters and the algorithm has a *time complexity*¹ of $O(m \times n)$. Conversely, if we impose a maximum limit k to the number of connections that each output can have, then the approach with sparse interactions requires a smaller number of parameters $k \times n$ with a reduced time complexity $O(k \times n)$.

For a better understanding, the concept of *sparse connectivity* is represented below in Figure 1.2 as a function of the “connections” (arrows) between a specific input and the outputs; while in Figure 1.3 as a function of the “connections” (arrows) between a specific output and the inputs.

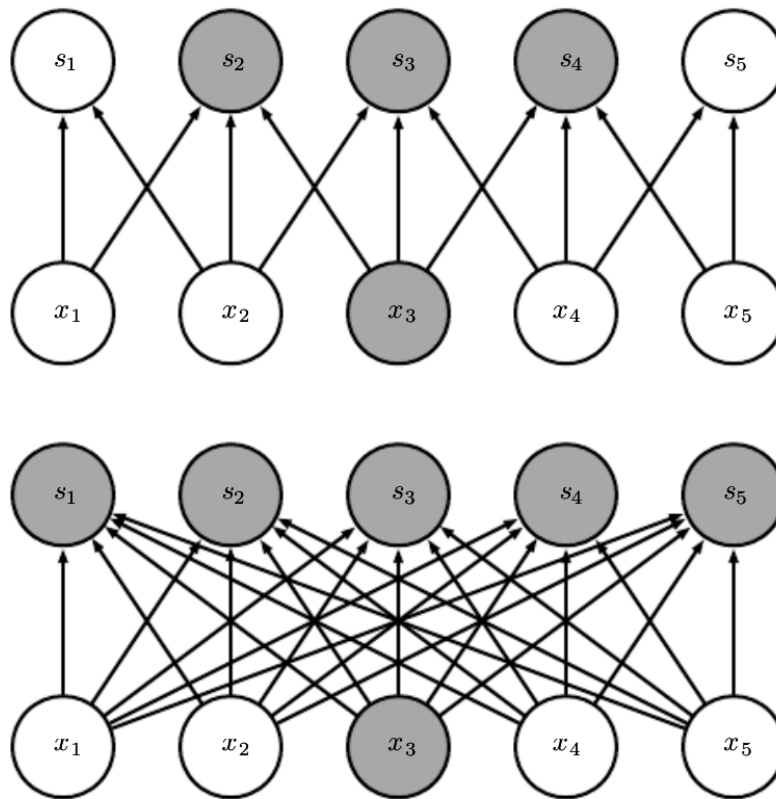


Figure 1.2. in the image the input node x_3 and the output nodes s linked to it are highlighted in dark gray. Above, when the output is obtained with *sparse connectivity*; below when it is obtained via matrix multiplication.

¹The *time complexity* of an algorithm quantifies the time required for execution.

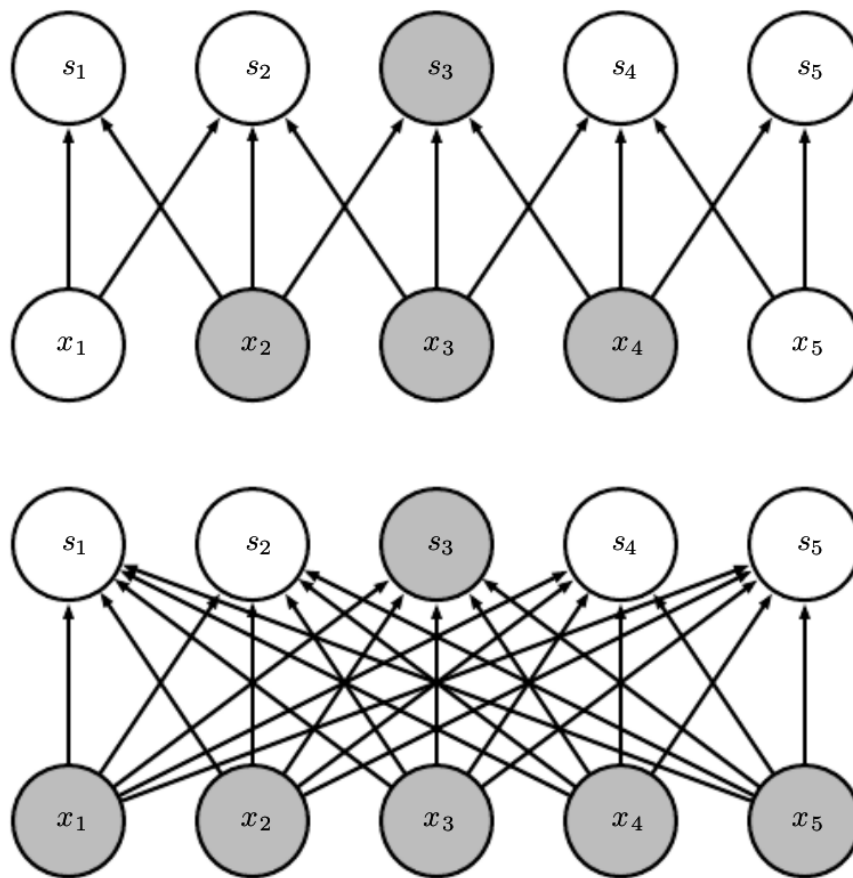


Figure 1.3. in the image the input nodes x and the output node s_3 are highlighted in dark gray. Above, when the output is obtained with *sparse connectivity*; below when it is obtained via matrix multiplication.

In particularly “deep” convolutional networks, that is, with a high number of layers, the nodes of the deepest layers can interact indirectly with a larger portion of the input nodes, as is graphically shown in [Figure 1.4](#). This allows the network to efficiently describe complicated interactions between multiple variables, by building interactions of this type starting from simple nodes, in which each describes sparse interactions.

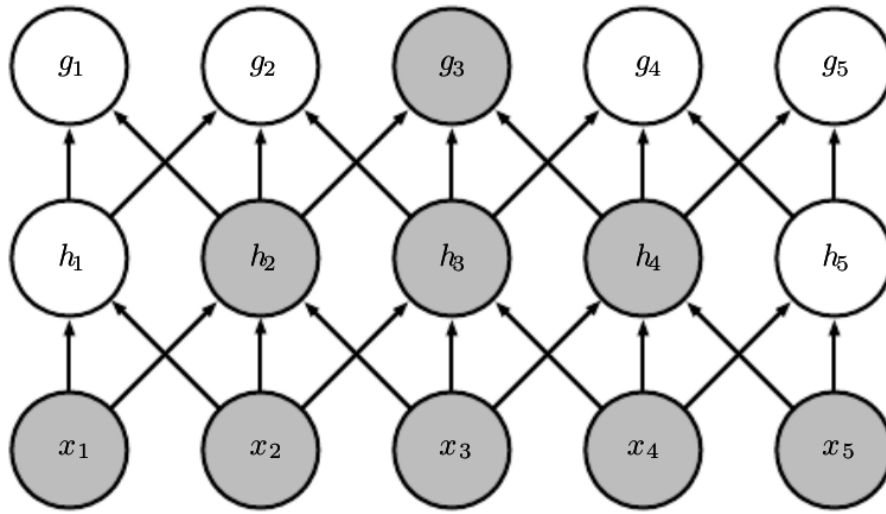


Figure 1.4. the image shows how the “receptive field” of the nodes of the deepest layers is greater than that of the most superficial layers. This means that, although in a convolutional network the connections are *sparse*, the nodes of the deepest layers can still be connected *indirectly with all or almost all the inputs*.

In a traditional network each element of the weight matrix is used exactly once in calculating the output of a certain layer: it is multiplied by an element of the input and never used again in the computation of the output.

In a convolutional network, on the other hand, “tied” weights (*tied weights*) are used; in fact, the value of a weight applied to a certain input is tied to the value of another weight applied to another input. This operation is called *parameter sharing* and prevents the network from having to learn different sets of parameters for every possible element, making it always use the same set of (shared) parameters for every element.

This feature does not change the execution time, which remains $O(k \times n)$, but reduces the space required in memory, given that it limits the number of parameters of the weight matrix to k . Generally k is several orders of magnitude smaller than m and, since m and n are approximately of the same order of magnitude, k is negligible compared to $m \times n$, thus considerably reducing the required memory.

From the feature of *parameter sharing* it also follows that convolutional networks also enjoy the property of “equivariance” (*equivariance*) with respect to translations.

A function is defined as “equivariant” if a variation of the input corresponds to a variation of the output *in the same way*. In more detail, it is meant that: given a function $f(x)$, this is *equivariant* with respect to a function $g(x)$ if

$$f(g(x)) = g(f(x))$$

If as function g we take any translation function of the input, then the convolution operation is equivariant with such g . For example, let I be a function that describes the brightness of an image in points with integer coordinates and let g be a function that maps one image with another, $I' = g(I)$ with $I'(x, y) = I(x - 1, y)$. The function g in this example moves all the pixels of image I by one unit to the right; if we apply this transformation to I and then apply the convolution on image I' , the result is the same as if we had applied the two operations in reverse order, that is, first the convolution on image I and then the function g :

$$w * g(I) = g(w * I)$$

In processing time-dependent series of data this translates into the possibility of having a *timeline* showing “when” a certain feature manifests itself in the inputs. In this case there is equivariance: in fact if we move an event of the inputs in time, its representation will appear in the outputs moved in the same way. Also for images, convolution produces a 2D map, which indicates in the inputs “where” certain features (or characteristics) appear and if we move the input image, its representation moves by the same amount.

Finally, attention should be paid to the fact that the convolution operation is not equivariant with every possible function g , but there are functions, such as scale variations or rotations of the input image, which are not.

1.3.3 Activation functions

The *activation function* g of a node i indicates a vector function that calculates the output of a node based on its inputs z and weights W :

$$(1.7) \quad f_i(z) = g_i(W_i z + b_i)$$

The weights W are represented in the form of a matrix, while the parameters b (real) in the form of a vector. $f_i(z)$ indicates the output produced by the i -th layer in response to the input vector z .

Such functions can be linear and non-linear; among the most used there is the *rectified linear unit* activation function (also used in the study object of this thesis, in [chapter 3](#)).

1.3.4 Rectified linear unit

The linear activation function *Rectified linear unit* (ReLU) is defined as follows:

$$(1.8) \quad g(z) = z^+ = \max(0, z) = \frac{z + |z|}{2} = \begin{cases} z & \text{if } z > 0, \\ 0 & \text{otherwise,} \end{cases}$$

as can be noticed the equation accepts real numerical values (scalars) as input. From this it is then possible to define the vector function $\mathbf{g}(\mathbf{z})$ as:

$$(1.9) \quad \mathbf{g}(\mathbf{z}) = [g(z_1), g(z_2), \dots, g(z_j), \dots, g(z_N)]$$

with N equal to the size of the input vector $\mathbf{z} = [z_1, z_2, \dots, z_j, \dots, z_N]$.

1.3.5 Pooling

A layer of a convolutional network usually consists of three levels. In the first, the layer executes a series of convolution operations in parallel to produce a set of linear activations. In the second, each linear activation is executed via a non-linear activation function, such as the *rectified linear unit* activation function.

In the third level, the layer executes the *pooling* function. A pooling function replaces an output node of a network with a statistical average of nearby nodes. For example, the *max pooling* function selects as output the node with the maximum value within a rectangle of pre-established dimensions. Other examples of pooling functions include the average over rectangular regions, the L^2 norm or the weighted average as a function of the distance from the central pixel of the selected region.

In any situation, the pooling operation improves the (approximate) *invariance* with respect to “small” translations of the inputs. By invariance for translations of the inputs we mean that if we translate the inputs by a “small” amount, the value of the output under pooling operation does not change.

Local invariance by translation turns out to be a particularly useful property in case one wants to verify the presence or not of certain features in the inputs, rather than their specific position. An example of application of this technique is *face detection*: when one wants to determine whether or not an image contains a face (or, in general, a certain object) and one does not want to determine its position within the image.

Chapter 2

CALorimetric Electron Telescope

This chapter provides a description of electromagnetic and hadronic showers, with some considerations on their lateral and longitudinal development. It then moves on to a description of the CALET calorimeter and the results obtained from it, in particular regarding helium nuclei in cosmic rays.

2.1 Particle showers

In high energy physics, a particle *shower* is a “cascade” of particles produced in a chain by the interaction of a *primary* particle with matter [6, 7, 8]. Starting from the interaction of the primary particle, other *secondary* particles are created, which in turn interact with matter and give rise to further particles: this process develops by creating an ever increasing number of particles, of progressively lower energy, which will eventually be absorbed by the matter.

There can be *electromagnetic* showers and *hadronic* showers, depending on whether the interaction phenomenon is mediated by the *electromagnetic interaction* or by the *strong interaction*.

2.1.1 Electromagnetic showers

An electromagnetic shower, such as the one shown in [Figure 2.1](#), originates when a high energy electron e^- , positron e^+ , or photon γ interacts with matter. At high energies (greater than a few tens of MeV) the main energy loss mechanisms are *pair production* for photons¹ and *bremsstrahlung* for electrons and positrons.

These two processes continue until the photons drop below the *threshold* energy for pair production and energy losses of electrons not due to bremsstrahlung begin to dominate.

¹At energies of the order of MeV and above, *pair production* is dominant over other interaction processes such as the *photoelectric effect* and the *Compton effect*.

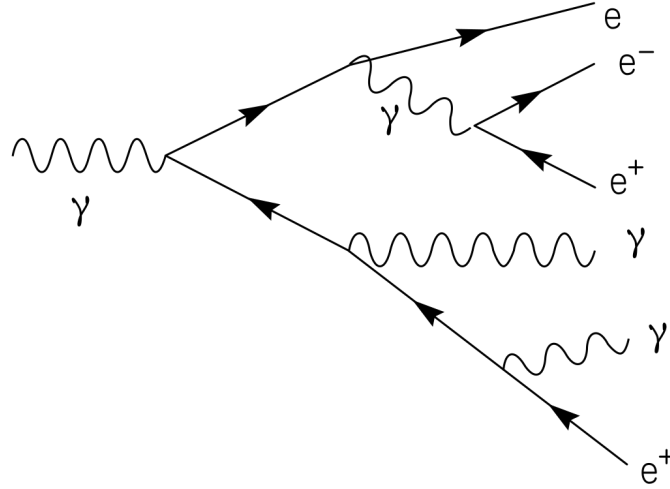


Figure 2.1. beginning of an electromagnetic shower represented with a Feynman diagram.

The characteristic length scale over which these phenomena develop is denoted by the *radiation length* X_0 . In the case of electrons, this quantity represents the thickness after which the electron's energy is $1/e$ of its initial value:

$$E(x) = E_0 e^{-\frac{x}{X_0}}$$

$$E(X_0) = \frac{E_0}{e}$$

whereas in the case of photons it is related to the *mean free path* λ_{pair} for pair production by:

$$\lambda_{\text{pair}} \approx \frac{9}{7} X_0$$

In what follows it will become clear that X_0 is also the length that characterizes the development of electromagnetic showers.

When a high energy photon or electron passes through matter, it gives rise to a shower of *secondary* photons and electrons, produced through *pair production* and *bremsstrahlung* respectively. At each subsequent interaction these processes repeat for an ever increasing number of particles and continue until the *critical energy* E_c is reached, below which the particle multiplication phenomenon stops and other energy loss mechanisms become dominant, such as the *photoelectric effect* and the *Compton effect* for photons and *collisional* energy loss for electrons.

Pair production

The pair production process, shown in Figure 2.2, is a reaction in which a photon interacts with matter, converting its energy into matter and antimatter, that is, producing an electron/positron pair (e^- , e^+).

Pair production can only occur in the coulomb field of the nucleus or of atomic electrons, according to the reactions:

- $\gamma + N \longrightarrow (e^- + e^+) + N$
- $\gamma + e^- \longrightarrow (e^- + e^+) + e^-$

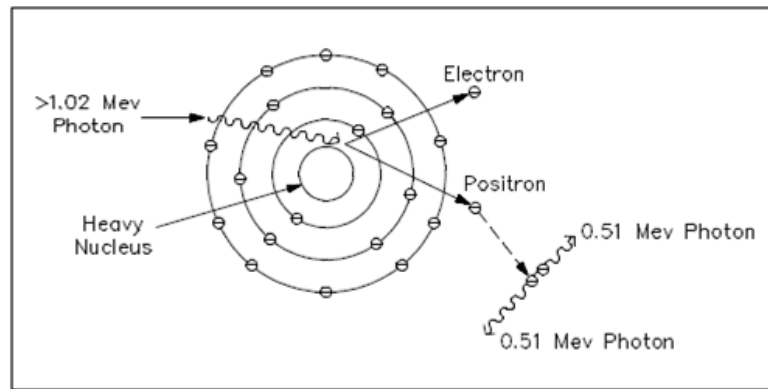


Figure 2.2. graphical description of the pair production process: a photon with energy equal to the threshold energy $E_Y^{\min}$ interacts with the coulomb field of a nucleus, producing an electron/positron pair with equally shared energy.

Energy is conserved in the process and therefore, in agreement with Einstein's mass-energy equivalence equation $E = mc^2$, where E indicates the total relativistic energy, m the relativistic mass and c the speed of light in vacuum.

The energy of the incident gamma ray is equally shared between the particle and its corresponding antiparticle, since they have the same mass. Furthermore, the production process is a *threshold* process, meaning that the energy of the incident particle must be greater than a certain value for the reaction to occur. In particular, the photon must have energy greater than the sum of the rest mass energies of an electron and a positron, so that:

$$E_Y^{\min} = 2m_e c^2 \left(1 + \frac{m_e}{M} \right)$$

- $E_Y^{\min} \approx 2m_e c^2 = 1.022 \text{ MeV}$ (on a nucleus with $M \gg m_e$)
- $E_Y^{\min} \approx 4m_e c^2 = 2.044 \text{ MeV}$ (on an electron $M = m_e$)

Bremsstrahlung

Bremsstrahlung (from the German, “braking radiation”) is an energy loss process due to radiation in the electric field of the nucleus and is particularly relevant for electrons. Figure 2.3 shows a graphical representation of the phenomenon. Electrons can lose considerable amounts of their kinetic energy through photon emission and be strongly deflected.

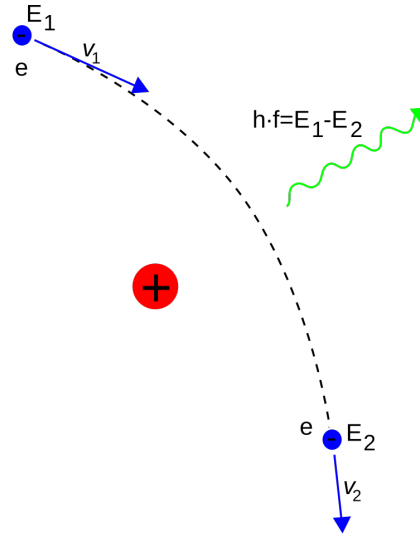


Figure 2.3. bremsstrahlung produced by a highly energetic electron deflected by the electric field of an atomic nucleus.

The deflected electron loses kinetic energy and, by the principle of energy conservation, therefore emits a photon (gamma radiation).

Photons emitted through bremsstrahlung are characterized by a continuous distribution that becomes more intense as the electron energy increases, and therefore have energy E_γ :

$$0 < E_\gamma \leq E_e = h\nu_{\max}$$

in the last equality, the relationship between the kinetic energy of the electron E_e and the maximum frequency of the emitted gamma radiation $\nu_{\max}$ is made explicit.

The probability of emission of a photon with energy E_γ has a distribution:

$$P(E_\gamma) \propto \frac{1}{E_\gamma}$$

Longitudinal development

Longitudinal development refers to the path of the secondaries in the direction of the primary particle. This can be described through the *radiation length* X_0 : indeed, the primary particle (e.g. a photon) will produce an electron/positron pair, each with half of its energy, after passing through a thickness of matter of about $\approx X_0$, more precisely $\frac{9}{7}X_0$. The secondary pair will travel through another thickness $\approx X_0$ and each particle will produce a photon through bremsstrahlung.

The thickness traversed x is therefore expressed in radiation lengths, $k = \frac{x}{X_0}$. This description of the development makes it possible to estimate the number of particles N at a depth of k radiation lengths as

$$N(k) \approx 2^k$$

and the energy of the secondary particles as a function of that of the primary as

$$E(k) \approx \frac{E_0}{N(k)} = \frac{E_0}{2^k}$$

Furthermore, knowing that the multiplication process is limited by the *critical* energy E_c , it is possible to determine at which depth $k_{\max}$ the shower reaches its maximum development:

$$E(k_{\max}) = E_c \Rightarrow k_{\max} = \frac{\log(E_0/E_c)}{\log(2)}$$

at which the number of secondaries is at its maximum

$$N(k_{\max}) = \frac{E_0}{E_c}$$

It is therefore possible to define the *total track length* as the sum of the ionization tracks:

$$TL \propto X_0 \frac{E_0}{E_c}$$

while the number of secondaries produced up to $k_{\max}$ is

$$N_{\text{tot}} = \sum_{k=0}^{k_{\max}} 2^k = 2^{(k_{\max}+1)} - 1 \approx 2 \frac{E_0}{E_c}$$

A more complex analytical method for describing the longitudinal development of electromagnetic showers, which takes into account the different characteristics of the interactions and introduces what is known as the “B approximation”, can be found in the article by Bruno Rossi [9].

Lateral development

Lateral development refers to the effect of those interaction processes that cause the shower to widen in the direction perpendicular to the direction of propagation.

The phenomenon has cylindrical symmetry around the propagation axis of the primary particle and is determined by multiple scattering processes in matter, which dominate over the negligible widening due to pair production and bremsstrahlung, characterized by an angle $\theta \sim \frac{m_e c^2}{E_s}$, with E_s the energy of the secondaries.

The widening cylinder is divided into two components: a narrow central one and a wider outer one.

The central structure is due to multiple coulomb scattering processes of those electrons that have enough energy to move away from the axis, but not enough to radiate photons. This component has a typical lateral size given by the *Molière radius*:

$$R_{\text{Mol}} = \frac{E_{\text{Mol}}}{E_c} X_0$$

$$\text{with } E_{\text{Mol}} = m_e c^2 \sqrt{\frac{4\pi}{\alpha}} = 21.2 \text{ MeV}$$

On average, 90% of the shower energy is contained within a cylinder with a radius equal to the Molière radius $R(90\%) = R_{\text{Mol}}$; while 95% is contained within a cylinder with a radius equal to twice the Molière radius $R(95\%) = 2R_{\text{Mol}}$.

The outer structure is due to the propagation of low energy photons, which therefore produce secondary electrons misaligned from the primary axis through the photoelectric effect and the Compton effect.

Often, to describe the widening of an electromagnetic shower on a plane perpendicular to the direction of propagation of the primary and at depth k , a *two-exponential parametrization* is used, one for the inner structure and the other for the outer one:

$$f(x, k) = A e^{-\frac{x}{b_1}} + B e^{-\frac{x}{b_2}}$$

with x the distance from the primary axis and b_1 and b_2 the characteristic inner and outer attenuation lengths, respectively.

2.1.2 Hadronic showers

Hadronic showers occur when the mechanisms of secondary particle production are mediated by the *strong interaction*. A hadronic shower is generated by a hadronic particle (that is, nucleons and other particles made up of quarks).

Unlike electromagnetic showers, the description of a hadronic shower is an extremely complex problem: there are in fact no analytical descriptions of this phenomenon, and quantitative approaches based on numerical methods and Monte Carlo simulations are used instead.

The interaction of energetic hadrons (charged or neutral, p, n, π , K) with the nuclei of a material is determined by inelastic nuclear processes that give rise to large fluctuations in the energy deposited by each shower.

The interaction takes place in two stages: initially there is a fast *intranuclear cascade* followed by a slow *evaporation* phase. In the first phase, there is an interaction between the hadron and a nucleon, which can be considered almost free within the nucleus. The nucleon acquires energy and collides with other nucleons: a cascade of fast nucleons, pions, and other hadrons is thus formed. Of the particles present in the cascade, some escape from the nucleus, while others remain and redistribute the acquired energy. In a typical hadronic interaction, about half of the incident energy is converted into the production of secondary particles such as charged and neutral pions.

An important fraction of the secondaries produced in a hadronic shower consists of neutral pions π^0 , which decay electromagnetically into two photons and thus produce an electromagnetic shower. It follows that a hadronic shower is always followed by an electromagnetic shower.

More specifically, if the energy of the primary is E (expressed in GeV) the average number of neutral pions produced is $n_0 \approx 5 \ln(E) - 4.6$. The fraction of hadronic energy deposited in the form of an electromagnetic cascade is on average $f_{\text{em}} \approx 1 - E^{-0.15}$.

The *intranuclear spallation cascade* consists of leading particles (p, n, π , K) with a direction roughly that of the incident particle and with high energy. The resulting intermediate nucleus de-excites and a number of particles are *evaporated* (p, n, α), i.e. emitted isotropically in the laboratory reference frame. The remaining energy is deposited through ionization (between 40% and 60%), neutron generation (between 10% and 15%), nuclear breakup and recoil of nuclear fragments (between 30% and 45%), and production of photons ($\sim 3\%$) of a few MeV.

Longitudinal development

At energies greater than a few GeV (high energies) the inelastic cross section depends little on the energy and type of the incident particle, but depends mainly on the target:

$$\begin{aligned}\sigma_{\text{anel}} &= 35A^{0.7} \text{ [mb]} \\ \sigma_{\text{tot}} &= \sigma_{\text{el}} + \sigma_{\text{anel}}\end{aligned}$$

with σ_{el} and σ_{anel} the elastic and inelastic cross sections on a nucleus of mass number A .

The longitudinal development of the hadronic shower is proportional to the *nuclear interaction length* λ_I :

$$\lambda_I = \frac{A}{N_A \rho \sigma_{\text{anel}}} \approx 35 \frac{A^{1/3}}{\rho} \text{ [cm]}$$

where ρ is the density of the material and N_A is Avogadro's number. The longitudinal development initially follows a rapid growth, also due to the electromagnetic contribution, before reaching a maximum and slowly decreasing through hadronic interactions.

The maximum development of the shower occurs at a depth (measured in *nuclear interaction lengths* λ_I):

$$k_{\text{max}} \approx 0.2 \ln(E) + 0.7$$

and 95% of the longitudinal containment of a hadronic shower is, on average, reached at the depth:

$$L_h(95\%) \approx k_{\text{max}} + 2.5 \lambda_{\text{att}}$$

where $\lambda_{\text{att}} \propto E^{0.13}$ is the *attenuation length*, describing the decay of the shower once it has passed the maximum k_{max} .

Typically, the X_0 values of electromagnetic showers are smaller than the λ_I values for hadronic showers, meaning the latter develop over greater lengths.

Materiale	X_0 [cm]	E_c [MeV]	R_M [cm]	R_M/X_0	λ_I [cm]
Be	35.28	113.70	6.60	0.19	42.10
Si	9.37	40.19	4.94	0.53	46.52
Al	8.90	42.70	4.42	0.50	39.70
Fe	1.76	21.68	1.72	0.98	16.77
BGO	1.12	10.50	2.26	2.02	22.32
Cu	1.44	19.42	1.57	1.09	15.32
W	0.35	7.97	0.93	2.66	9.95
Pb	0.56	7.43	1.60	2.86	17.59
U	0.32	6.65	1.00	3.13	11.03
PWO	0.89	9.64	1.96	2.20	20.27
PVT	42.54	94.11	9.59	0.23	78.80
PS	41.31	93.11	9.41	0.23	77.07

Lateral development

Lateral development is also greater in a hadronic shower than in an electromagnetic one, due to the wide angles at which secondary particles are produced. Thus, a significant fraction of the primary's energy is deposited far from the shower axis.

Similarly to the electromagnetic case, in this case too two structures can be identified, an inner and an outer one, describable by the sum of two exponentials. The radial

containment of 95% of the energy deposited by the shower is reached within a cylinder of radius equal to:

$$R_h(95\%) \approx \lambda_I$$

Finally, as seen for the longitudinal development, due to the differences between X_0 and λ_I , the lateral development of a hadronic shower turns out to be greater than that of an electromagnetic shower, meaning they are ‘wider’.

2.1.3 Calorimetry

Instruments called *calorimeters* are used to measure the energy deposited by particle showers passing through portions of matter. A calorimeter is a particle detector that measures the energy of a particle: the energy can be measured internally, requiring total containment of the particle shower, or only partially.

Calorimeters are usually segmented both transversely, in order to acquire information not only on the particle’s energy but also on its direction of travel, and longitudinally, in order to carry out measurements aimed at particle identification.

The total deposited energy is measured as the sum of the signals (for example, scintillation light) produced by the shower secondaries in the active elements of the calorimeter, and is proportional to the energy of the incident particle.

In order to allow the particle to deposit all of its energy, the calorimeter must be made of material with high density and thickness; since the depth needed to fully stop the particle grows logarithmically with increasing energy, it is possible to build very compact instruments even for analyzing high energies.

Lateral containment is also particularly important at the design stage: the calorimeter must be correctly sized so that it is able to contain the multitude of secondary particles generated in the shower, which spread out at large angles with respect to the primary axis. Good sizing makes it possible to increase the energy resolution of the instrument.

Calorimeters can be *electromagnetic* or *hadronic*, depending on the nature of the shower to be detected. In the first case, particles such as electrons, positrons, and photons that interact electromagnetically with matter; while in the second, protons, pions, and nuclei, which interact with matter through the strong interaction.

These two types of calorimeter can in turn be divided into two further subcategories, depending on whether the calorimeter is *homogeneous* or *sampling*. The former consist of a single, homogeneous block of sensitive material, the advantage of which is excellent energy resolution, since the entire volume is sensitive material; the latter, on the other hand, consist of layers of dense material in which the shower is produced and layers of sensitive material in which the deposited energy is measured. The advantage is that

they can be made more compact than a homogeneous calorimeter, at the cost of poorer energy resolution.

2.2 The CALET mission

The CALET mission is a space experiment funded by the *Japan Aerospace Exploration Agency* (JAXA), the *Italian Space Agency* (ASI), and the *National Aeronautics and Space Administration* (NASA). The Italian universities of Siena (the lead institution for Italian participation in the mission), Pisa, Florence, and Rome Tor Vergata take part in the international collaboration.

CALET was launched into orbit on 19 August 2015 aboard the Japanese *H-II Transfer Vehicle* (HTV-5) rocket and installed on the *Japanese Experiment Module Exposure Facility* (JEM-EF), the external platform of the Japanese module (KIBO) [10, 11] of the *International Space Station* (ISS) on 24 August, as shown in [Figure 2.4](#). The JEM-EF is a $4 \times 6 \text{ m}^2$ platform directly exposed to outer space, able to host up to 10 experiment modules at the same time and to supply them with electrical power, heat exchangers, and data transmission systems; CALET is positioned at port number 9.

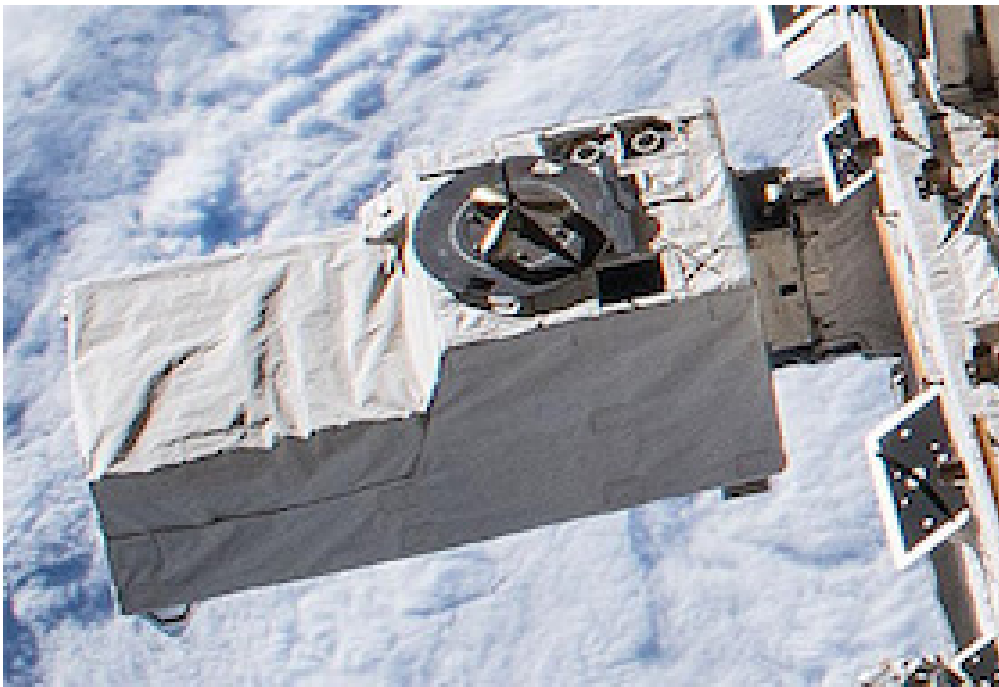


Figure 2.4. the CALET calorimeter at port number 9 of the ISS JEM-EF platform.

The main goal of the mission is to search for clues to the presence of astrophysical sources of high energy electrons near Earth, or clues to dark matter, through the

measurement of the $e^- + e^+$ spectrum in the energy range from 1 GeV up to about 20 TeV. CALET also studies the acceleration and propagation of cosmic rays (CR) in the Galaxy, carrying out direct measurements of the energy spectra and elemental composition of the nuclei contained in CR. The main characteristics of the instrument are excellent energy resolution and excellent capability for separating hadrons from electrons.

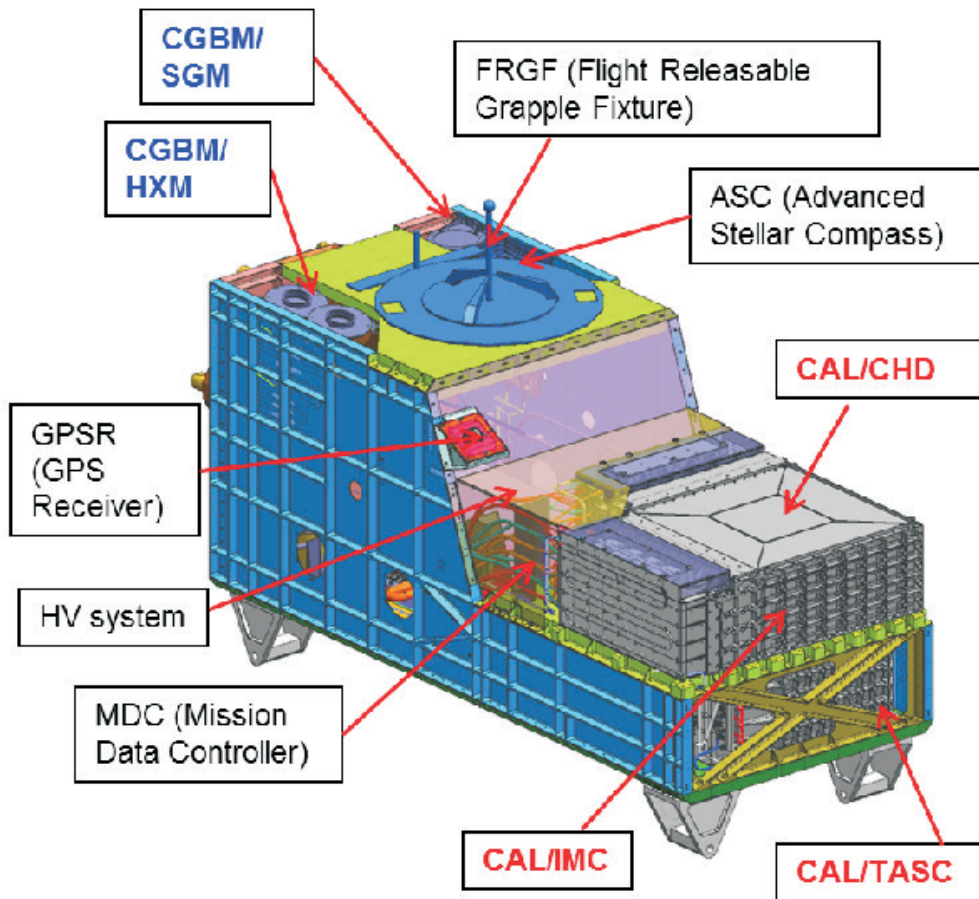


Figure 2.5. diagram of the CALET instrument.

The instrument is made up of the main detector, the calorimeter, and further systems, whose position is shown in [Figure 2.5](#). The main ones are listed below:

- *Flight Releasable Grapple Fixture* (FRGF), an anchoring point for the robotic arm and astronauts;
- *Mission Data Controller* (MDC), a computer for managing the detector and the data;

- *Advance Stellar Compass* (ASC) and GPS receiver (GPSR) for pointing and position control;
- *CALET Gamma ray Burst Monitor* (CGBM), a sub-detector for measuring X-rays and gamma rays in an energy range outside the sensitivity of the main telescope.

2.2.1 The CALET calorimeter

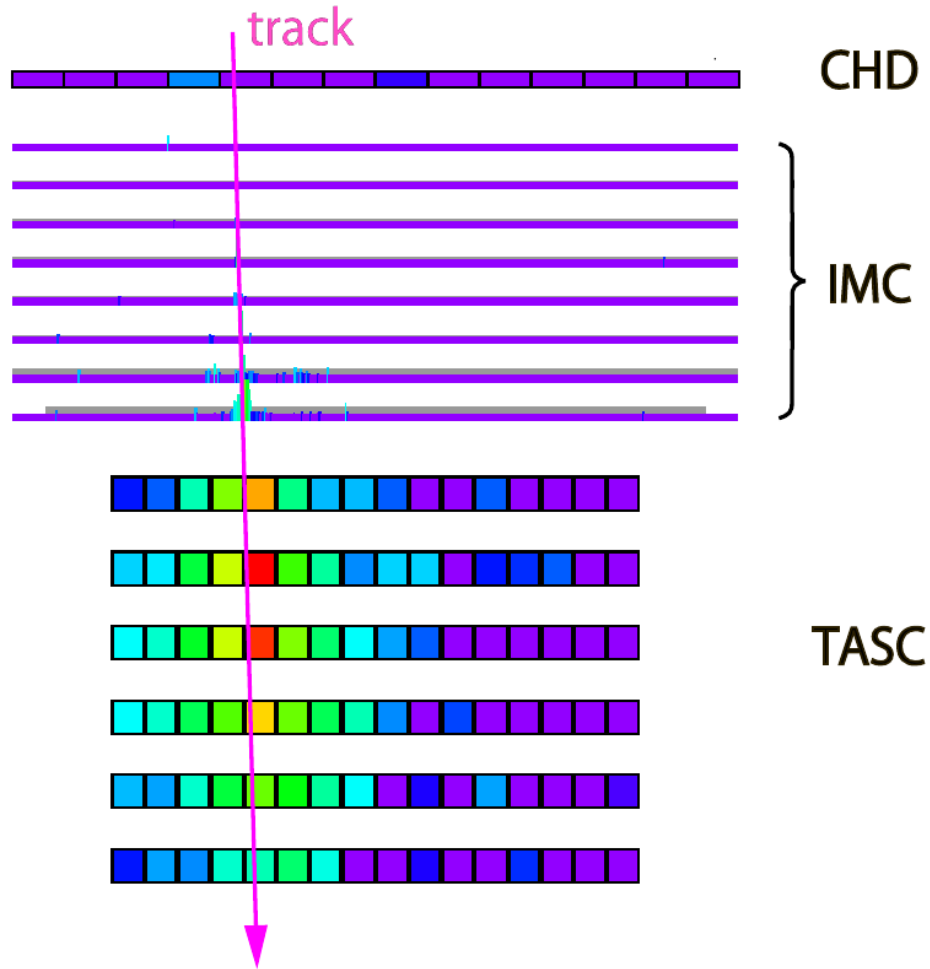


Figure 2.6. diagram of a two-dimensional view of the calorimeter.

The main instrument of CALET is a calorimetric detector with a thickness of 30 radiation lengths. The field of view is about 45 degrees from the zenith and the geometric factor is $1200 \text{ cm}^2 \text{ sr}$. The calorimeter, as shown in Figure 2.6 and in greater detail in Figure 2.7, is divided into three detector subsystems arranged one above the other [12]:

- *CHarge Detector* (CHD), positioned at the top of CALET, measures the electric charge Z of the incident particle, that is, it performs the identification of the chemical elements present in cosmic rays;
- *IMaging Calorimeter* (IMC), placed below the CHD and above the TASC, provides an image of the initial profile of the shower and, through 16 planes of scintillating fibers, reconstructs the trajectory of the incident particle. It is also able to measure the charge of the particle, contributing to its identification;
- *Total AbSorption Calorimeter* (TASC), measures the total energy of the shower generated by the primary particle and makes it possible to distinguish between electromagnetic and hadronic showers.

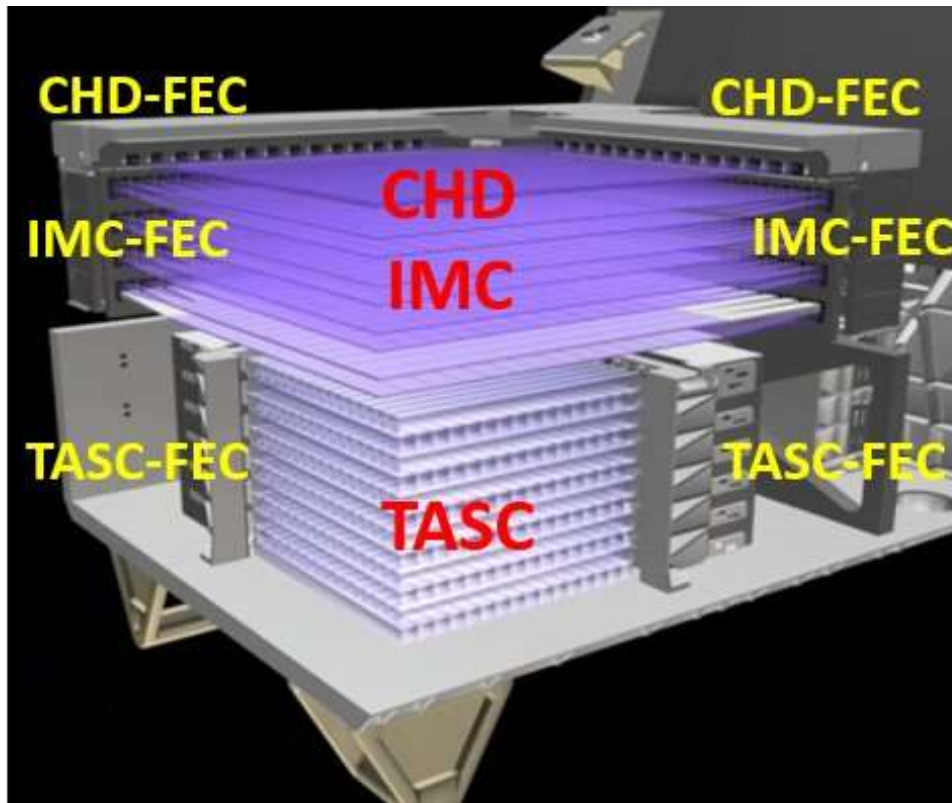


Figure 2.7. three-dimensional diagram of the calorimeter and its detector subsystems.

CHarge Detector (CHD)

The *CHarge Detector* (CHD) is the first detector traversed by the incident particle. Its function is to measure the absolute value of the charge of the primary particle, in order

to identify it. The charge measurement is based on the Z^2 dependence of the measured value of the specific ionization of ultra-relativistic ions with atomic number Z (see the *Bethe-Bloch formula*).

Therefore, since the *Bethe-Bloch formula* allows the value of Z^2 to be derived, what is obtained is the absolute value of the particle's charge, but not its sign: the CHD therefore does not make it possible to distinguish between particles with equal charge in magnitude but opposite sign, for example electrons and positrons.

The charge resolution is expressed in terms of the elementary unit of charge e . The CHD resolution values range between $0.15e$ and $0.3e$ for nuclei with $2 < Z < 26$.

The CHD is made up of two orthogonal layers of 14 strips (*paddles*) of plastic scintillators ($32 \text{ mm} \times 450 \text{ mm} \times 10 \text{ mm}$ each). The orthogonality of the two planes, that is, the fact that they are rotated by 90° relative to each other, makes it possible to estimate the incidence position of the cosmic rays, the (x, y) coordinates on the CHD plane. The segmentation into strips makes it possible to reduce the contribution to the signal due to the energy deposits of *backscattering* particles produced by the shower developing in the IMC and TASC.

Backscattering is an issue common to all calorimetric instruments at high energies: in CALET, this problem is addressed by exploiting the track reconstruction performed in the IMC, so as to select in the CHD only the strip actually struck by the primary particle, avoiding confusing it with those struck instead by backscattered secondary particles [13].

IMaging Calorimeter (IMC)

The *IMaging Calorimeter* (IMC) is a *sampling* calorimeter, that is, one divided into layers of dense material in which the shower is allowed to develop and layers of sensitive material in which the deposited energy is measured; it is made up of 8 planes (*layers*) of scintillating fibers (*SciFi*) made of polystyrene (PS), alternated with 7 tungsten (W) absorber planes.

In each of the 8 planes there are 2 layers of scintillating fibers, rotated 90° relative to each other, that is, oriented along the orthogonal axes x and y , thus providing two transverse views of the xz and yz planes respectively.

Each layer contains 448 fibers, 448 mm long with a cross-sectional diameter of 1 mm. The dimensions of the fiber planes are therefore $448 \text{ mm} \times 448 \text{ mm} \times 1 \text{ mm}$ each, and two successive planes are located at a distance of about 2 cm from each other. With the exception of the first fiber plane, each of the other planes lies immediately below one of the 7 tungsten planes, which have different thicknesses depending on the depth. Moving from top to bottom, the first five planes have a thickness of 0.07 cm (corresponding to $0.2 X_0$), while the last two have a thickness of 0.35 cm (corresponding to $1 X_0$). The total thickness of the IMC is therefore about $\sim 3X_0$ radiation lengths.

Thanks to this structure, the IMC calorimeter allows the cosmic ray to begin and develop its shower in the tungsten planes, while the SciFi planes provide images of the shower with high spatial resolution, useful for identifying and tracking the primary particle.

The IMC is in fact used as a *preshower*: within it, primary photons and electrons are induced to shower by the tungsten planes, while the SciFi fibers produce an image of the initial part of the shower's development; conversely, if the incident primary particles are protons, only a small fraction of them interact and shower within the IMC planes, due to the limited depth of the instrument in terms of interaction length.

The characteristics described above can be summarized by saying that the IMC makes it possible to:

- reconstruct the trajectory of the incident particle;
- determine the starting point of the electromagnetic shower;
- discriminate the incident primary particle from backscattered ones.

The IMC's ability to reconstruct the trajectory of the incident particle, and thus the starting point of the shower, is of considerable importance for carrying out measurements aimed at identifying electrons against the 'background' of photons and protons. Indeed, in the calorimeter, showers originating from electrons (or positrons) and from photons are similar in many respects and differ only in the fact that, in shower development events originating from photons, the fibers preceding the starting point of the development do not record an energy release compatible with that of a minimum ionizing charged particle. This makes it possible to identify showers originating from photons.

In the case of protons, on the other hand, the start of the shower occurs at depths greater than the total thickness of the IMC [14, 15].

Total AbSorption Charge (TASC)

The *Total AbSorption Calorimeter* (TASC) is a *homogeneous* calorimeter designed to fully contain the development of the showers, that is, it consists of a single, homogeneous block of sensitive material, the advantage of which is excellent energy resolution, since the entire volume is sensitive material. Thanks to this structure, the TASC therefore measures the total energy of the shower generated by the primary particle and makes it possible to distinguish between electromagnetic and hadronic showers, separating events due to electrons from the hadronic background, which consists almost exclusively of protons.

Inside it there are 12 planes of lead tungstate (PbWO_4) crystal bars, shaped as square-based rectangular parallelepipeds (20 mm $\times$ 20 mm $\times$ 320 mm each). The planes

are alternately rotated by 90° relative to each other, to provide the two xz and yz views of the shower's development.

Each of the 12 planes contains 16 bars, for a total of 192 lead tungstate crystals inside the TASC calorimeter.

The TASC has a depth of 24 cm, corresponding to $27 X_0$ radiation lengths and $1.2 \lambda_I$ nuclear interaction lengths.

This implies almost total containment for electromagnetic showers and, conversely, considerable energy loss from the bottom of the calorimeter for hadronic showers, since their longitudinal containment is only partial. This fact is exploited to achieve a high rejection of the proton background in electron flux measurements.

In order to measure electrons up to tens of TeV, a particularly wide dynamic range is required: the system must in fact be able to detect energy deposits from 0.5 MIP (Minimum Ionizing Particle) up to about 10^5 MIP for 10 TeV electrons. This dynamic range therefore spans about 6 orders of magnitude and is covered by using two photodiodes with different sensitive areas for reading each crystal, and dual-gain readout electronics for each photodetector [14]. [Figure 2.8](#) shows the interior of the calorimeter, specifying the dimensions of its subsystems.

Note that CALET is a purely calorimetric instrument and is not equipped with a spectrometer: both the measurement of the deposited energy and the rejection of the proton background are carried out by the TASC, which, for this reason, constitutes the 'heart' of the whole instrument.

CALET is able to measure the electron spectrum with an energy resolution of the order of $2 \sim 3\%$ and with a background rejection power of the order of 10^5 [16].

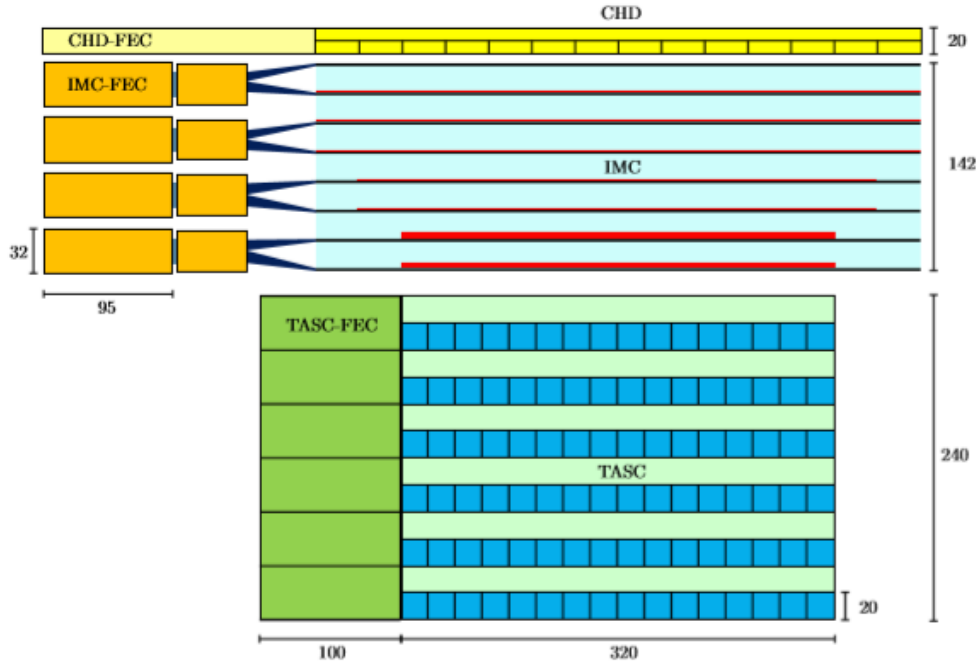


Figure 2.8. dimensions and positions of the sub-detectors in the CALET calorimeter, measurements in mm, yz view.

Trigger systems

In CALET, several hardware triggers are used for data acquisition and storage, configurable through threshold discriminators in the three previously described detectors (CHD, IMC, and TASC).

The main trigger modes used to carry out data collection in CALET are:

- *High Energy* shower trigger (HE), to detect with an efficiency of $\epsilon > 95\%$ electrons with energy $E > 10$ GeV. This trigger makes it possible to select those events that give rise to showers inside the calorimeter;
- *Low Energy* shower trigger (LE), to detect electrons with energy $(1 < E < 10)$ GeV in the polar regions and low energy gamma rays ($E > 1$) GeV at low latitudes;
- *Single* trigger, used to calibrate the instrument by acquiring data from non-interacting particles.

2.3 CALET results: helium nuclei spectrum

From “Direct Measurement of the Cosmic-Ray Helium Spectrum from 40 GeV to 250 TeV with the Calorimetric Electron Telescope on the International Space Station” [17] and its *supplemental material*. The main results presented in the article are reported below.

The flight data analyzed in the study cited above were collected from 13 October 2015 to 30 April 2022, (2392 days). The total direct observation time amounts to 48459.7 hours, and the fraction of direct live time relative to the total time is 84.4%. The data analysis procedure is similar to that used for protons [18, 19], carbon C and oxygen O [20], iron Fe [21], and nickel Ni [22].

A high-efficiency reconstruction of the direction of hadronic tracks is of primary importance for the flux measurement. For this reason, as in the analysis of the proton spectrum, the combinatorial Kalman filter algorithm (KF, in [section A.1](#)) is also used for the helium nuclei spectrum, as it provides good results for reconstructing the direction of the tracks of the incident particles.

The energy of the shower generated by each event is calculated from the sum of the energy deposits in the TASC (E_{TASC}).

Monte Carlo simulations were carried out, based on the [EPICS](#) simulation package [23], to estimate the reconstruction efficiencies. The instrument was calibrated in a series of *beam tests* at CERN-SPS [24, 25].

The trigger efficiency and energy response obtained from the MC simulations were tuned using the results of the 2015 beam test, carried out with ion beams of energy equal to 13, 19, and 150 GeV/n. For helium nuclei showers, a correction of 10.4% was applied at 13 GeV/n and of 8% at 19 GeV/n, while 3.2% at energies $E > 150$ GeV/n. The correction factors for intermediate energy values (not measured at CERN) were calculated by performing a logarithmic-linear interpolation. No corrections were instead applied to the trigger efficiency, since the beam test measurements were found to be consistent with the MC simulations.

In the analysis of hadrons, particularly at high energies, where it is not possible to carry out a calibration beam test of the instrument, it is necessary to compare different MC models. For this purpose, further MC simulations were carried out using [FLUKA](#) [26] and compared with those carried out using EPICS.

A preselection of ‘well reconstructed’ and ‘well contained’ events within the calorimeter is then applied, even before proceeding with charge identification, so as to minimize background contamination of the selected helium nuclei.

The main selection criteria (“cuts”) applied to the data are reported below:

- *Trigger*: only those events that ‘fired’ the High Energy shower trigger (HE) are

selected. Consistency between flight data (FD) and simulated MC data for events selected with the HE trigger is achieved by applying an offline trigger that requires stricter conditions than the onboard trigger. This makes it possible to remove residual errors due to positional and temporal variations in detector gain.

- *Track quality*: the selected events are required to have a ‘good’ primary track reconstructed in both views with the KF algorithm. In detail, a certain minimum number of reconstructed points is required for each track segment, and a cut is also applied on the value of the fit’s χ^2 . In this way, an angular resolution for helium nuclei of $\sim 0.1^\circ$ is achieved, along with a spatial resolution on the impact point (IP) of $\sim 400 \mu\text{m}$ on the upper plane of the CHD.
- *Geometric conditions*: the selected events are required to have “acceptance A”, that is, to pass through the entire calorimeter from the upper plane of the CHD to the bottom of the TASC, and within 2 cm of the lateral walls of the TASC.
- *Electron rejection*: an electron rejection cut is applied, based on the fractional quantity known as the ‘Molière concentration along the track’, calculated by first summing all the energy deposits within a Molière radius around each IMC fiber crossed by the track, and then normalizing with respect to the sum of the total energy deposits in the IMC. By requiring this quantity to be < 0.75 and the fraction of TASC energy deposited in the last layer to be > 0.01 , more than 90% of electrons are rejected, while maintaining a very high efficiency for helium nuclei ($> 99.9\%$ for $E > 50 \text{ GeV}$).

The width of the energy bins was chosen to be proportional to the TASC resolution. In this way, the energy spectrum, shown in [Figure 2.9](#), is obtained from the energy distribution as follows:

$$\Phi(E) = \frac{N(E)}{\Delta E \times \epsilon(E) \times S\Omega \times T}$$

$$N(E) = U[N_{\text{obs}}(E_{\text{TASC}}) - N_{\text{bg}}(E_{\text{TASC}})]$$

where ΔE denotes the width of the energy bins; E is the kinetic energy of the particle, calculated as the geometric mean of the lower and upper bounds of the bin; $N(E)$ is the number of counts in the bin corresponding to a given E ; $\epsilon(E)$ is the overall efficiency parameter; T is the instrument’s livetime; $S\Omega$ is the geometric acceptance; U is the *unfolding* procedure; $N_{\text{obs}}(E_{\text{TASC}})$ is the content of the bin of the observed energy distribution (including background); and $N_{\text{bg}}(E_{\text{TASC}})$ is the background events in the same bin.

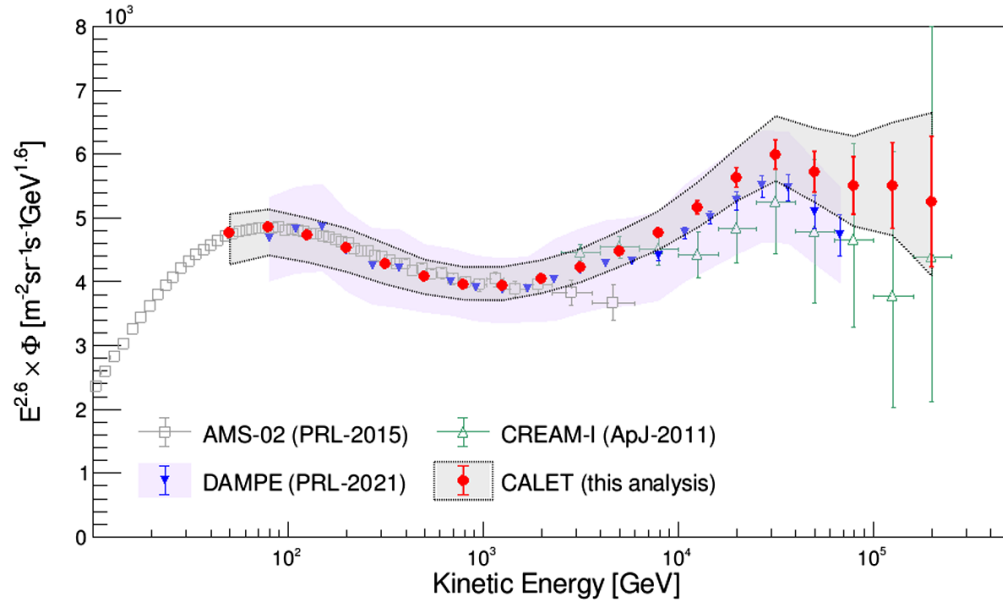


Figure 2.9. cosmic-ray helium nuclei spectrum measured by CALET, shown in *red*, compared with previous direct observations [27, 28, 29]. The error bars represent only the statistical error; the gray band indicates the quadratic sum of systematic and statistical errors, while the purple band shows the systematic uncertainties.

The data measured by CALET were fitted in the range from 60 GeV to 250 GeV using a *double broken power law* (DBPL) function:

$$\Phi(E) = C \left(\frac{E}{\text{GeV}} \right)^{\gamma} \left[1 + \left(\frac{E}{E_0} \right)^S \right]^{\frac{\Delta\gamma}{S}} \left[1 + \left(\frac{E}{E_1} \right)^{S_1} \right]^{\frac{\Delta\gamma_1}{S_1}}$$

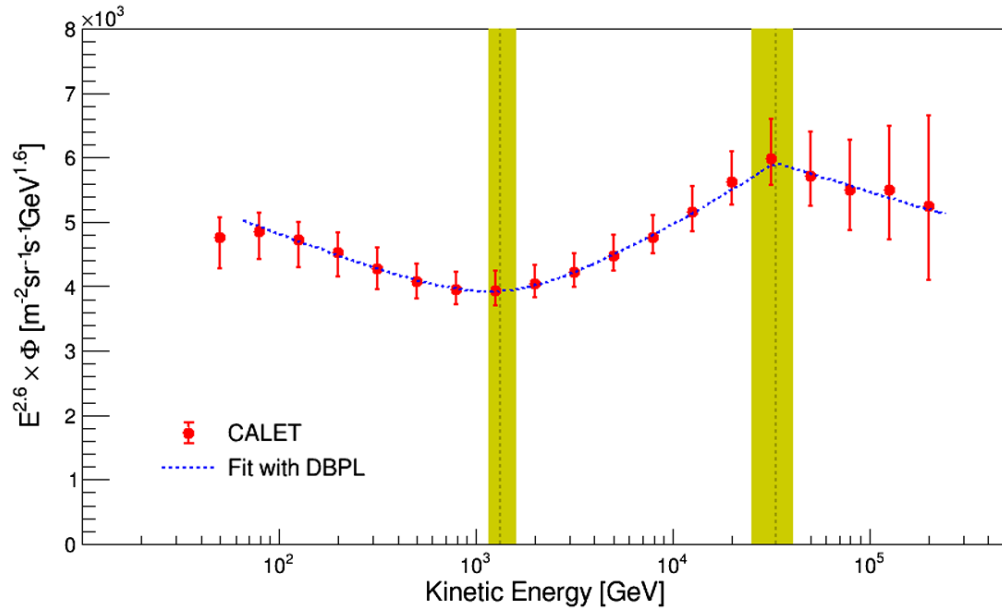


Figure 2.10. fit of the data recorded by CALET with a DBPL function. The results agree, within the margin of error, with other recent measurements [27]. Both systematic and statistical errors were taken into account.

Finally, two significant variations in the spectral index are observed in the spectrum, showing the presence of *hardening* at an energy of about 1319 GeV and *softening* at about 33.2 TeV, highlighted by the yellow bars in Figure 2.10. (For numerical values of the fit, see [17]).

The next chapter ([chapter 3](#)) explains the procedure followed for building and training the CNN model subsequently used to analyze CALET flight data.

Chapter 3

Structure and training of the neural network

3.1 Introduction

In [chapter 1](#), neural networks were introduced. For a generic network, we can identify its structure as a hierarchy that relates the following elements:

- *layers*;
- corresponding *input data* and *targets*;
- *loss function*;
- *optimizer*.

The *layers* are the ‘layers’ that form the structure of the network and to which the *input data* and the corresponding *targets* are passed to perform the training. During this last phase, a *loss function* is used to have a feedback signal, i.e., a measure of success of the network: the lower the value of this function, the better the agreement between predictions and targets; it is therefore the quantity to be minimized during learning. Finally, an *optimizer* is used to determine the learning process to follow and therefore how to modify the weights of the layers in order to minimize the loss function. This process is schematically shown in [Figure 3.1](#).

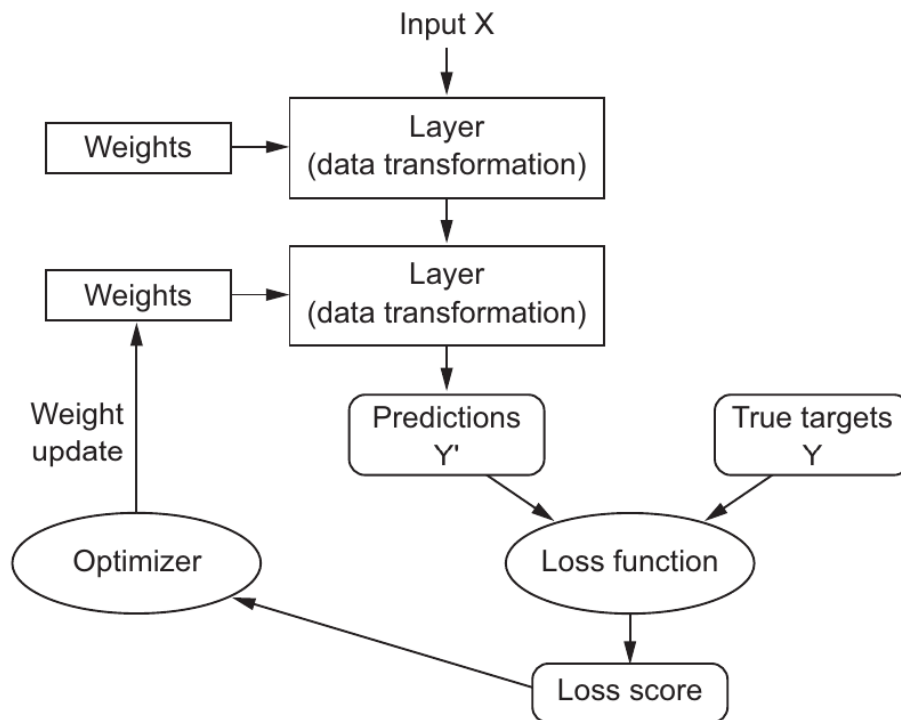


Figure 3.1. in the flowchart it is possible to appreciate schematically the relationship between these elements: there is a succession of layers, where each, with its own weights, performs a series of operations on the input data and maps them to the predictions; then the distance of these from the targets (true values) is measured via the loss function and then the update of the weights is performed via the optimizer.

The layer constitutes the fundamental structure of a neural network; the data it receives as input are represented via tensors and are transformed according to a series of operations. These are regulated by the weights of the layer, which perform the function of returning a new tensor (or more than one) as output, modified in a direction that increases the ‘knowledge’ of the network and therefore improves its ability to make accurate predictions.

3.2 Description of the problem

The work carried out in this thesis focused on measuring the performance of a CNN neural network applied to the problem of reconstructing the trajectory of charged cosmic rays in the CALET experiment. Some characteristics of the selected events and the strategy adopted to perform the study are explained in greater detail below.

3.2.1 Acceptance

Inside the calorimeter we can identify 3 main subsystems: *CHarge Detector* (CHD), *IMaging Calorimeter* (IMC), *Total AbSorption Calorimeter* (TASC). Each time an event is recorded, it is possible to reconstruct the trajectory followed by the primary particle using algorithms (e.g., *Kalman filter* in [section A.1](#), *shower axis method* in [section A.2](#)) that analyze the ionization signals it releases in the crossed CHD scintillators, IMC fibers, and TASC crystals. Events are classified into 4 categories of geometric acceptance (A, B, C, D) as shown in [Figure 3.2](#).

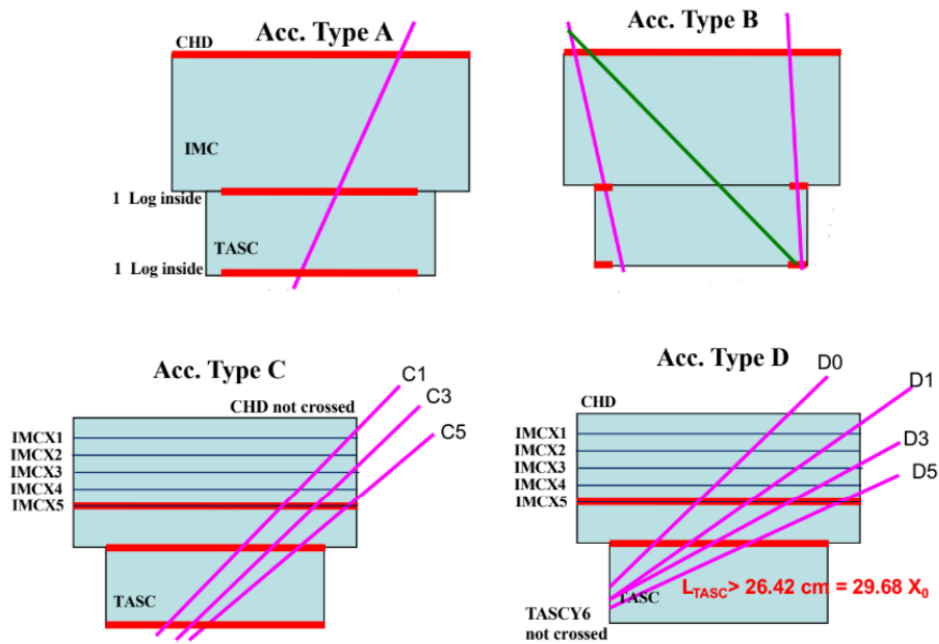


Figure 3.2. schematization of the CALET calorimeter in which the direction of particles classified into the different acceptance categories is shown in *purple*.

Acceptance A events are those ‘best’ contained in the detector, i.e., those that enter the upper plane of the CHD, cross the first plane of the TASC and reach its bottom

while remaining within 2 cm from the lateral edges.

On the other hand, if the particle follows the same trajectory, but reaches the bottom of the TASC at a distance of less than 2 cm from the lateral edges, we speak of events with acceptance B.

Then, events whose trajectory does not cross the CHD and enter the calorimeter from one side of the IMC to then continue well contained in the TASC have acceptance C.

Finally, acceptance D events are those that develop a shower longer than 27 cm (about 30 radiation lengths) in the TASC, but do not reach its top floor, exiting from one of the lateral edges.

In this study, acceptance A events were processed and analyzed.
(`tree.fAccType[0] == 1` in the `.root` file)

3.2.2 Strategy and procedure

The procedure followed to arrive at the ultimate goal of this work, namely reconstructing the direction of cosmic rays with a neural network applied to CALET data and evaluating the spatial and angular resolution of the method, is schematically illustrated below:

- event simulation with the CALET Monte Carlo (MC) software;
- writing the **Python** code of the CNN;
- training of the CNN on the MC simulation data;
- analysis of the resolution obtained on the simulated MC data and comparison with the shower axis method ([chapter 4](#));
- trajectory reconstruction of flight data via CNN;
- analysis of the resolution obtained with the CNN on flight data and comparison with that of the Kalman filter ([chapter 5](#)).

The variables under study are therefore the pairs of coordinates (x_0, y_0) , (x_1, y_1) of the impact points respectively on the first plane of the CHD and on the bottom of the TASC, and the *angular variables* (θ_x, θ_y) . The angular variables indicate the angle formed by the direction of the particle's trajectory ([Figure 3.2](#)) with the calorimeter planes, respectively in the x and y views.

The value of these variables can be reconstructed from the impact points as follows:

$$\theta_x = \arctan \left[\frac{x_1 - x_0}{z_1 - z_0} \right] \quad \theta_y = \arctan \left[\frac{y_1 - y_0}{z_1 - z_0} \right]$$

These variables allow the reconstruction of the calorimeter crossing trajectory.

Note that in the study of the CNN resolution on simulated data it is possible to make a comparison between predicted and true values (as the truth is provided by the simulation); whereas in the study of the CNN resolution on flight data this is not possible (also because, if it were not so, it would mean already having knowledge of the impact point and it would therefore not be necessary to reconstruct it). In this case, we therefore proceeded to measure the resolution of the CNN by comparing the predictions with the reconstructions of the impact points performed via Kalman filter. The resolution of the KF on the MC data was then measured and, finally, the effective resolution of the CNN on the FD was estimated by subtracting that of the KF in quadrature.

3.2.3 Simulated MC data

The MC simulations of CALET crossing events are performed using the Monte Carlo software **EPICS** [23], which allows reproducing in a virtual environment the geometry of the calorimeter, the radiation-matter interaction processes, and the signals that are processed in the various detectors.

The interactions of hadronic showers in the calorimeter are modeled using the software **DPMJET-III** [30].

In the MC program, particles are generated isotropically on a sphere of radius 1 m with the center placed in the middle of the upper plane of the CHD. Those events leaving the sphere that have a trajectory passing through the calorimeter are propagated and the energy deposits in the detectors are written into a **TTree** inside a `.root` file [31].

The kinetic energy of the primary particle is chosen randomly from a power-law distribution E^{-1} and subsequently the data are reweighted to simulate the observed spectra which have power laws with a spectral index between $(-2.7, -2.6)$.

Finally, the simulated data are processed by CREC, the official program of the Italian CALET collaboration, which produces the `L2.root` file inside which there is a **TTree** containing the reconstructed information for each event.

For this thesis work, the following variables of the simulated MC data were used:

- `fTASCElayer` [12], the energy deposited in each plane of the TASC;
- `fAccType`, the acceptance type calculated using the event generation parameters;
- `fIP (x0, y0, x1, y1)`, the true values of the simulated impact points;
- `fTASCparx` and `fTASCpary`, the particle direction parameters reconstructed with the shower axis method;

- fTASCIing, images of hadronic showers in the TASC, like those in [Figure 3.3](#), whose pixels are colored with an 8-bit grayscale and are proportional to the deposited energy (input data to the CNN).

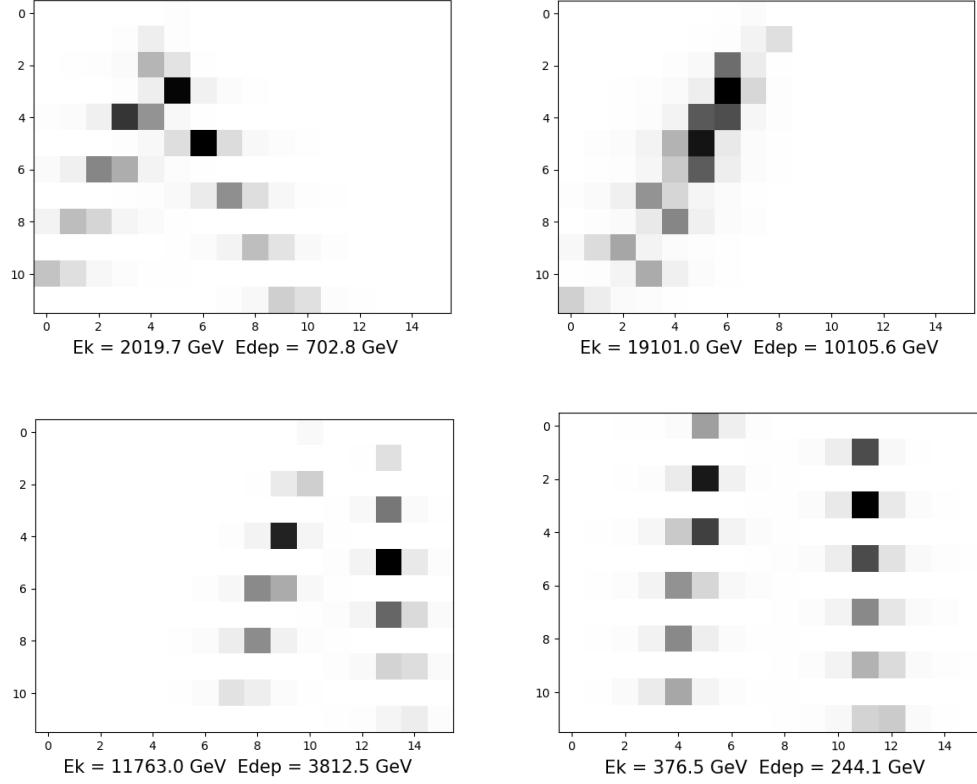


Figure 3.3. images of 4 events in the TASC, displayed in both views. Starting from the top, the odd-numbered images represent the xz view, the even-numbered ones the yz view. For each event, the kinetic energy of the incident particle and the energy deposited in the TASC are reported.

3.3 Coding and network development

The environment used for the development and training of the network was the deep-learning framework for Python, [Keras](#) [32].

It was chosen to use Keras because it allows running the code on both CPU and GPU, furthermore it provides a user-friendly API that simplifies the design operations of a network model and also has built-in support for convolutional networks.

Keras is distributed under the MIT license, which means it can be freely used in commercial and research projects; it is compatible with every version of Python starting from 2.7.

3.3.1 Network model with Keras

The Keras framework was used to define the structure of the neural network and perform the training.

The Python coding operations performed to produce the code with which to train the model are reported below:

1. definition of training data: modified in order to have *input* and *target* in the form of tensors;
2. definition of the network structure;
3. setting up the learning process, choosing a loss function and an optimizer for measuring and correcting the error on the predictions;
4. iteration of the procedure on the training data via the Keras `fit()` method.

3.3.2 Workstation

The neural network model was trained exploiting the computational power provided by a workstation with

CPU: INTEL CORE I7 at 3.60 GHz

GPU: [Asus NVIDIA GeForce RTX3050 8 GB](#).

In fact, it is highly recommended, although not strictly necessary, to run the code on an NVIDIA graphics card: some deep learning applications, in particular image processing via convolutional networks, turn out to be particularly slow if executed on a CPU. The execution speed on a GPU is a factor of 5 or 10 times higher than that on a CPU.

3.3.3 Network structure

The neural network used in this study was trained with images of the xz and yz views of hadronic showers in the TASC with dimensions of 12×16 pixels, where these are colored according to an 8-bit grayscale and are proportional to the deposited energy ($fTASCimg$).

As output, the network returns tensors of size $(4 \times nev)$, i.e., there is a ‘row’ of four variables indicating the coordinates (x_0, y_0) , (x_1, y_1) of the impact points reconstructed by the network, respectively on the first plane of the CHD and the last of the TASC. Usually, the 4D tensor containing the event images is processed via 2D convolutional layers (Conv2D). The reconstructed impact points were then used to evaluate the spatial and angular resolution of the network, comparing them with the ‘true’ values of the MC, i.e., with the reference *targets*. The structure of the network used in this study is reported below.

```

1  inputs=Input(shape=(12, 16, 1))
2
3  x=layers.Conv2D(256, (3, 3), activation='relu')(inputs)
4  x=layers.Conv2D(64, (3, 3), activation='relu')(x)
5  x=layers.Conv2D(32, (3, 3), activation='relu')(x)
6  x=layers.Flatten()(x)
7  x=layers.Dense(100, activation='relu')(x)
8  x=layers.Dense(50, activation='relu')(x)
9
10 outputs = layers.Dense(4)(x)

```

The structure is to be read from top to bottom. There are therefore three convolutional layers Conv2D that return outputs of size 256, 64, 32 respectively and receive inputs with a size equal to the output of the previous layer, or of the data themselves in *inputs*. A Flatten layer follows, which makes the output one-dimensional, and three Dense layers that provide a different representation of the input tensor via the *rectified linear unit* (relu) activation function.

This structure was chosen taking into account the results obtained in [33]. In ‘Study 1: comparison between CNNs with different structures’, the number of training epochs is optimized to avoid overfitting and underfitting phenomena, and the performances of networks with different structures (number of layers) are compared.

The optimal number of epochs turns out to be 40, while the best structure is the one with 3 layers.

Layer (type)	Output Shape	Param #
input_1 (InputLayer)	[(None, 12, 16, 1)]	0
conv2d (Conv2D)	(None, 10, 14, 256)	2560
conv2d_1 (Conv2D)	(None, 8, 12, 64)	147520
conv2d (Conv2D)	(None, 6, 10, 32)	18464
flatten (Flatten)	(None, 1920)	0
dense (Dense)	(None, 100)	192100
dense_1 (Dense)	(None, 50)	5050
dense_2 (Dense)	(None, 4)	204

Once the structure of the network is defined, one proceeds by specifying which loss function and optimizer to use, i.e., declaring the optimizer and metric with which to supervise the training phase.

```
1 model.compile(optimizer=optimizers.RMSprop(learning_rate=1e-4),
    loss='mse', metrics=['mae'])
```

As loss function, the *mean squared error* (mse) was chosen, that is, the root of the mean value of the squared distance between targets and predictions of the CNN.

As metric, the *mean absolute error* (mae) was chosen, which represents the absolute value of the difference between targets and predictions.

As optimizer, the *root mean squared propagation* (RMSprop) algorithm was chosen with a *learning rate* equal to 10^{-4} .

3.4 Network training

In this section, the procedure followed to perform the training of the neural network is described, focusing in particular on the model and the data analyzed in this study.

3.4.1 Data preparation

The data from the MC simulations are contained in .root files, inside a TTree.

The files were divided into energy ranges: for protons four files are produced (20-200, $200-2 \times 10^3$, $2 \times 10^3-2 \times 10^4$, $2 \times 10^4-10^6$) [GeV]; while for helium nuclei three are produced (20-200, $200-2 \times 10^3$, $2 \times 10^3-2 \times 10^4$) [GeV].

The variables extracted from each file are saved in `np.array()` tensors of the Python `NumPy` library [34] listed below:

- nev, number of selected events in the TTree;
- data, selected input data;

- targets, targets corresponding to the input data;
- TASCip, impact points in the TASC;
- TASCedep, energy deposited in the TASC;
- AccType, acceptance parameter;
- probe, parameter to distinguish whether the event relates to a proton or a helium nucleus.

The event selection process takes place sequentially, analyzing each file one after the other. An event selection filter is applied to each file: only those satisfying the following condition are extracted and saved in the corresponding tensors.

```
1 if (tree.fAccType[0]==1
2   and tree.fTASCElayer[12]<EDEPTH
3   and tree.fTASCElayer[12]>20
4   and emaxlay[0]<0.45
5   and emaxlay[1]<0.45):
```

Where, with `tree.fAccType[0]==1`, ‘acceptance A’ events are selected; with `tree.fTASCElayer[12]<EDEPTH` and `tree.fTASCElayer[12]>20`, the value of the energy deposited in the TASC, i.e., (E_{TASC}), is required to be in the range $(20, 2 \times 10^4)$ [GeV].

The data variable is a 4-dimensional tensor in which the values of the inputs selected via two for loops on the variables `j` and `k` are saved.

```
1 data[nev, j, k, 0]=tree.fTASCimg[j*16+k]
```

This allows creating (visually) ‘stacks’ of matrices of unit ‘thickness’, where each matrix coincides with an image (of the impact points) contained in the `.root` file and each element of a matrix coincides with the pixel at position (j, k) of the image. The height of each ‘stack’ depends on the number of stacked matrices, i.e., on the number of images saved in the tensor and therefore on the number of events selected in each file (`nev`).

3.4.2 Training and saving predictions

Once the data of interest have been extracted from the `.root` file, we proceed with their subdivision into three categories: *train*, *validation*, *test*.

In the study in question, it was chosen to distribute them according to the percentages:

```
1 train=0.8
2 val=0.1
3 test=0.1
```

where the `train` variable indicates the percentage of data to be used as data to perform the training of the network; while the `val` variable indicates the percentage of data to be used to perform the validation of the model, i.e., to measure the accuracy of the training operation; finally the `test` variable indicates the percentage of data on which not to train the network to test its ability on a set of ‘new’ data, i.e., data for which the network does not know the truth.

We then continue with the training phase, calling the `model.fit()` function and passing the `test` and `val` data as input:

```
1 history=model.fit(train_data, train_targets, epochs=epoche,
    batch_size=32, validation_data=(val_data, val_targets),
    callbacks=None)
```

This string launches and performs the training of the model for the number of epochs passed as input (40, in our study). Subsequently, the newly trained model is tested on the portion of data dedicated to the test, to measure its accuracy also on data not coming from the training dataset:

```
1 test_pred = model.predict(test_data)
2 model.save(path_model_save, overwrite=True)
```

Then, via the `model.save()` command reported in the second line, the obtained model is saved, so that it can be used to produce predictions on other datasets without having to perform the training again every time.

Finally, the predictions produced by the network and the values of `mae` and `mse` measured during the training phase are saved in an output `.root` file. This file is subsequently used to perform the analysis of the data obtained and produce the relative graphs. ([Appendix B](#)).

3.5 Application of the CNN to data

In this thesis work, the model was trained on two datasets: simulated MC data of helium nuclei and combined data of protons and helium nuclei.

The two models were then used to make predictions on the simulated MC data of helium nuclei and evaluate which of the two was the most precise, i.e., had better

spatial resolution.

Subsequently, both models were used to reconstruct the impact points of a dataset of ‘true’ data, for which the truth is therefore not known: for this reason, in order to estimate the resolution of the method in this case, it was necessary to compare the predictions of the latter with those of another method, the *combinatorial Kalman filter* algorithm (of which more details are in [section A.1](#)).

In the next chapters, [chapter 4](#) and [chapter 5](#), the analyses of the results obtained by the CNN on the MC and FD data respectively are performed.

Chapter 4

Analysis of MC simulation data

The neural network model previously described was tested on CALET simulation data in order to measure its performance and, in particular, the spatial and angular resolution on the reconstructed direction of the particles. This allowed for a comparison with the results obtained using the shower axis reconstruction method (in [section A.2](#)).

In particular, two models were trained, both with the same structure, but on different datasets: one on simulated MC data of helium nuclei and the other on combined data of protons and helium nuclei.

4.1 Introduction

Below are the results obtained from the data analysis performed on the output data produced by supplying the two datasets to the CNN described in [subsection 3.3.3](#).

Operationally, to analyze these data, a ROOT ‘macro’ was written that performs the following operations:

- accepts as input the files processed by the CNN model;
- calculates the residual distributions of the spatial variables, i.e., the difference between the value predicted by the network and the true value from the Monte Carlo simulation for each variable of the impact point tensor (x_0, y_0) , (x_1, y_1) ;
- calculates the residual distributions of the angular variables¹ (θ_x, θ_y) ;
- calculates the spatial resolution;
- calculates the angular resolution.

The distributions were analyzed in 14 intervals of deposited energy in the TASC (E_{TASC}). For the data analysis of the results produced by the CNN trained on combined p and He data, only the predictions relating to helium nuclei were selected.

¹Defined in [subsection 3.2.2](#).

Range [GeV]			Events				
20	↔	31	1827	411	↔	632	3244
31	↔	47	2215	632	↔	974	3286
47	↔	73	2621	974	↔	1500	3353
73	↔	112	2722	1500	↔	2310	3297
112	↔	173	2961	2310	↔	3557	3120
173	↔	267	3041	3557	↔	5477	2481
267	↔	411	3128	5477	↔	2×10^4	932

The criterion adopted in dividing the energy range ($20, 2 \times 10^4$) [GeV] is as follows: intervals of constant width on a logarithmic scale were used, i.e., such that $\log_{10}(E_{i+1}/E_i) = \text{const}$ for all intervals except the last one. From 5477 to 2×10^4 [GeV], it was decided to extend the interval to collect all the remaining events.

4.2 Data analysis

The events produced by the CALET MC simulation were saved in three files, depending on the energy range they belong to ($20\text{--}200$, $200\text{--}2 \times 10^3$, $2 \times 10^3\text{--}2 \times 10^4$) [GeV] and, as explained in [chapter 3](#), the CNN was first *trained* and *validated* on 90% of the data present in the dataset, while it was then *tested* on 10% of them.

To evaluate the efficiency of the network in predicting the impact points (spatial variables) and angular variables, and therefore the spatial and angular resolution, the residual distributions between these values and the Monte Carlo truth were calculated.

As explained in greater detail on [page 36](#), the data present in the dataset are images of hadronic showers in the TASC, from which high and low energy p and He were selected. Some examples of these images are shown in [Figure 3.3](#).

In order to concretely evaluate ‘how well’ the network performs, it is necessary to know the ‘true’ value and measure how far apart it and the predicted value are. Indeed, it would not have been possible to use the CNN directly to make predictions on CALET flight data (FD) since the ‘true’ value is not available for them—which is precisely what we want to estimate—and therefore it would not have been possible to have an estimate of the error and resolution of the network in making this prediction.

Therefore, it was first of all necessary to evaluate the resolution of the CNN on simulated data and only subsequently apply it to flight data.

Below are some plots of the residual distributions and resolution for both the CNN trained on helium nuclei and the CNN trained on combined p and He data. The most important features are highlighted for each plot.

(Only the plots relating to the spatial variable x_0 and angular variable θ_x have been included in this chapter. For the others, see [Appendix B](#)).

4.2.1 Training, *mae* and *mse*

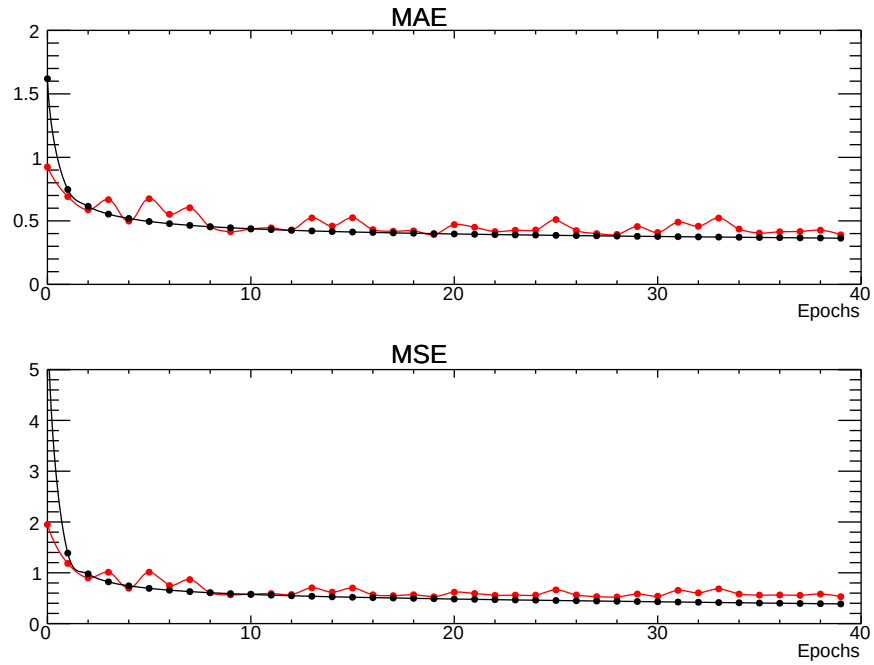


Figure 4.1. plots of the *mean absolute error* (*mae*) and *mean square error* (*mse*) relating to the CNN trained on helium nuclei. In *black*, the training curve; in *red*, the validation curve.

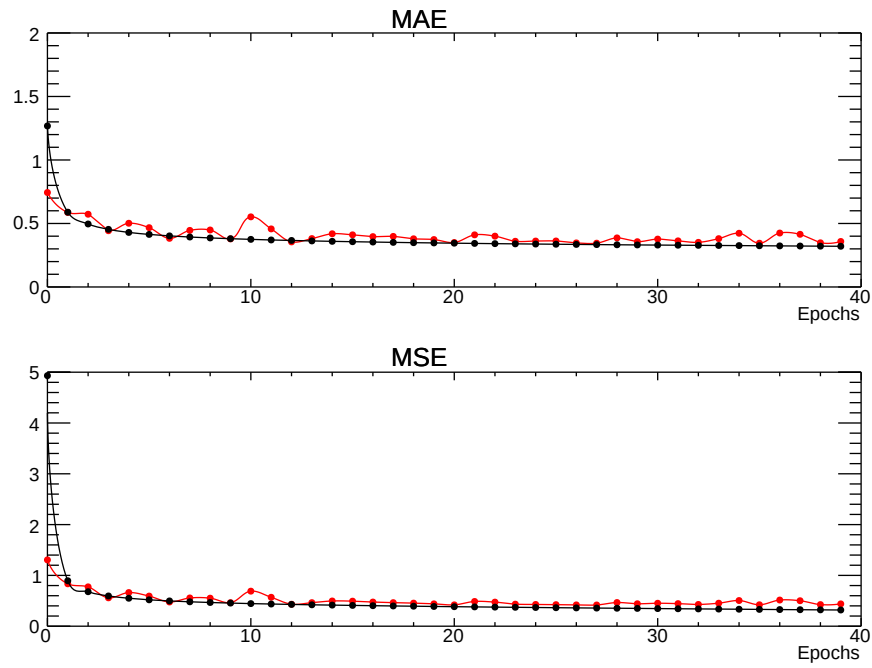


Figure 4.2. plots of the *mean absolute error* (mae) and *mean square error* (mse) relating to the CNN trained on the combined dataset of p and He. In *black*, the training curve; in *red*, the validation curve.

It is observed that training for 40 epochs stabilizes both the mae and the mse at values close to 0.5 and prevents overfitting problems.

4.2.2 Residual distributions

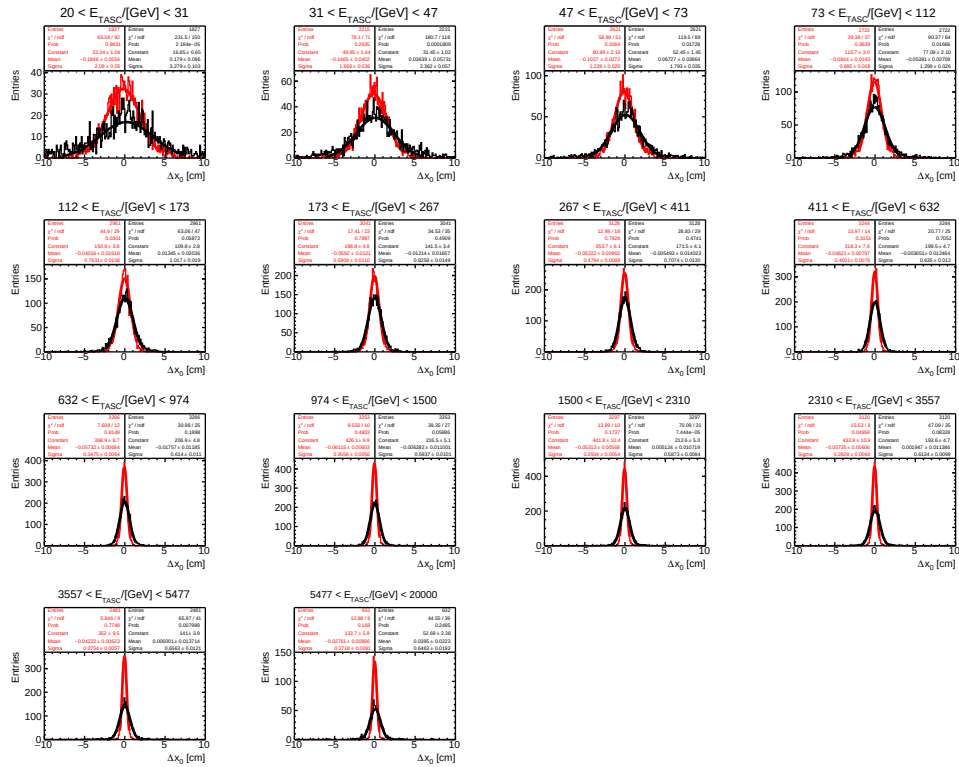


Figure 4.3. residual distribution of the impact point variable x_0 predicted by the CNN trained on helium nuclei (histogram in *red*) and by the shower axis method (histogram in *black*). The curves superimposed on the histograms represent the Gaussian fits to the histograms of the same color.

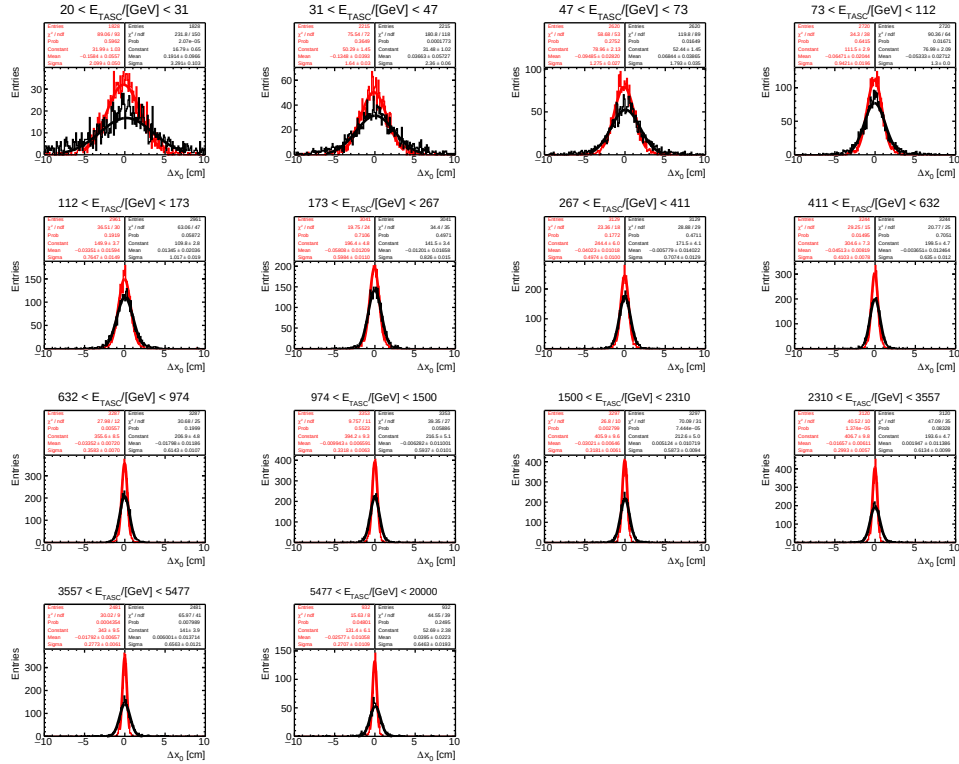


Figure 4.4. residual distribution of the impact point variable x_0 predicted by the CNN trained with the combined dataset of p and He (histogram in *red*) and by the shower axis method (histogram in *black*). The curves superimposed on the histograms represent the Gaussian fits to the histograms of the same color.

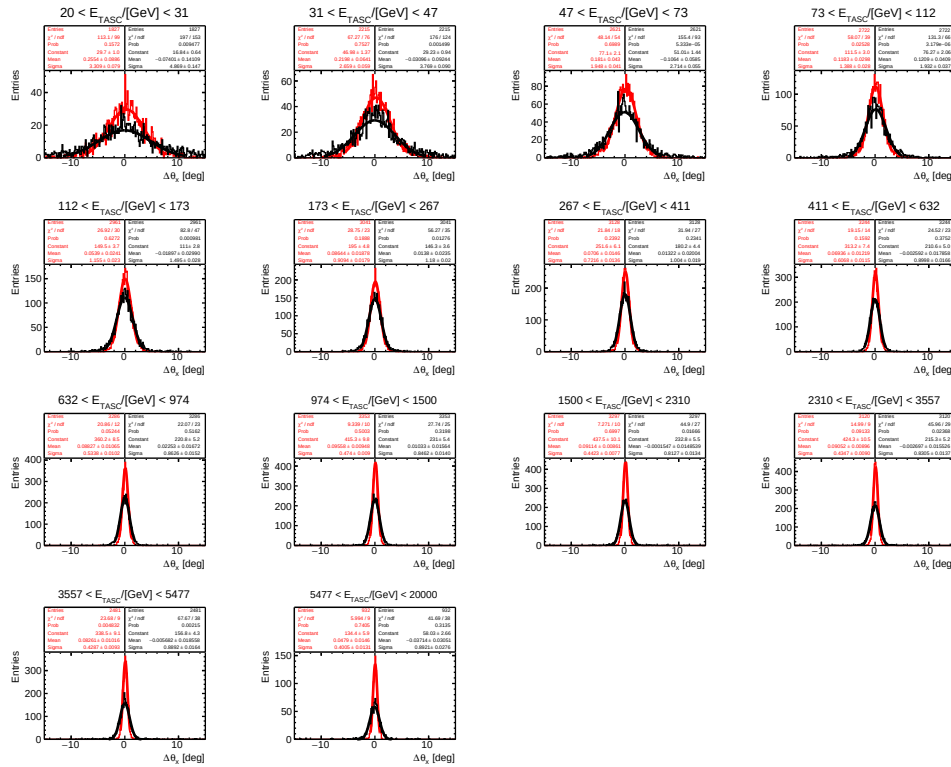


Figure 4.5. residual distribution of the angular variable θ_x predicted by the CNN trained on helium nuclei (histogram in *red*) and by the shower axis method (histogram in *black*). The curves superimposed on the histograms represent the Gaussian fits to the histograms of the same color.

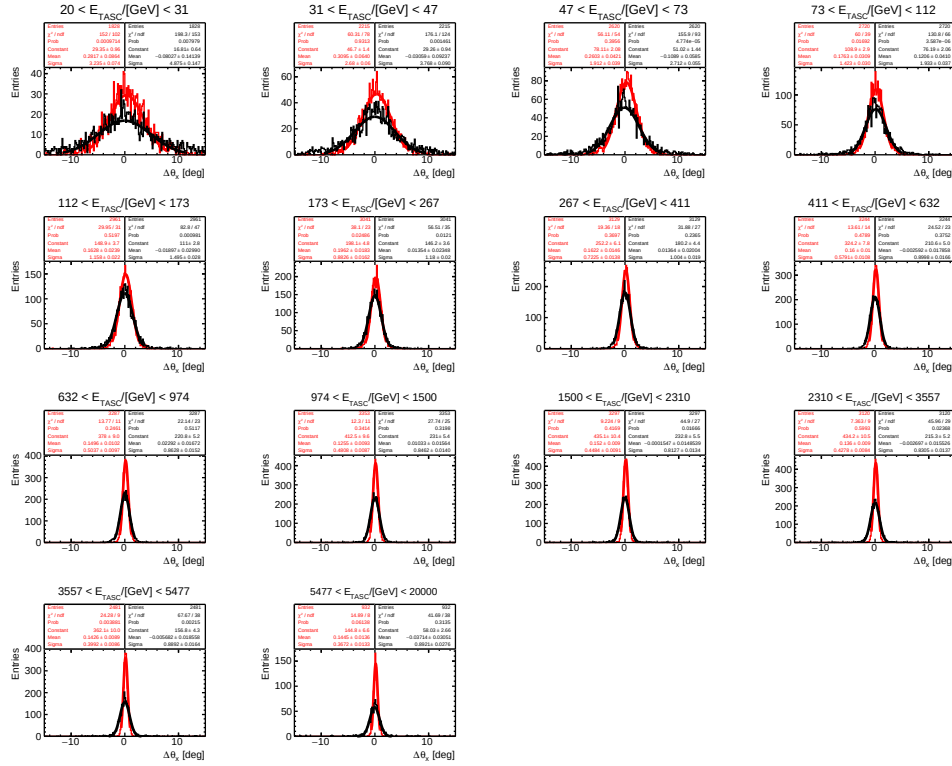


Figure 4.6. residual distribution of the angular variable θ_x predicted by the CNN trained with the combined dataset of p and He (histogram in *red*) and by the shower axis method (histogram in *black*). The curves superimposed on the histograms represent the Gaussian fits to the histograms of the same color.

4.2.3 Mean of residuals and spatial resolution

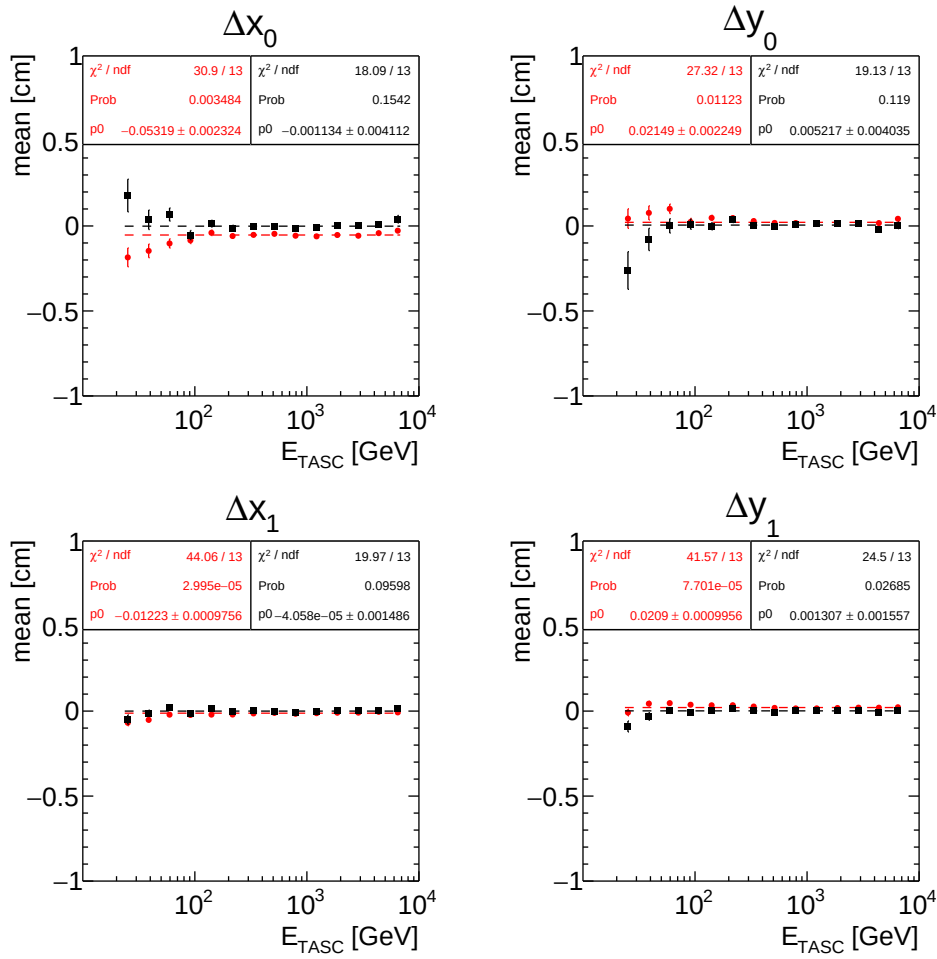


Figure 4.7. mean value of the residual distribution of the impact points for the CNN trained on helium nuclei (in *red*) and the shower axis method (in *black*), as a function of the average deposited energy in each range.

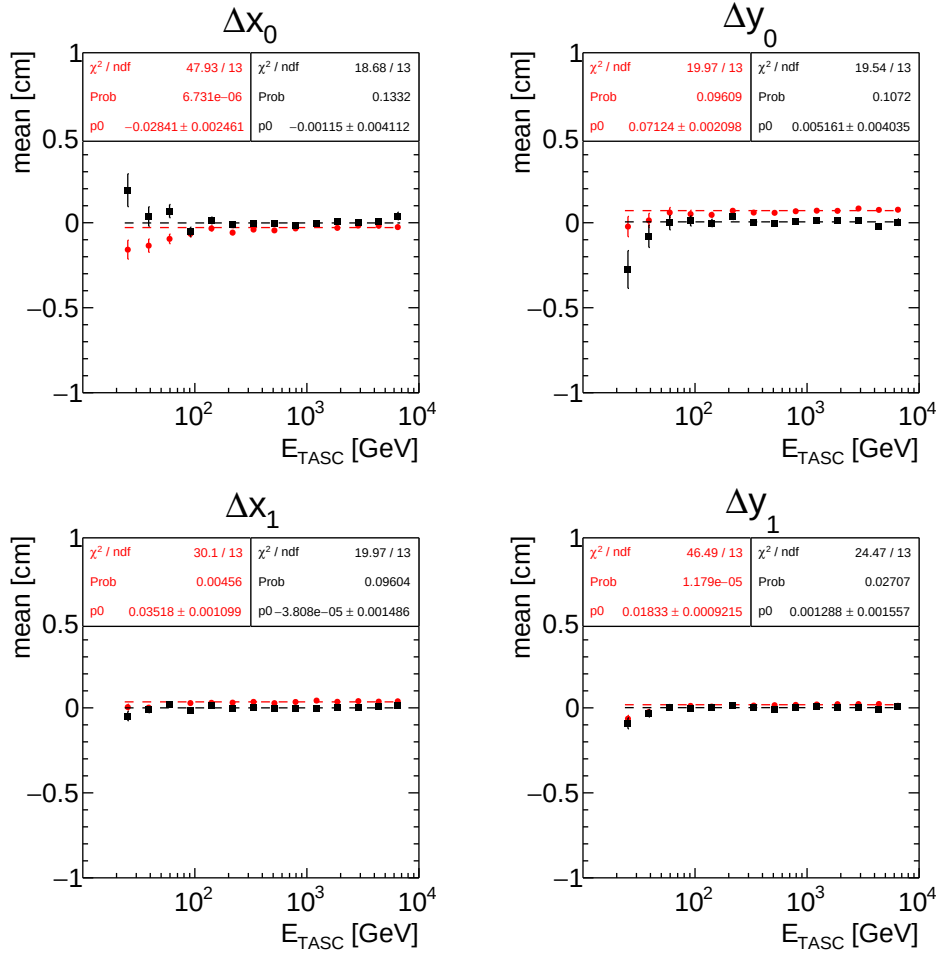


Figure 4.8. mean value of the residual distribution of the impact points for the CNN trained with the combined dataset of p and He (in *red*) and the shower axis method (in *black*), as a function of the average deposited energy in each range.

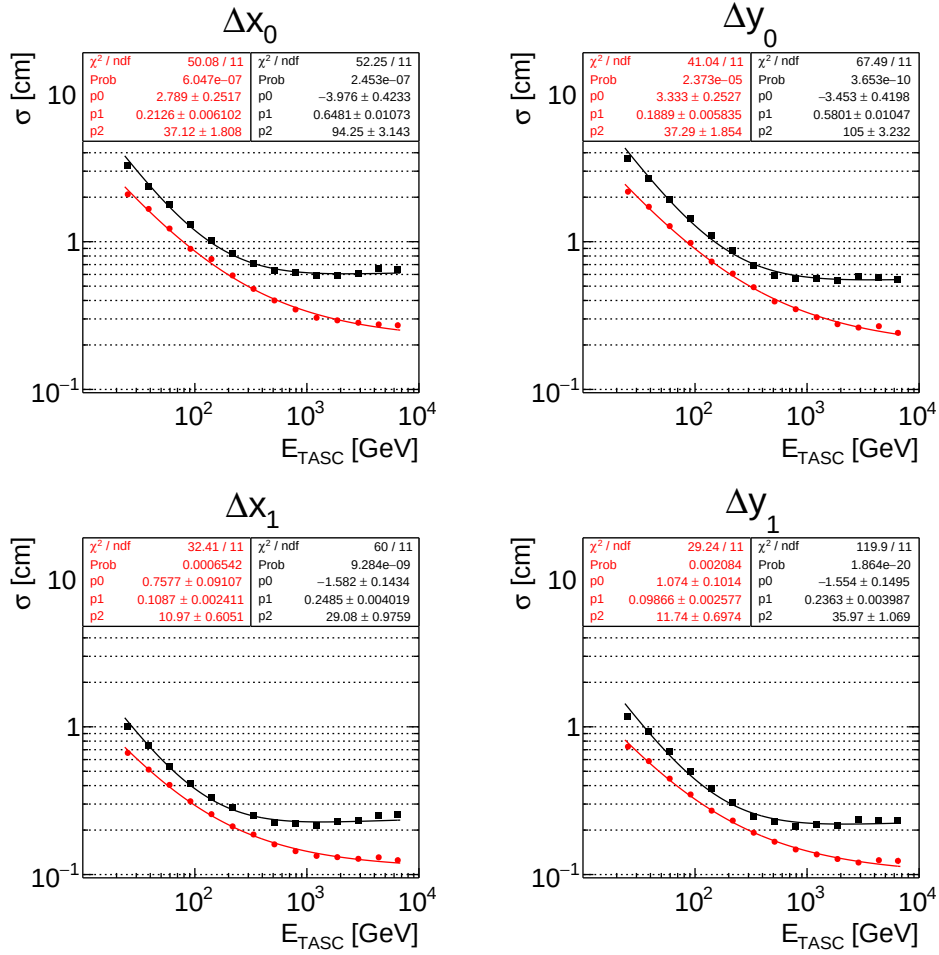


Figure 4.9. spatial resolution of the residual distribution of the impact points for the CNN trained on helium nuclei (in *red*) and the shower axis method (in *black*), as a function of the average deposited energy in each range.

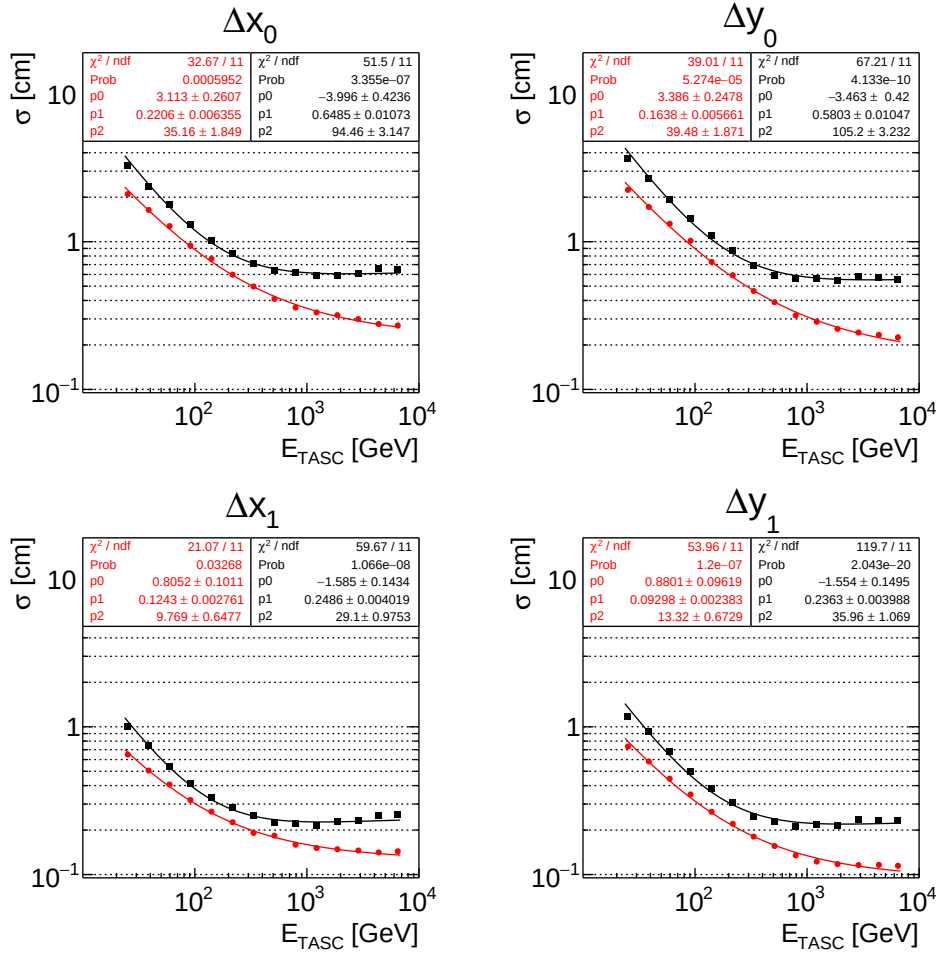


Figure 4.10. spatial resolution of the residual distribution of the impact points for the CNN trained with the combined dataset of p and He (in red) and the shower axis method (in black), as a function of the average deposited energy in each range.

4.2.4 Mean of residuals and angular resolution

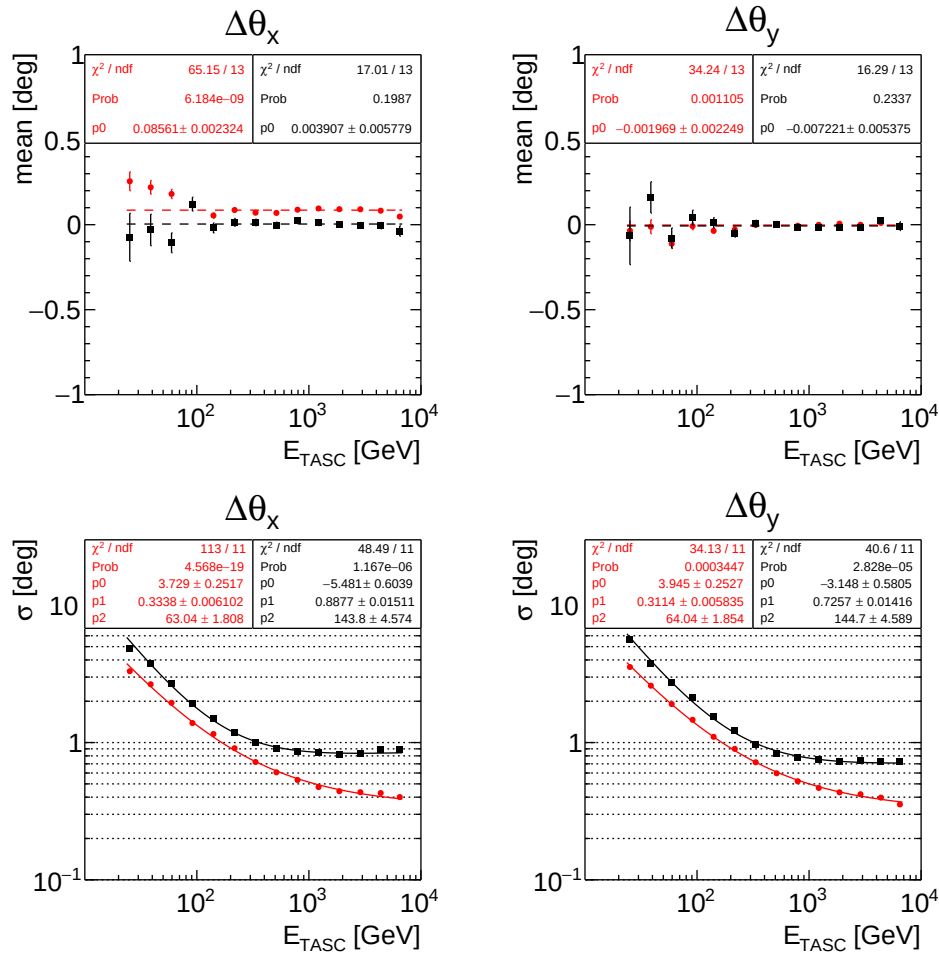


Figure 4.11. in the plots *above*, the mean value of the residual distribution; *below*, the angular resolution of the residual distribution of the angular variables for the CNN trained on helium nuclei (in *red*) and the shower axis method (in *black*), as a function of the average deposited energy in each range.

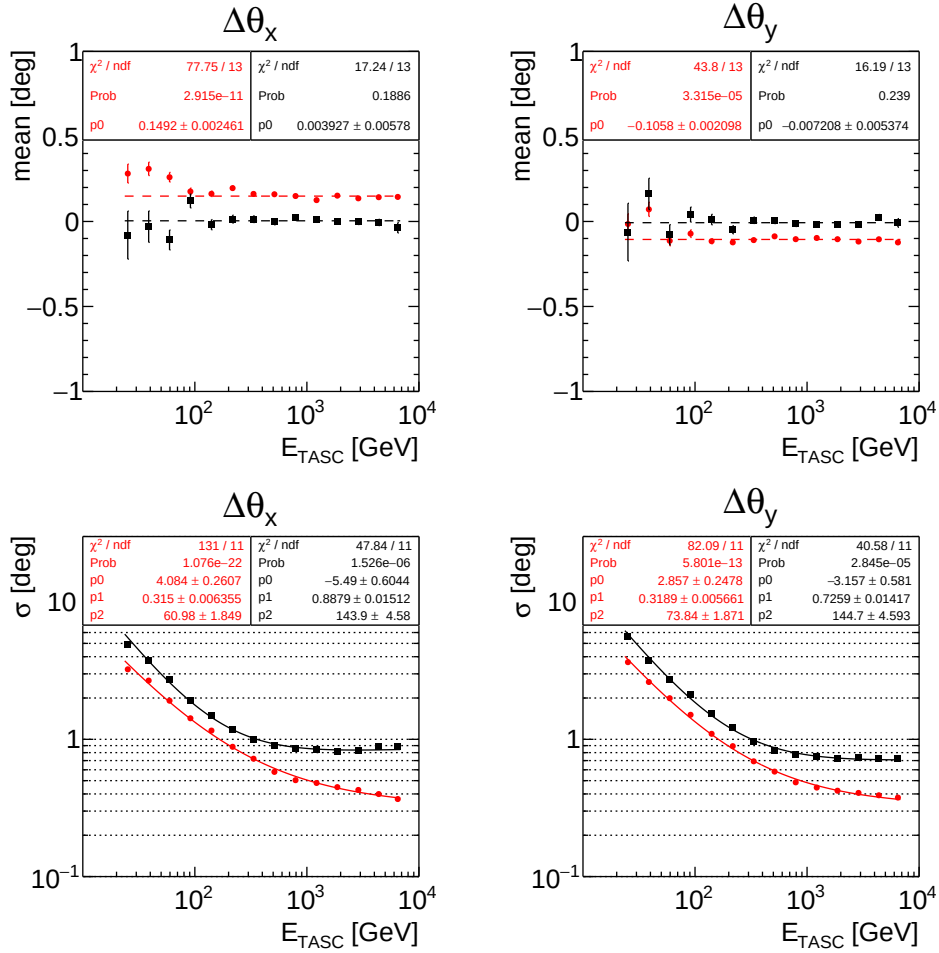


Figure 4.12. in the plots *above*, the mean value of the residual distribution; *below*, the angular resolution of the residual distribution of the angular variables for the CNN trained with the combined dataset of p and He (in *red*) and the shower axis method (in *black*), as a function of the average deposited energy in each range.

4.3 Quantitative comparison between CNN and shower axis method

Plots [Figure 4.3](#) and [Figure 4.4](#) show the residual distributions relating to the predictions on the impact variable x_0 made by the network trained on helium nuclei and the one trained on the p and He sample.

The error made by the network in making the prediction is expected to be *random*, meaning it is a *random variable*. From statistical theory on *central limit theorems* [35], we know that the (normalized) sum of a large number of random variables is distributed approximately like a normal random variable, i.e., it follows a *Gaussian* distribution.

For this reason, the plots of the residuals were fitted with Gaussian curves: the mean μ represents the distance of the predictions from the true value of the MC simulation, and therefore, the closer it is to zero, the better the prediction; the σ indicates the “resolution” of the method, i.e., how much the error is dispersed around the mean, and therefore the width of the ‘bell’ under which a certain percentage of the residuals falls—for example, 68.3% within one sigma ($\mu - 1\sigma, \mu + 1\sigma$). Having a smaller sigma, meaning a narrow and tall ‘bell’, translates to a better resolution.

That being said, it is clear how from the fit of the residual distributions of the impact points ([Figure 4.3](#), [Figure 4.4](#)) and of the angular variables ([Figure 4.5](#), [Figure 4.6](#)), it is possible to calculate the mean value of the residual distribution ([Figure 4.7](#), [Figure 4.8](#), [Figure 4.11](#)), and the spatial ([Figure 4.9](#), [Figure 4.10](#)) and angular resolution ([Figure 4.12](#)), and thus the residual distribution and the angular resolution ([Figure 4.11](#), [Figure 4.12](#)) of the two neural network models as a function of the energy deposited in the TASC, and compare it with the results produced by the shower axis method. Indeed, we proceeded to extract from each energy range the mean value of the energy deposited in the TASC (x -axis in the plots) and then to plot (y -axis) the mean value and the resolution, extracted from the Gaussian fit, as a function of the mean energy of each range.

From the analysis of the plots, we can therefore make the following observations:

No noticeable differences are observed between the results obtained with the dataset containing only helium nuclei compared to the one with the combined p and He dataset, in any of the considered parameters. Therefore, it is concluded that the use of the combined p and He dataset introduces neither improvements nor worsenings in the model’s resolution compared to the one trained on the dataset containing only helium nuclei.

Therefore, since there are no appreciable differences between the two datasets, only the one with He nuclei alone was considered in the following.

The situation is different, however, regarding the comparison between the CNN and the shower axis method: first of all, it is observed that in every energy range, the spatial and angular resolution of the CNN is better than that of the shower axis method. Furthermore, it is also observed that the resolution increases as the energy increases.

In fact, while at low energies (20, 31) GeV, the x_0 coordinate of the CNN has a resolution of $\sigma_{x_0} = (2.09 \pm 0.05)$ cm and the θ_x coordinate has a resolution of $\sigma_{\theta_x} = (3.31 \pm 0.08)^\circ$, it is noted that, as the energy increases, the value of σ_{x_0} progressively decreases—meaning the width of the Gaussian bell shrinks—reaching high energies (5477, 2×10^4) GeV where $\sigma_{x_0} = (0.27 \pm 0.01)$ cm and the θ_x coordinate has a resolution of $\sigma_{\theta_x} = (0.40 \pm 0.01)^\circ$. Thus, as the energy increases, there is an increase in resolution by a factor of $\simeq 10$ on the spatial resolution and a factor of $\simeq 8$ on the angular resolution.

On the other hand, the shower axis method has a resolution at low energies equal to $\sigma_{x_0} = (3.28 \pm 0.11)$ cm and $\sigma_{\theta_x} = (4.87 \pm 0.15)^\circ$, while at high energies it is equal to $\sigma_{x_0} = (0.65 \pm 0.02)$ cm and $\sigma_{\theta_x} = (0.89 \pm 0.03)^\circ$.

Therefore, the CNN has a better resolution than the shower axis method by $\Delta\sigma_{x_0} = (1.19 \pm 0.16)$ cm and $\Delta\sigma_{\theta_x} = (1.56 \pm 0.23)^\circ$ at low energies; and by $\Delta\sigma_{x_0} = (0.38 \pm 0.03)$ cm and $\Delta\sigma_{\theta_x} = (0.49 \pm 0.04)^\circ$ at high energies.

It has been observed that, in some cases, there is a systematic error in the reconstruction of the impact points that causes the mean value of the residuals to be ‘shifted’ positively or negatively with respect to zero. This error is probably due to a non-optimal choice of weights during training by the CNN and, to be resolved, one could try to perform the training again on a dataset with a higher number of events.

To fit the mean value of the residual distributions, a constant function was used, while for the fit of the spatial and angular resolution, the function used is:

$$y = \frac{a}{\sqrt{x} + b + \frac{c}{x}}$$

with a, b, c being the fit parameters. "[0]/TMath::Sqrt(x)+[1]+[2]/x" in the ROOT macro.

4.4 Data analysis with Kalman filter

In addition to the CNN and the shower axis method, the helium nuclei dataset produced by the CALET MC simulation was also passed as input to the combinatorial Kalman filter algorithm, reported in the appendix ([section A.1](#)). This algorithm, used in CALET primarily to distinguish between electromagnetic and hadronic showers, also

allows for estimating the direction and impact point of the primary shower, thus constituting a reliable analysis method with which to compare the performance of the CNN.

The combinatorial Kalman filter algorithm of CALET is implemented in the MC simulation program. Therefore, operationally, it was not necessary to implement it again or call it later on the dataset obtained as output from the simulation: already during the simulation itself, the CALET program reconstructs the simulated data using the KF. Therefore, the values reconstructed using this algorithm are provided directly as output by the MC simulation.

The choice made for the division of the energy range will be justified in [chapter 5](#), where the KF predictions will be used to evaluate the resolution of the CNN on flight data.

Below are some plots of the residual distributions between the value predicted by the KF and the true MC value, as well as the spatial and angular resolution of this reconstruction method. Only the plots relating to the spatial variable x_0 and angular variable θ_x have been included in this chapter. For the others, see [Appendix B](#).

4.4.1 Residual distributions

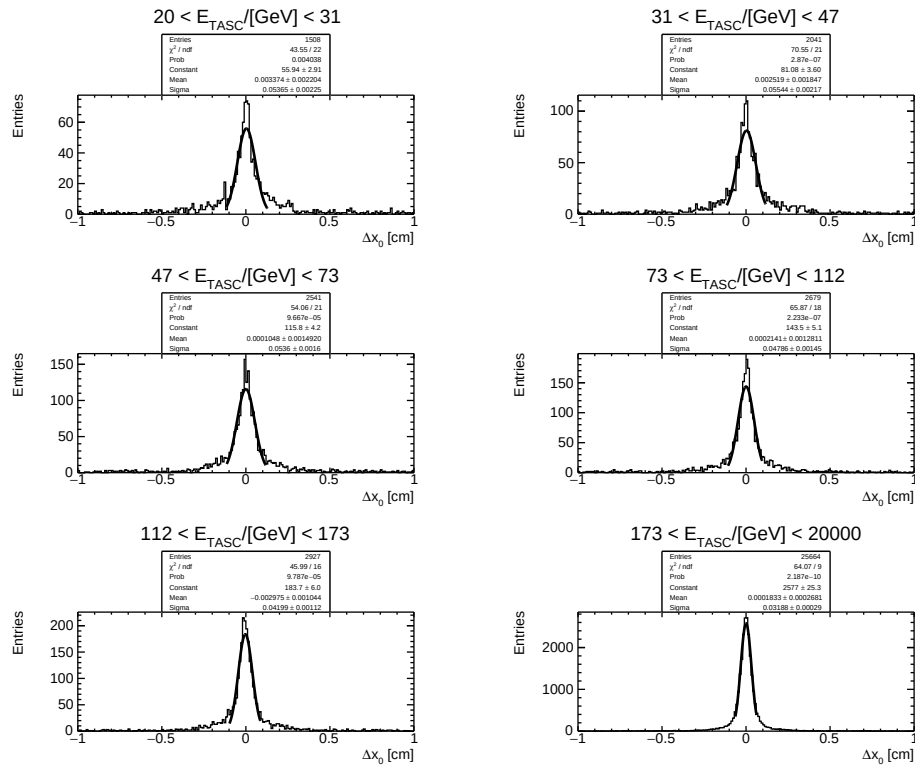


Figure 4.13. residual distribution of the impact point variable x_0 predicted by the Kalman filter (histogram in *black*) on the simulated helium MC data. The curve superimposed on the histograms represents the Gaussian fit to the histograms.

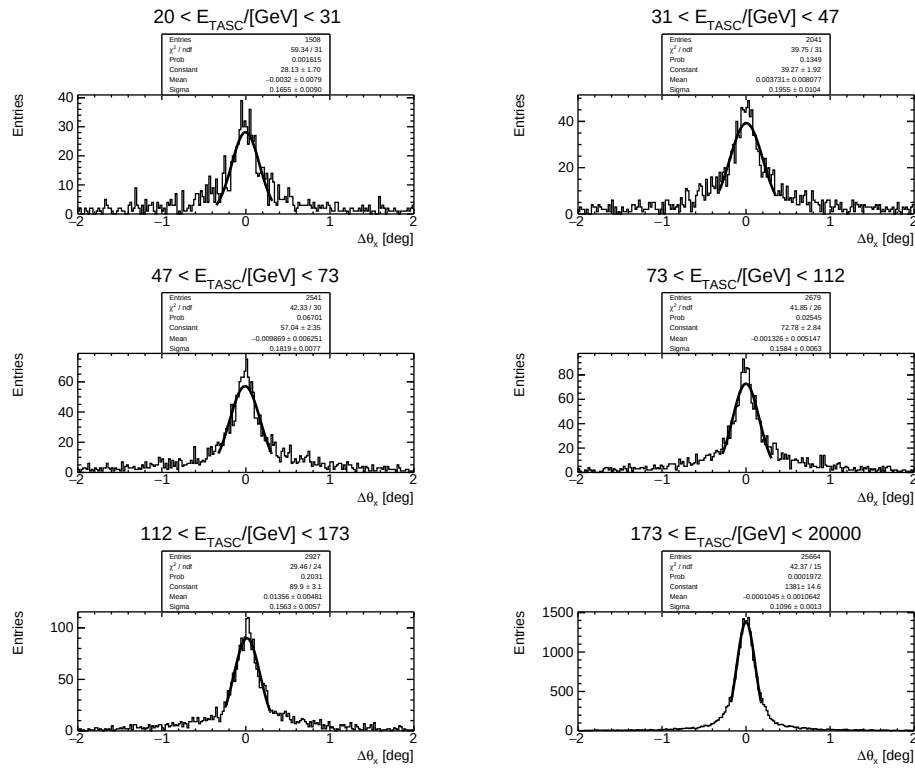


Figure 4.14. residual distribution of the angular variable θ_x predicted by the Kalman filter (histogram in *black*) on the simulated helium MC data. The curve superimposed on the histograms represents the Gaussian fit to the histograms.

4.4.2 Mean of residuals and spatial resolution

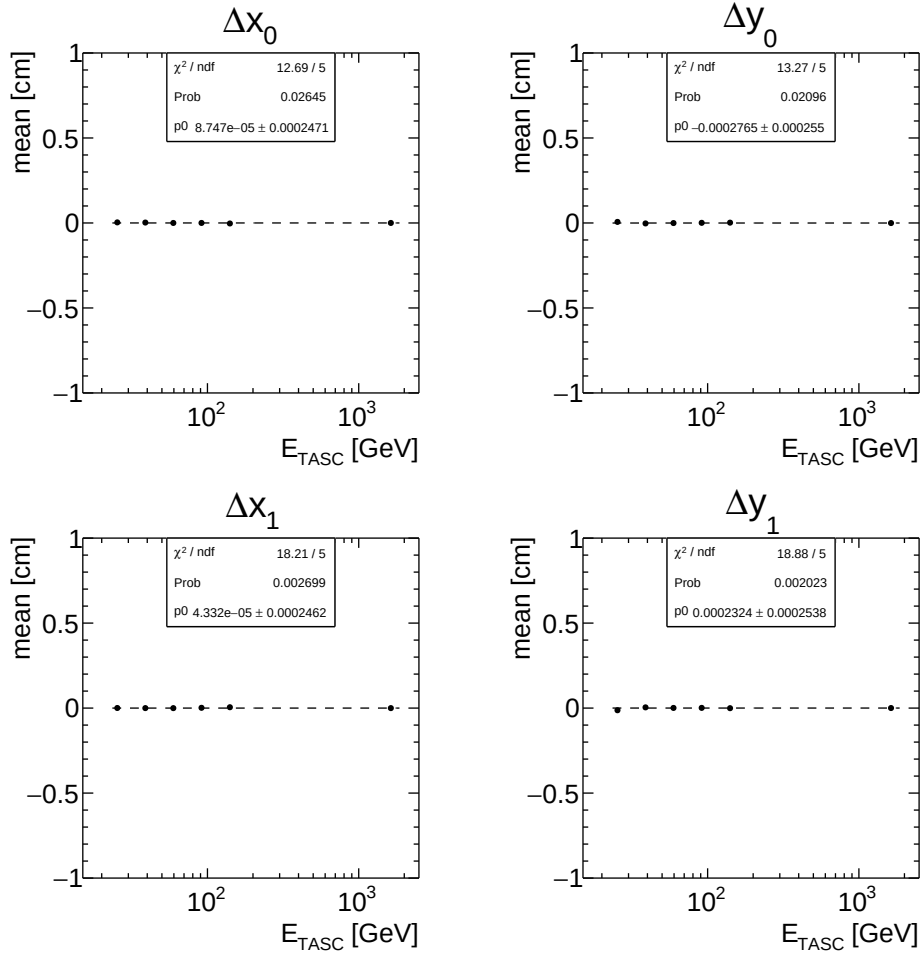


Figure 4.15. mean value of the residual distribution of the impact points predicted by the Kalman filter (in *black*) on the simulated helium MC data, as a function of the average deposited energy in each range.

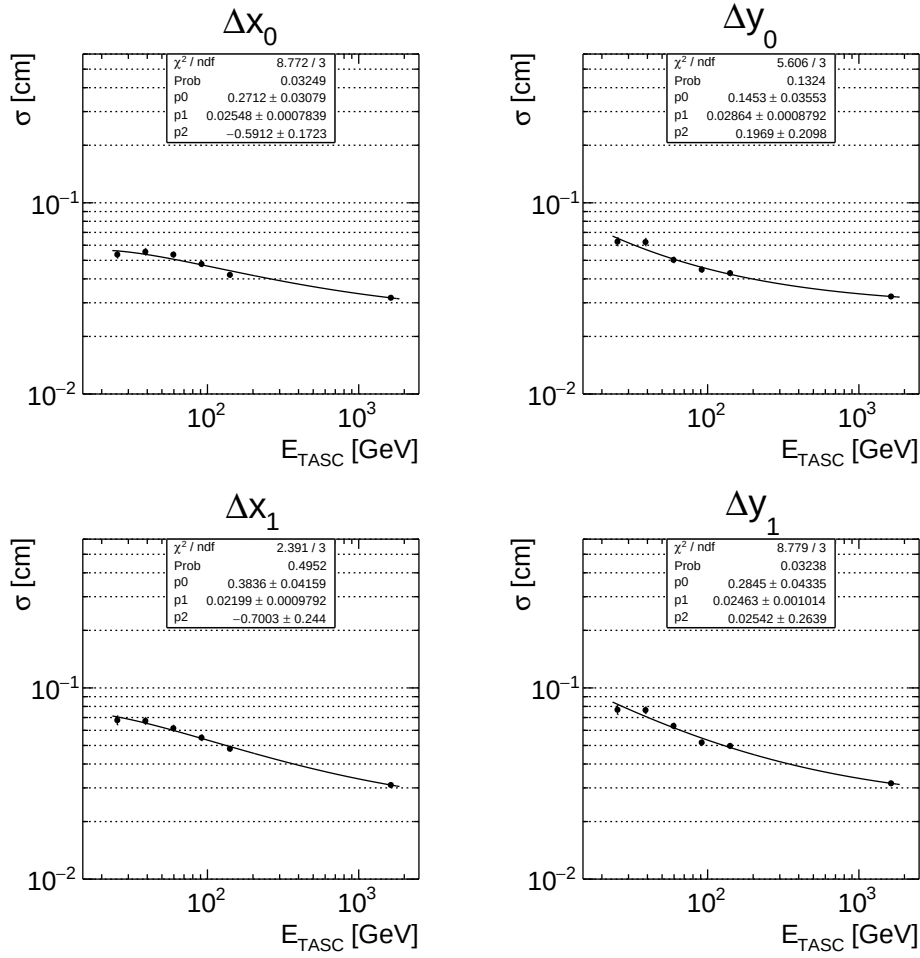


Figure 4.16. spatial resolution of the Kalman filter (in black) on the simulated helium MC data, as a function of the average deposited energy in each range.

4.4.3 Mean of residuals and angular resolution

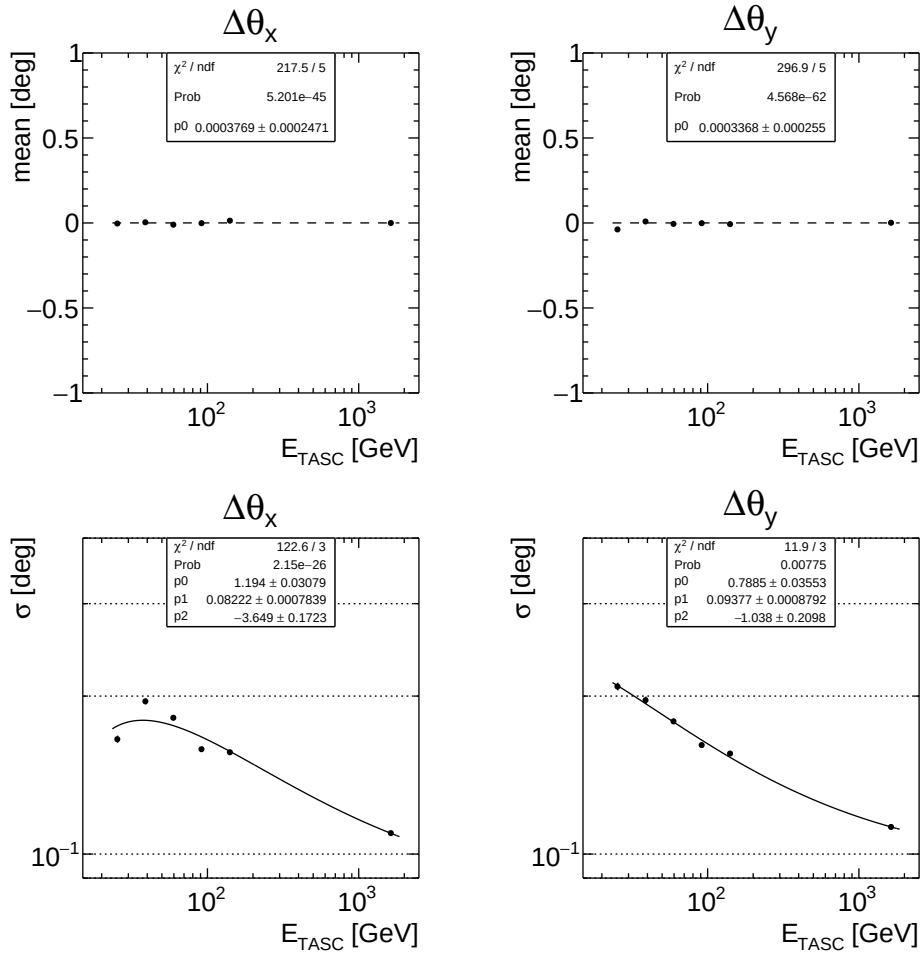


Figure 4.17. in the plots *above*, the mean value of the residual distribution; *below*, the angular resolution of the Kalman filter (in *black*) on the simulated helium MC data, as a function of the average deposited energy in each range.

4.5 Kalman filter: qualitative comparison

From the plots shown above and the others in the appendix ([section A.1](#)), the following observations can be made:

- the resolution of the KF improves with increasing energy, with an increase of $\simeq 2$ between low energies (20, 31) GeV and high energies (173, 2×10^4) GeV for the spatial variable x_0 and angular variable θ_x .
- the resolution of the Kalman filter algorithm is better than that of the CNN reconstruction method; the ratio between the two resolutions is approximately a factor of $\simeq 10$ at both low and high energies. This confirms the reliability of the KF algorithm. However, it must also be pointed out that the Kalman filter, unlike the CNN, reconstructs the trajectory starting from measurements taken not in the TASC, but in the IMC, which has a much finer segmentation and therefore greater precision, as explained in [chapter 2](#).

In the next chapter ([chapter 5](#)), the results of the KF method on the CALET MC simulation data are exploited to calculate the resolution of the CNN on *flight data*, i.e., real data from events recorded by the CALET calorimeter.

Chapter 5

Flight data analysis

5.1 Introduction

In this chapter, the flight data analysis of the events recorded by CALET in the period between January and March 2019 is carried out.

Since the flight data contains events associated with different particles, it was necessary to select only the data relating to the ones to be analyzed: in our case, helium nuclei.

Subsequently, the track reconstruction performance was compared between the shower axis method and the neural network, in two distinct situations: a network trained only on helium nuclei, and a network trained on the combined dataset of p and He.

(Since in [chapter 4](#) no significant differences were observed between the predictions made by the CNN trained on helium nuclei and the one trained on the combined dataset of p and He, this chapter presents only the results obtained from the CNN trained on helium nuclei. For the flight data results of the CNN trained on the combined dataset of p and He, see [Appendix B](#)).

5.2 Particle selection

5.2.1 Charge measurement

The data selection process is now explained, along with the criteria used to record events and the measurements used to reconstruct the particles' charge.

A cosmic ray crossing the calorimeter is recorded as an event in the flight data only if it satisfies a series of control conditions. It must in fact 'fire' the *high energy trigger* (HET), to which the offline trigger confirmation is then applied, that is, the second control condition: the energy deposited in the first plane of the TASC must be greater than 100 equivalent MIP (*Minimum Ionizing Power*).

Of these events, all those in which the following occurs are then discarded:
either the KF algorithm failed to reconstruct the particle's direction because the

number of track points in the IMC in one or both views is < 4 , or the direction found by KF is incompatible with the shower axis reconstructed in the TASC.

Finally, the ‘acceptance A’ condition is applied, that is, only those events in which the particle crosses the upper face of the CHD and reaches the lower face of the TASC while staying within 2 cm of the edges are selected (this condition makes it possible to select the events with the best containment of the hadronic shower).

To reconstruct the charge of the primary particle, its track is first extrapolated in the IMC and backward through the two planes of the CHD, in order to identify the strips crossed by the cosmic ray and measure the energy loss per unit length (dE/dx).

By exploiting the two energy loss per unit length measurements, it is possible to reconstruct the charge of the primary particle, as shown in [Figure 5.1](#) and [Figure 5.2](#), where all nuclei from protons to iron can be distinguished.

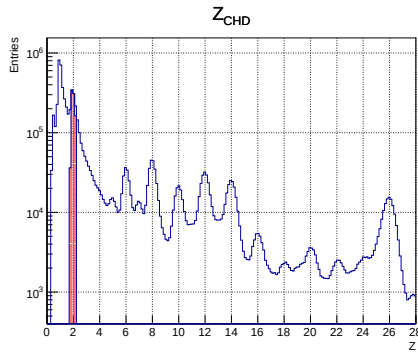


Figure 5.1. in *blue*, the charge reconstructed in the two CHD planes for values between 0 and 28; in *red*, the helium nuclei selected in the range (1.8, 2.2).

Peaks can be observed corresponding to the nuclei (p, He, C, O, Ne, Mg, Si, S, Ar, Ca, Ti, Cr, Fe).

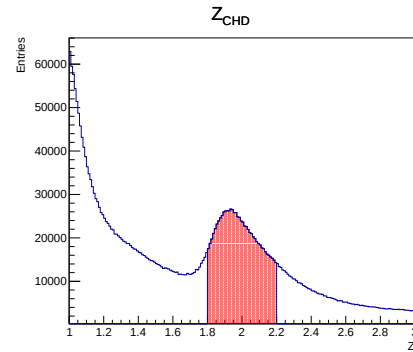


Figure 5.2. in *blue*, a zoom of the charge reconstructed in the two CHD planes for values between 1 and 3; in *red*, the helium nuclei selected in the range (1.8, 2.2).

(The plots of the charge measured by the CHD in the xz and yz views are shown in [section B.5: Sample of p and He nuclei from flight data](#))

The flight data analysis was carried out on helium nuclei, selected by requiring that the charge Z reconstructed in the two CHD planes, using the procedure described above, be $1.8 < Z < 2.2$.

5.2.2 Power-law spectral fit

Plots of the distribution of the energy deposited in the TASC were also produced, both with and without the selection on helium nuclei.

Both plots were drawn choosing $\log_{10}(E_{\text{TASC}})$ as the values for the x-axis; this makes it possible to observe that, starting from $\log_{10}(E_{\text{TASC}}) \approx 1.77$, that is, $E_{\text{TASC}} \approx 58.89$ [GeV], the energy distribution follows a *power-law* trend. This highlights a proportionality between the kinetic energy of the cosmic particles crossing the detector and the deposited energy.

This proportionality was verified by fitting the distributions of both plots using the function

$$N(E_{\text{TASC}}) = N_0 10^{(1-\gamma)E_{\text{TASC}}}$$

with N_0 the *normalization factor* and γ the *spectral index*. "[0]*pow(10, (1-[1])*x)" in the ROOT macro. The plots obtained from the fit are shown in [Figure 5.3](#) and [Figure 5.4](#).

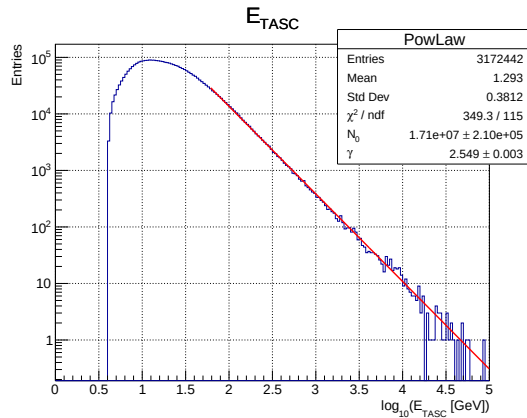


Figure 5.3. in *blue*, the histogram of the energy distribution deposited in the TASC by all particles; in *red*, the power-law fit.

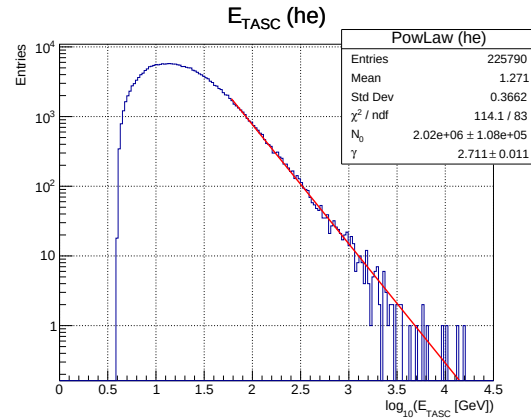


Figure 5.4. in *blue*, the histogram of the energy distribution deposited in the TASC by helium nuclei; in *red*, the power-law fit.

Note that the plots do not provide a measurement of the cosmic-ray energy spectrum (flux versus kinetic energy, dN/dE), but rather show the count dN of events recorded in each range of energy dE deposited in the TASC, essentially the integral of the flux.

From the fit of the energy distribution deposited by all particles (left) and by helium nuclei only (right), values of $\gamma = 2.549 \pm 0.003$ and $\gamma = 2.711 \pm 0.011$ are obtained respectively, in agreement with the values observed in CALET.

5.3 CNN results on flight data

The selected data were therefore fed as input to the two CNNs previously described (chapter 3), and the values predicted by these were compared with those reconstructed using the shower axis method.

For both CNNs, that is, both the one trained only on helium nuclei and the one trained on protons and helium nuclei, the residual distributions of each coordinate of the impact points (x_0, y_0, x_1, y_1) and of the angular variables (θ_x, θ_y) were calculated and plotted.

The distributions were analyzed in 6 deposited-energy intervals E_{TASC} , as described in section 5.2, selecting particles with charge $1.8 < Z < 2.2$ (he nuclei).

	Range [GeV]		$\langle E_{TASC} \rangle$ [GeV]	Events
20	$\longleftrightarrow$	31	24.79	38696
31	$\longleftrightarrow$	47	37.83	24960
47	$\longleftrightarrow$	73	57.66	15413
73	$\longleftrightarrow$	112	88.79	7865
112	$\longleftrightarrow$	173	136.19	3810
173	$\longleftrightarrow$	2×10^4	378.17	3487

The values reported in the table, relating to the number of events and the average value of the energy deposited by helium nuclei in each range, were calculated using the ROOT software. Histograms with 200 bins were produced, and the values reported above were extracted from these.

The criterion adopted for dividing the energy range (20, 2×10^4) [GeV] is as follows: intervals of constant width on a logarithmic scale were used, that is, such that $\log_{10}(E_{i+1}/E_i) = \text{const}$ for all intervals except the last one. Indeed, from 173 to 2×10^4 [GeV], in order to have statistics comparable to those of the previous intervals, it was chosen to extend the interval until all remaining events were collected, thus reaching a number of entries comparable to that of the interval from 112 to 173 [GeV].

The analysis was restricted to events with $E_{TASC} > 20$ [GeV] in order to avoid considering non-interacting helium nuclei, which behave as minimum ionizing particles when crossing the calorimeter.

The value $\langle E_{TASC} \rangle$ represents the average energy deposited in the TASC planes for each range into which the sample to be analyzed was divided. This average value was extracted directly from the distribution of the variable E_{TASC} in the intervals specified above.

In the next two sections, ([section 5.4](#)) and ([section 5.5](#)), the plots of the residual distributions of the impact point variables and of the angular variables are presented. The plots were produced using the ROOT software.

(In this chapter, only the plots relating to the spatial variable x_0 and the angular variable θ_x have been included. For the others, see [Appendix B](#)).

Note that, unlike the analysis carried out in [chapter 4](#), in which it was possible to compare the values predicted by the network with the *true* values from the MC simulation, in this second analysis this is no longer possible. Indeed, the data analyzed are not simulated data, but flight data recorded by CALET, for which we do not have a *true* value; on the contrary, this is exactly what we are trying to reconstruct.

In order to obtain an estimate of the resolution of the CNNs and of the shower axis method on the flight data, a comparison was therefore made with the values reconstructed by the *Kalman filter* (KF) method, applied to the same events.

It follows that the measured (*mis*) resolution of the CNN in this analysis will be equal to the square root of the sum in quadrature of the KF resolution and the *effective* resolution of the CNN on the flight data; that is

$$\sigma_{\text{mis}} = \sqrt{\sigma_{\text{CNN}}^2 + \sigma_{\text{KF}}^2}$$

A comparison between the effective resolution of the CNN on flight data and on simulated MC data is carried out in [section 5.6](#).

5.4 Residual distributions

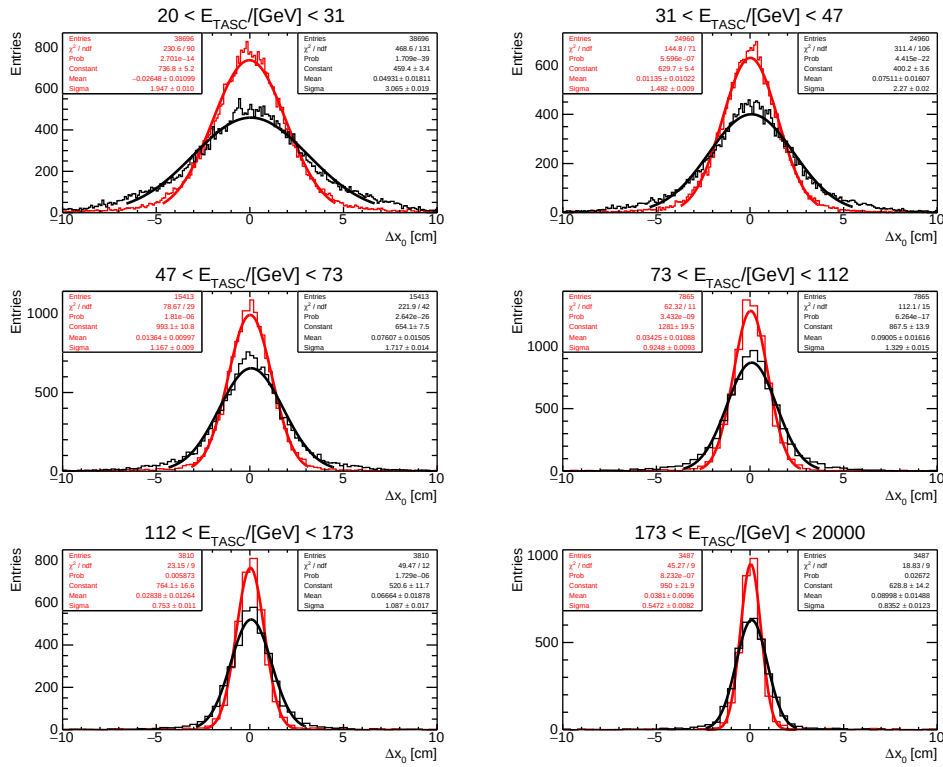


Figure 5.5. distribution of the residuals of the impact point variable x_0 predicted by the CNN trained on helium nuclei (histogram in *red*) and by the shower axis method (histogram in *black*). The curves superimposed on the histograms are the gaussian fits to the histograms of the same color.

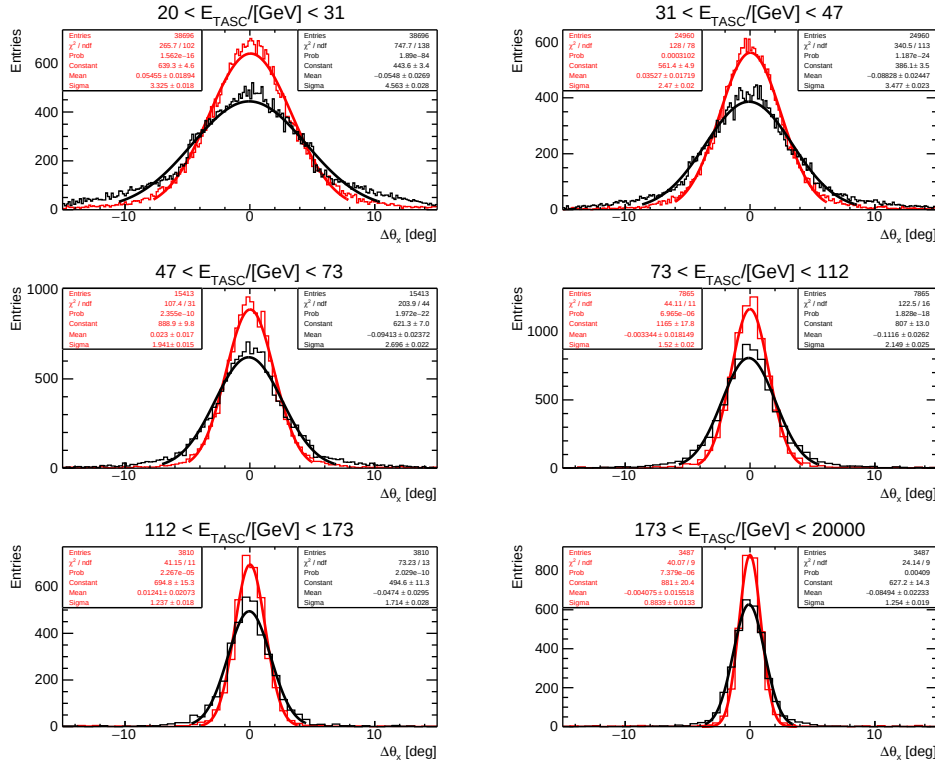


Figure 5.6. distribution of the residuals of the angular variable θ_x predicted by the CNN trained on helium nuclei (histogram in *red*) and by the shower axis method (histogram in *black*). The curves superimposed on the histograms are the gaussian fits to the histograms of the same color.

5.5 Mean of the residuals and spatial resolution

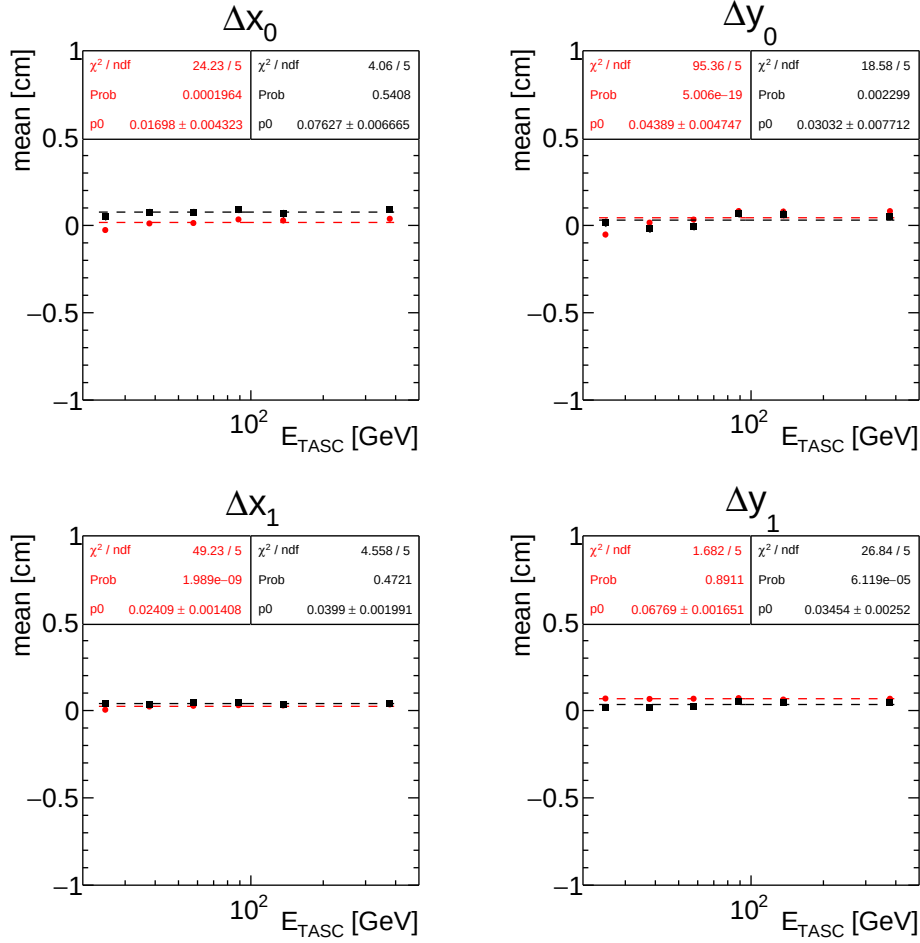


Figure 5.7. mean value of the residual distribution of the impact points of the CNN trained on helium nuclei (in *red*) and of the shower axis method (in *black*), as a function of the average energy deposited in each range.

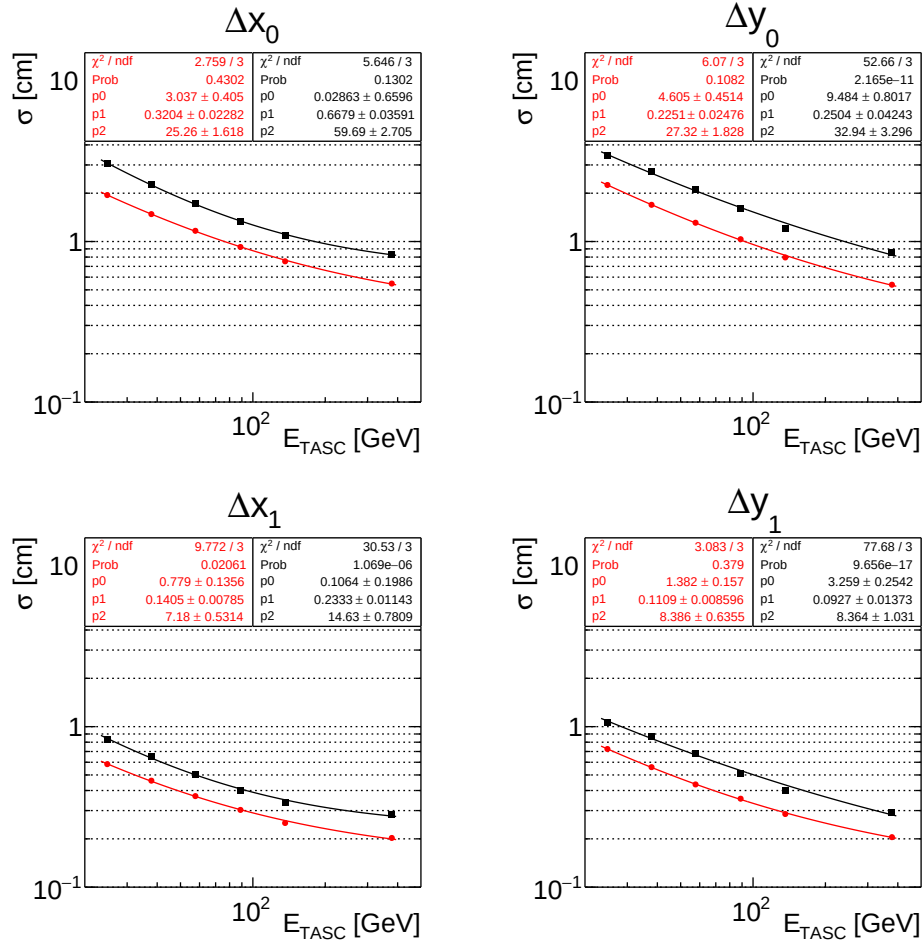


Figure 5.8. spatial resolution of the CNN trained on helium nuclei (in *red*) and of the shower axis method (in *black*), as a function of the average energy deposited in each range.

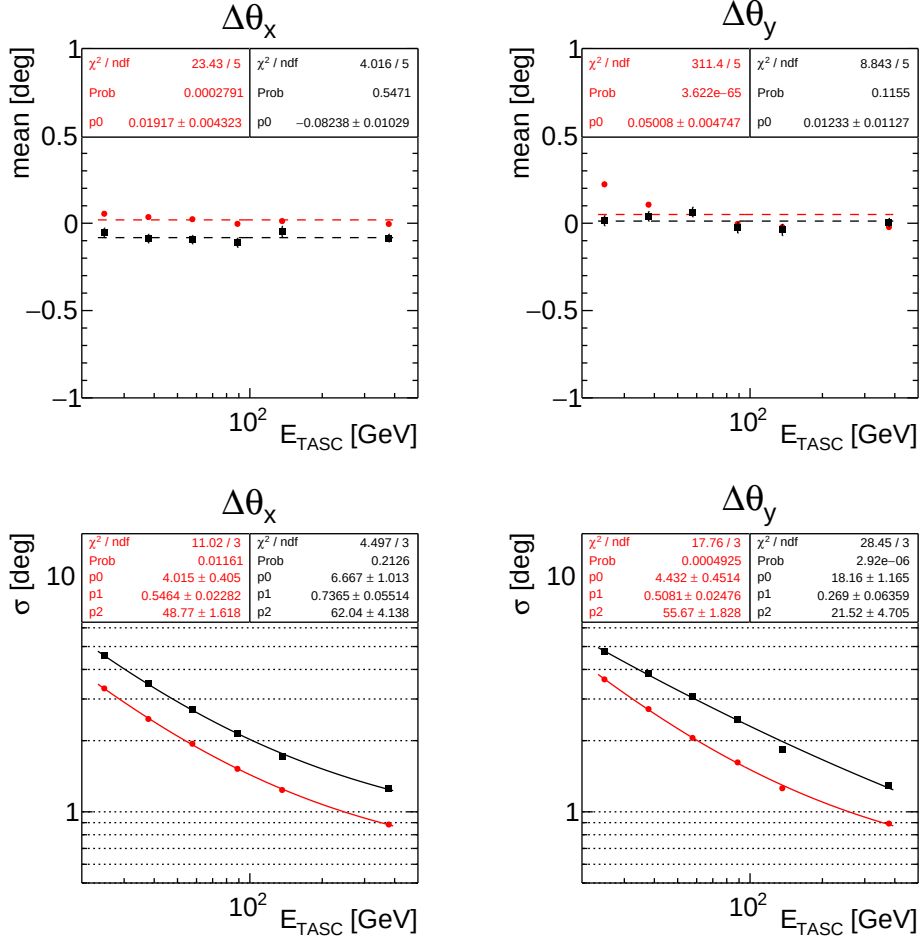


Figure 5.9. in the plots *above*, the mean value of the residual distribution; *below*, the angular resolution of the CNN trained on helium nuclei (in *red*) and of the shower axis method (in *black*), as a function of the average energy deposited in each range.

It can be observed that, as for the simulated MC data, for the flight data (FD) too the CNN residuals are smaller than those of the shower axis, and therefore a better resolution is obtained, which increases as the average energy of the range increases.

It should also be noted that, for the FD, the CNN resolution is comparable to that obtained on the MC data: this indicates the ‘stability’ of the network and a good predictive capability, given that the FD do not belong to the training dataset, so a worsening of the resolution would have been reasonable to expect.

5.6 Flight data: quantitative comparison

From the plots reported above, we can make the following observations:

Comparison between the CNN and the shower axis method: first of all, it can be observed that in every energy range the spatial and angular resolution of the CNN is better than that of the shower axis method. It can also be observed that the resolution improves as the energy increases.

Indeed, if at low energies (20, 31) GeV the coordinate x_0 and the angular variable θ_x of the CNN have resolution $\sigma_{x_0} = (1.95 \pm 0.01)$ cm and $\sigma_{\theta_x} = (3.33 \pm 0.02)^\circ$, it can be seen that, as the energy increases, the values of σ_{x_0} and σ_{θ_x} progressively decrease, that is, the width of the gaussian bell curve narrows, down to high energies (5477, 2×10^4) GeV, where $\sigma_{x_0} = (0.55 \pm 0.01)$ cm and $\sigma_{\theta_x} = (0.88 \pm 0.01)^\circ$. Thus, as the energy increases, there is an improvement in spatial resolution by a factor of $\simeq 4$ and in angular resolution by a factor of about $\simeq 3.7$.

As in the data analysis carried out for the training on simulated data, in this case too it was observed that there is a systematic error in the reconstruction of the impact points, which causes the mean value of the residuals to be ‘shifted’ positively or negatively with respect to zero. This error is probably due to a non-optimal choice of weights during training by the CNN and, to resolve it, one could try to perform the training again on a dataset with a larger number of events.

To fit the mean value of the residual distributions, a constant function $y = mx + q$ was used, "pol0" in the ROOT macro; while the function used to fit the spatial and angular resolution is:

$$y = \frac{a}{\sqrt{x} + b + \frac{c}{x}}$$

with a, b, c fit parameters. "[0]/TMath::Sqrt(x)+[1]+[2]/x" in the ROOT macro.

In [section 5.3](#) it was observed that the resolution estimated for the CNN is actually an overestimate of the *effective* resolution, since

$$\sigma_{\text{mis}} = \sqrt{\sigma_{\text{CNN}}^2 + \sigma_{\text{KF}}^2}$$

This follows from the fact that this resolution was calculated by comparing the impact points reconstructed by the CNN and by the Kalman filter: that is, the value against which the comparison was made, unlike in the MC data analysis, is not a ‘true’ value, but is itself a reconstructed value, and therefore also has an associated reconstruction uncertainty, namely the resolution of the combinatorial Kalman filter algorithm on the MC simulation data.

$$\sigma_{\text{CNN}} = \sqrt{\sigma_{\text{mis}}^2 - \sigma_{\text{KF}}^2}$$

It follows then that, in order to determine the effective resolution of the CNN on the FD, it is necessary to estimate the resolution of the Kalman filter on the MC data.

These values were therefore estimated, and their plots are shown in [Appendix B](#), and subtracted in quadrature from σ_{mis} .

[Figure 5.10](#) shows the spatial resolution of the CNN on the simulated MC data (in *red*) and of the CNN on the flight data FD (in *black*). [Figure 5.11](#) shows the angular resolution, with the same color scheme for the resolution on the simulated MC data and on the flight data FD.

Note that the number of resolution measurements is different (14 for the simulated MC data and 6 for the flight data FD): this depends on how the energy range ($20, 2 \times 10^4$) GeV was chosen to be divided. It follows that the values of the average energy deposited in each range (that is, the x coordinates in the plots of [Figure 5.10](#) and [Figure 5.11](#)) do not coincide. The resolution of the CNN on the flight data FD is therefore estimated only up to energy values of $E_{\text{TASC}} \simeq 378$ GeV, and not beyond. In order to estimate the resolution on the flight data FD at higher energy values (*high energies*), a larger number of data points measured by CALET than those used in this study would be needed, so as to improve the statistics also at higher energy values. (See [chapter 4](#) and [chapter 5](#) for further details on the choice of how the energy range was divided).

Note that the resolution values σ_{CNN} of the CNN on the flight data are of the same order as the resolution values σ_{MC} on the MC simulation data. This observation is of considerable importance: the fact that the resolution does not worsen on a dataset of real data is an indication of the stability and reliability of the neural network. It means that the network has learned to recognize the features useful for tracking, that is, it has developed a good ability to make reliable predictions even on ‘new’ events, that is, events not belonging to the training dataset.

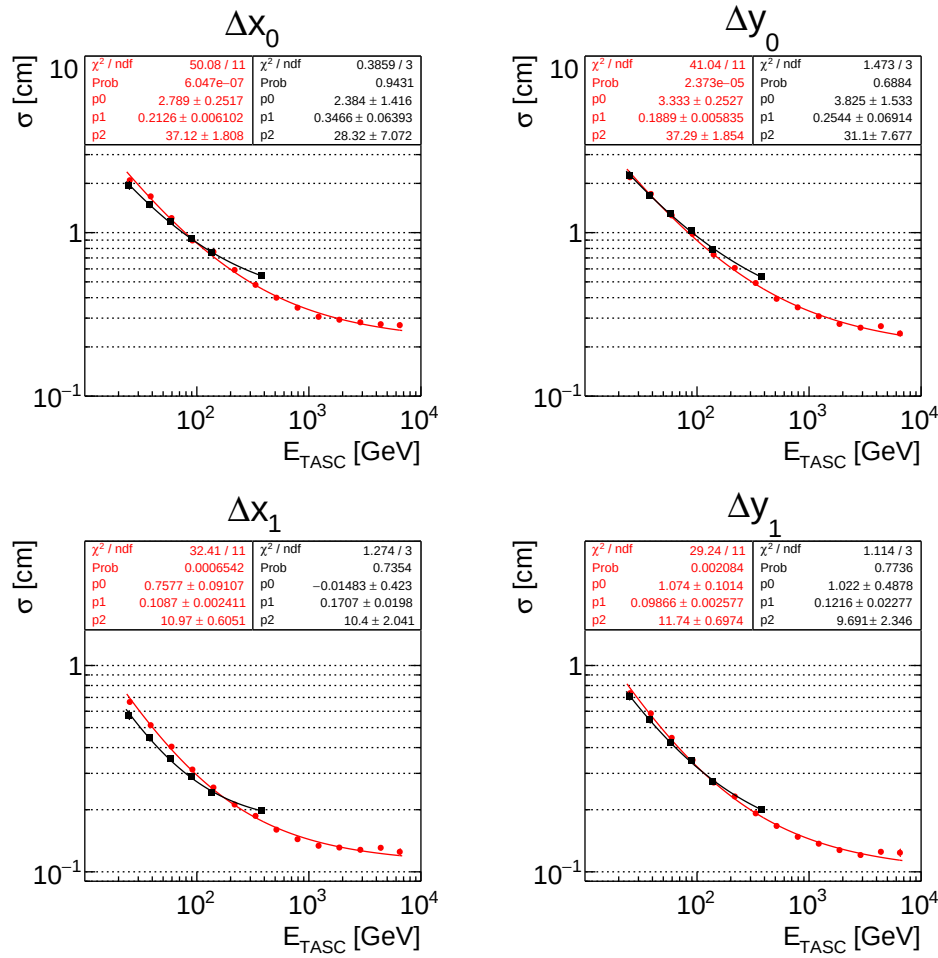


Figure 5.10. comparison between the spatial resolution of the CNN on the simulated MC data (in red) and on the flight data FD (in black), as a function of the average energy deposited in each interval.

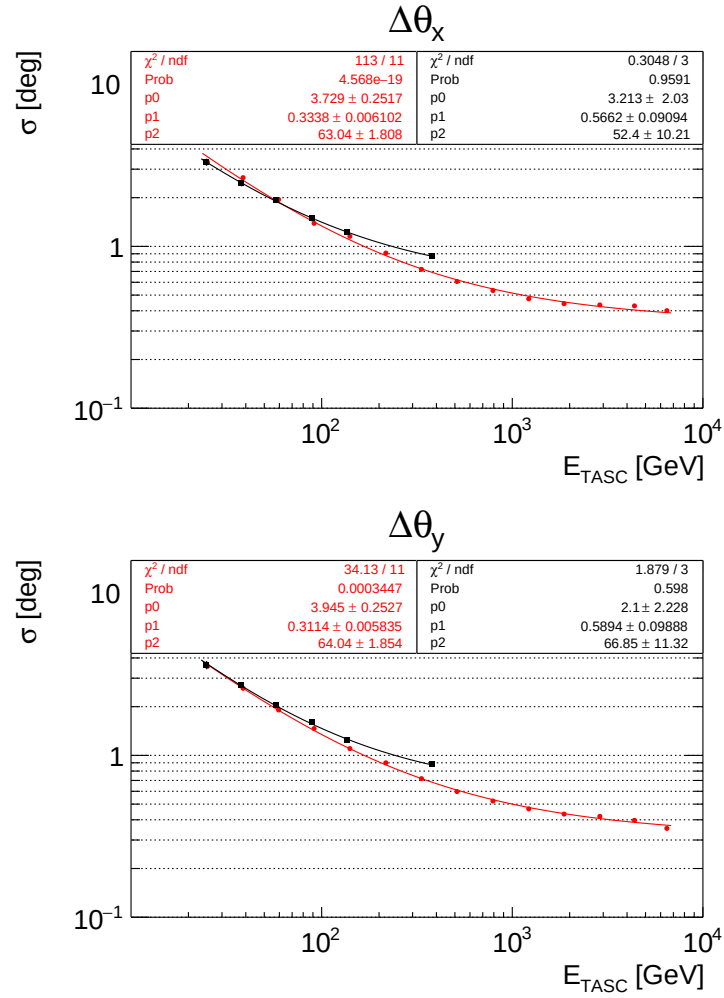


Figure 5.11. comparison between the angular resolution of the CNN on the simulated MC data (in *red*) and on the flight data FD (in *black*), as a function of the average energy deposited in each interval.

Conclusions

The convolutional neural network (CNN) model developed in this thesis work was therefore applied to CALET flight data, and the analysis of the resolution of the predictions made provided promising results. The performance obtained with the CNN was then compared with the shower axis method and with the combinatorial Kalman filter (KF) algorithm, implemented in the CALET experiment's reconstruction software.

The CNN was trained on two distinct datasets of simulated MC events: one containing events relating to helium nuclei only, and the other containing events relating to both protons and helium nuclei. The difference between the predictions of the two models turned out to be negligible, thus showing how CNNs make it possible to develop and train more 'flexible' models: that is, it is possible to produce a single neural network model capable of tracking both protons and helium nuclei at the same time, without losing resolution compared to the case of two separate models, one trained to track protons and the other to track helium nuclei.

Furthermore, from the comparison of the results obtained with the different methods used in the study, it emerged that the neural network model developed constitutes a more accurate trajectory reconstruction method than the shower axis method, but, as expected, less accurate than the KF. Indeed, as explained in [section 4.5](#), the KF performs tracking based on the measurements taken in the IMC rather than in the TASC.

Spatial resolutions of about ~ 2 cm were obtained on the impact point in the upper plane of the CHD and of about ~ 0.6 cm in the upper plane of the TASC at low energies, while about ~ 0.5 cm on the impact point in the upper plane of the CHD and about ~ 0.2 cm in the upper plane of the TASC at high energies. As for the angular resolution on the incidence direction of cosmic rays in the two projections of the detector (xz and yz), the CNN achieves about $\sim 3^\circ$ at low energies and about $\sim 0.8^\circ$ at high energies.

Finally, no significant worsening of the spatial and angular resolution was observed when applying the CNN to the flight data FD. Indeed, the resolution of the CNN on the flight data FD turns out to be of the same order of magnitude as that estimated on the simulated MC data, although the energy range over which it was measured is smaller than that of the simulated MC data, as explained in [section 5.6](#).

The resolution measured on the flight data thus confirms the reliability of the model in making trustworthy predictions on real data recorded by the CALET calorimeter.

Appendix A

Tracking methods used in CALET

A.1 Kalman filter

Track identification and reconstruction in CALET is of the utmost importance, since accurate tracking is necessary in order to:

- identify which *paddles* (strips) of the CHD and which fibers of the IMC were crossed by the incoming cosmic ray particle, in order to measure its charge and identify the starting point of the shower;
- define the geometric acceptance and calculate the calorimeter's topological variables, such as the width of the shower relative to the particle's trajectory, which are used to perform electron-hadron discrimination.

These operations can be particularly computationally expensive due to the large number of secondary particle tracks in the TASC and the many 'hits' recorded in the IMC. The hits must be divided and grouped into groups of 'candidate' tracks, before performing a fit. The large number of spurious hits to be fitted, combined with possible detector inefficiencies, makes it difficult to identify the real track of the primary particle.

CALET uses a track reconstruction algorithm based on a combinatorial Kalman filter [36]: this is a hybrid data analysis technique for the simultaneous reconstruction of tracks (*track fitting*) and the fitting of vertices (*vertex fitting*), widely used in high energy accelerator experiments, but never before used in a calorimetric experiment such as CALET.

The algorithm is able to handle a large number of hits, as occurs in the IMC for high energy showers, and to reconstruct multiple track candidates with reduced computation time, even in the presence of detector inefficiencies.

The problem of trajectory reconstruction in CALET can be reduced to two independent two-dimensional problems, in the xz and yz views respectively, since the IMC layers are oriented in either the x direction or the y direction and are positioned at different heights z .

The process starts from the top layer of the IMC: hits close to one another are grouped into a cluster for each IMC layer, whose centroid is passed to the KF as a possible track point. A track candidate is then created for every possible combination of clusters in the first two layers, including tracks with missing hits. Each candidate track is fitted with a straight line, which is evolved separately from the others, that is, the angle of incidence and the impact point (which form the *state vector*) on the next layer ($k + 1$) are calculated using only the information (state vector and covariance matrix) from the previous layer (k).

This predicted state can be used to identify which cluster to assign to the track in layer ($k + 1$): one possible criterion is to associate the cluster closest to the predicted

measurement. In any case, this does not constitute a reliable algorithm, for example one that is resilient to incorrect initial hypotheses. If the starting track is estimated incorrectly, there is a risk that an incorrect measurement will be associated with the track, such as a noise cluster or a point from another track.

The combinatorial formulation of the Kalman filter solves this problem by creating a new candidate track for each cluster in layer $(k + 1)$ that is sufficiently close to the predicted state. The candidate track is thus ‘branched’ into many candidates, each of which can be evolved separately.

For each new branch, the information regarding the track point in layer $(k + 1)$ is combined with the predicted state, to produce a ‘filtered’ estimate of the state vector and of the covariance matrix of level $(k + 1)$; this filtered state is used to fit the track again and update the value of χ^2 .

The procedure is then iterated over layer $(k + 2)$ and so on, up to the last layer.

Obviously, a combinatorial algorithm of this type can produce a large number of candidate tracks, which can considerably slow down the computation. To avoid running into problems of this kind, a candidate track can be discarded if, at any point in its evolution, it has a χ^2 greater than a certain predefined value (typically 10) or too large a number of missing points (typically > 2).

For events that generate high energy showers in the TASC, resulting in a large number of hits in the IMC due to *backscattering*, in order to reduce the number of track combinations and the time needed to process them, only the IMC clusters within a ROI (*Region Of Interest*) are passed to the algorithm; this ROI is defined by the set of hits in each layer that lie within ± 5 cm of the intersection point with the extrapolation of the shower axis reconstructed in the TASC.

Finally, among the tracks reconstructed by the algorithm, the one chosen as the most probable (real) track of the primary cosmic ray is the one that passes closest to the *core* of the shower in the TASC, that is, the central region with the maximum energy release [37].

A.2 Shower axis method

The *shower axis method* [38] is an algorithm originally developed to separate electronic and hadronic showers in calorimeters.

In CALET it is mainly used to identify electrons within the TASC; it is also used to reconstruct the direction of travel and the impact points on the detector planes.

The algorithm follows the procedure outlined below, repeated event by event:

1. the coordinates (x_{ij}, z_{ij}) of the xz (yz) plane of the TASC are ‘weighted’ by the deposited energy E_{ij} ;
2. using the values obtained, the *covariance* matrices are built for the x view and for the y view;
3. the matrices are diagonalized and, once the eigenvalues are obtained, the corresponding eigenvectors are derived, which provide an estimate of the particle’s direction.

Below are the ‘weighted’ coordinates (x_{ij}, z_{ij})

$$\begin{aligned}\bar{x} &= \frac{\sum_{ij} x_{ij} E_{ij}}{\sum_{ij} E_{ij}} & \bar{z} &= \frac{\sum_{ij} z_{ij} E_{ij}}{\sum_{ij} E_{ij}} & \bar{xz} &= \frac{\sum_{ij} x_{ij} z_{ij} E_{ij}}{\sum_{ij} E_{ij}} \\ \overline{x^2} &= \frac{\sum_{ij} x_{ij}^2 E_{ij}}{\sum_{ij} E_{ij}} & \overline{z^2} &= \frac{\sum_{ij} z_{ij}^2 E_{ij}}{\sum_{ij} E_{ij}}\end{aligned}$$

which are used to calculate the covariance matrices of the x (y) centroids of the event under consideration

$$\text{Cov}(x, z) = \begin{pmatrix} \overline{x^2} - \bar{x}^2 & \bar{xz} - \bar{x}\bar{z} \\ \bar{xz} - \bar{x}\bar{z} & \overline{z^2} - \bar{z}^2 \end{pmatrix}$$

Where E_{ij} are the energies deposited in the TASC crystals, the index i represents the bar number (0, 15) and j the layer (0, 5) of the xz (yz) view.

From the covariance matrix, the eigenvalues are calculated

$$\begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$$

along with the corresponding eigenvectors $\vec{u}$ and $\vec{v}$. The eigenvector $\vec{v}$ provides an estimate of the shower axis and therefore gives a reconstruction of the particle’s direction inside the calorimeter.

The impact point is finally obtained by extrapolating to $z = 0$ (upper plane of the CHD) the line with direction parallel to $\vec{v}$ that passes through the centroid $(\bar{x}, \bar{z})$.

Appendix B

Plots

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- **section B.1: Kalman filter, MC simulated data sample**, plots of the residuals of the impact point variables and of the angular variables, mean value of the residual distribution of the spatial variables and spatial resolution, mean value of the residual distribution of the angular variables and angular resolution, calculated on simulated MC helium data using the Kalman filter method;
- **section B.2: Sample of MC simulated He nuclei**, plots of the *mean absolute error* (mae) and *mean square error* (mse), residuals of the impact point variables and of the angular variables, mean value of the residual distribution of the spatial variables and spatial resolution, mean value of the residual distribution of the angular variables and angular resolution, calculated on simulated MC data by the CNN trained on helium nuclei;
- **section B.3: Sample of MC simulated p and He nuclei**, plots of the *mean absolute error* (mae) and *mean square error* (mse), residuals of the impact point variables and of the angular variables, mean value of the residual distribution of the spatial variables and spatial resolution, mean value of the residual distribution of the angular variables and angular resolution, calculated on simulated MC data by the CNN trained on the combined p and He dataset;
- **section B.4: Sample of He nuclei from flight data**, plots of the residuals of the impact point variables and of the angular variables, mean value of the residual distribution of the spatial variables and spatial resolution, mean value of the residual distribution of the angular variables and angular resolution, calculated on flight data FD by the CNN trained on helium nuclei;

- **section B.5: Sample of p and He nuclei from flight data**, plots of the residuals of the impact point variables and of the angular variables, mean value of the residual distribution of the spatial variables and spatial resolution, mean value of the residual distribution of the angular variables and angular resolution, calculated on flight data FD by the CNN trained on the combined p and He dataset;
- **section B.6: Nuclei charge selection**, charge reconstruction plots in the CHD, with the sum of the two views and the individual xz and yz views.

B.1 Kalman filter, MC simulated data sample

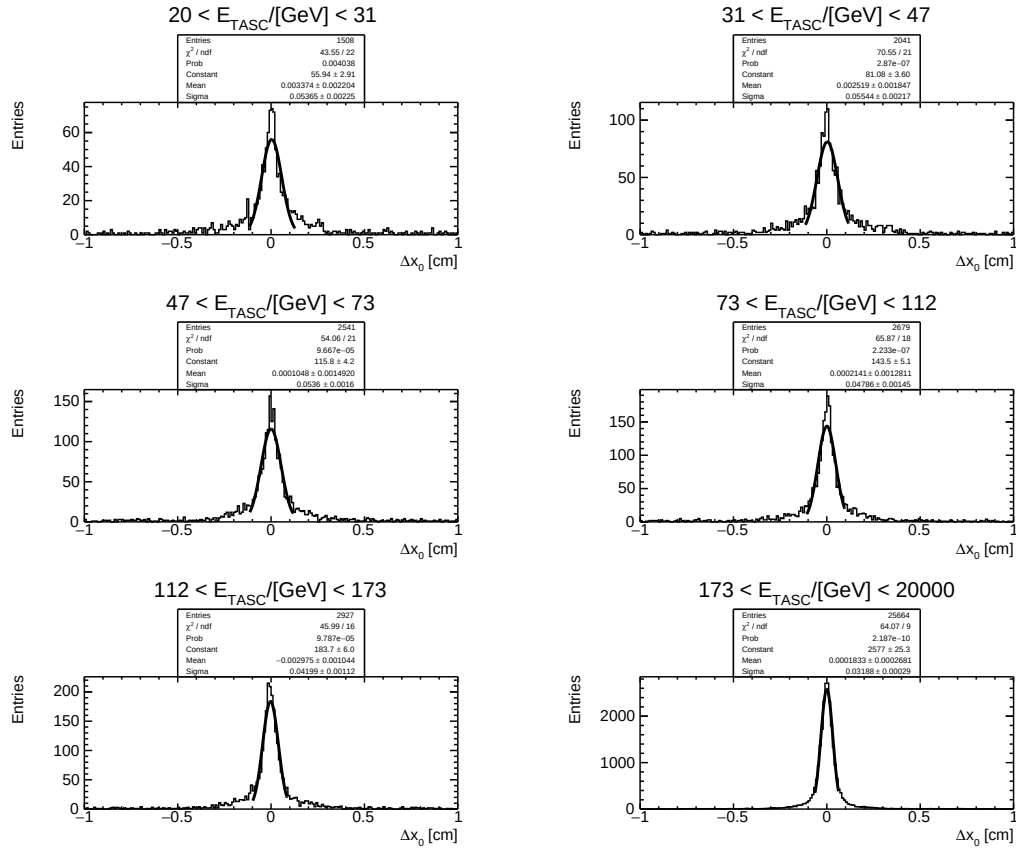


Figure B.1. residual distribution of the impact point variable x_0 predicted by the Kalman filter (*black* histogram) on the MC simulated helium data. The curve superimposed on the histograms constitutes the Gaussian fit to the histograms.

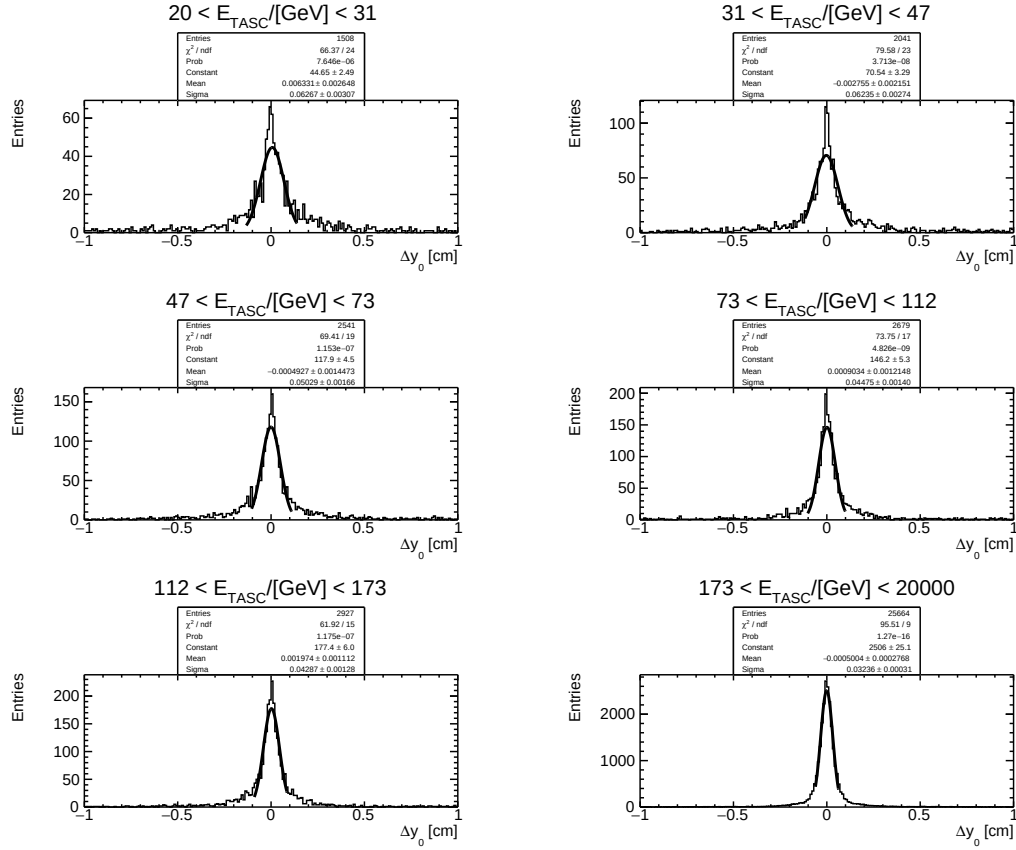


Figure B.2. residual distribution of the impact point variable y_0 predicted by the Kalman filter (*black* histogram) on the MC simulated helium data. The curve superimposed on the histograms constitutes the Gaussian fit to the histograms.

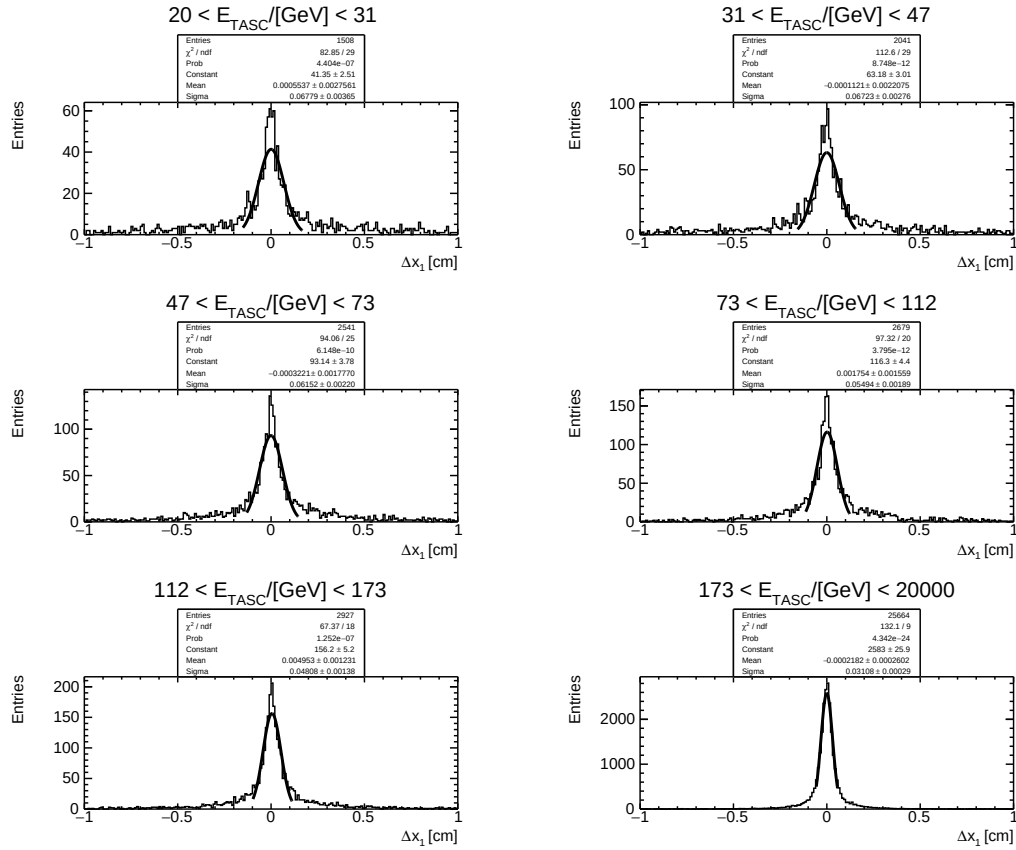


Figure B.3. residual distribution of the impact point variable x_1 predicted by the Kalman filter (*black* histogram) on the MC simulated helium data. The curve superimposed on the histograms constitutes the Gaussian fit to the histograms.

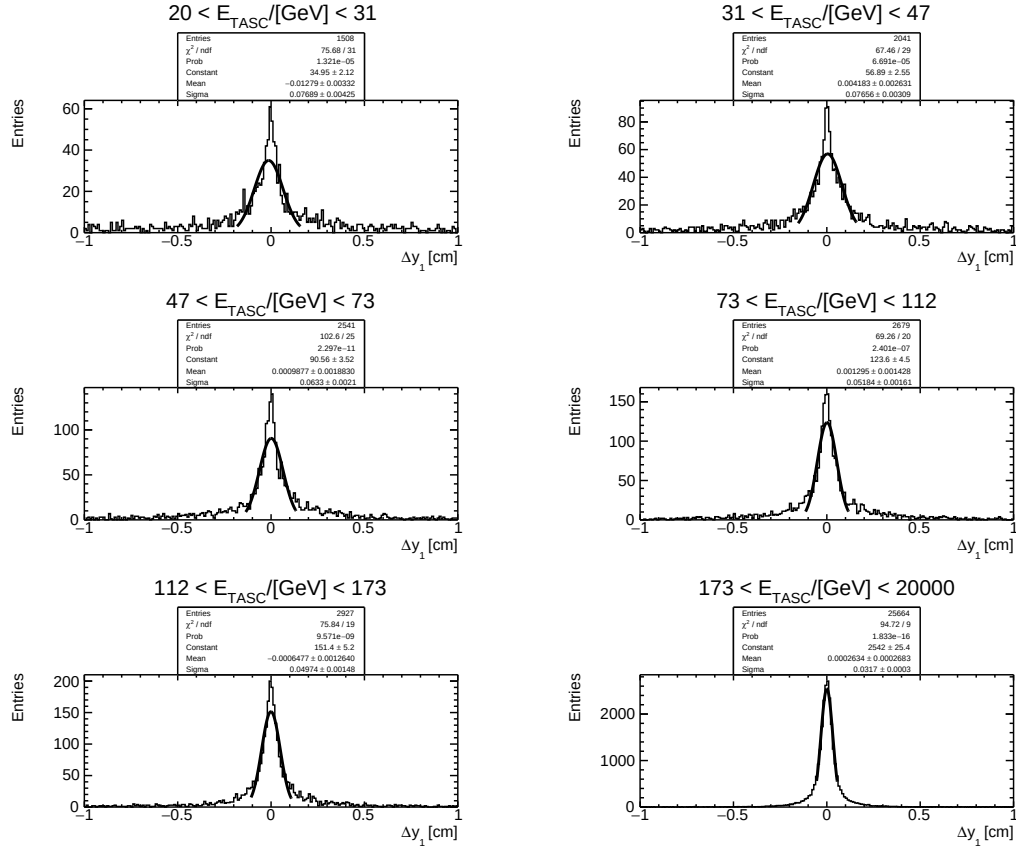


Figure B.4. residual distribution of the impact point variable y_1 predicted by the Kalman filter (*black* histogram) on the MC simulated helium data. The curve superimposed on the histograms constitutes the Gaussian fit to the histograms.

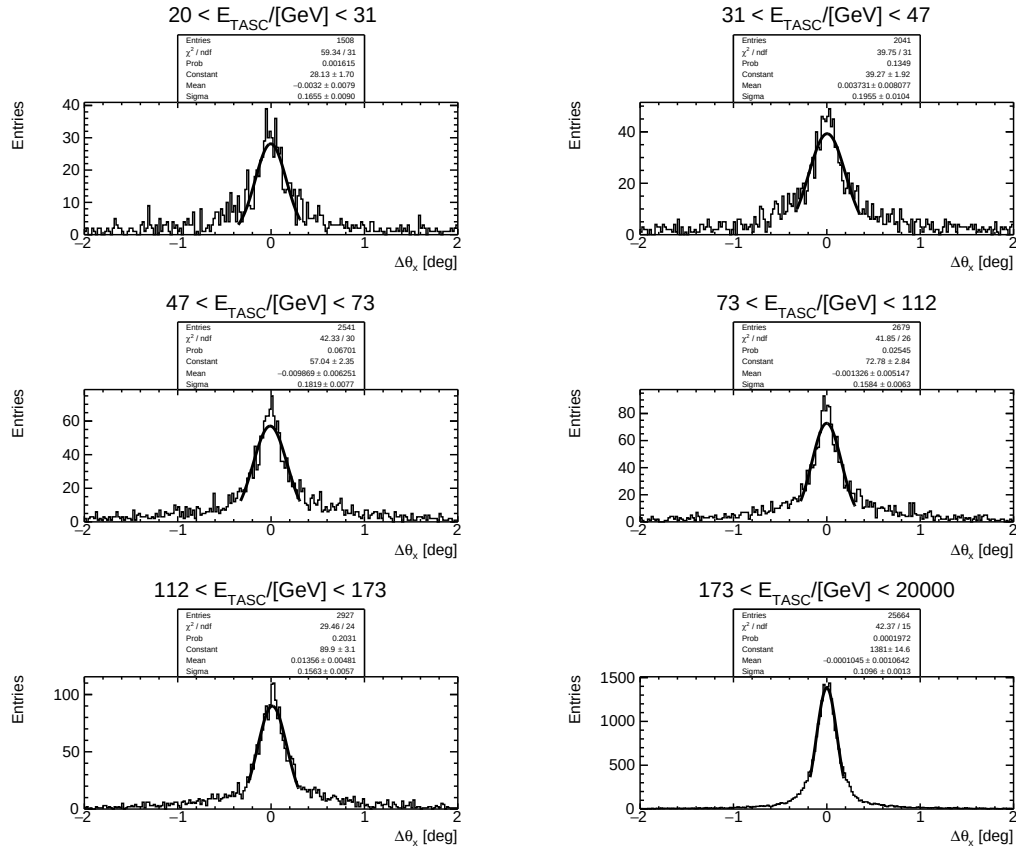


Figure B.5. residual distribution of the angular variable θ_x predicted by the Kalman filter (*black* histogram) on the MC simulated helium data. The curve superimposed on the histograms constitutes the Gaussian fit to the histograms.

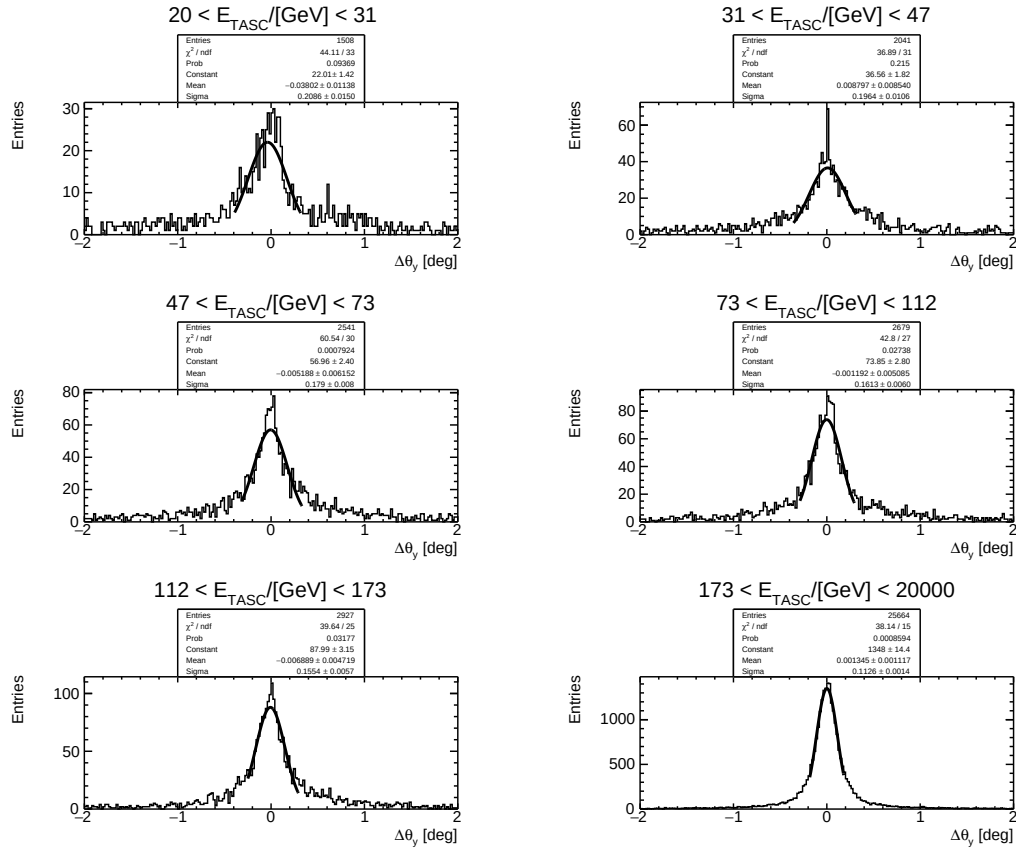


Figure B.6. residual distribution of the angular variable θ_y predicted by the Kalman filter (*black* histogram) on the MC simulated helium data. The curve superimposed on the histograms constitutes the Gaussian fit to the histograms.

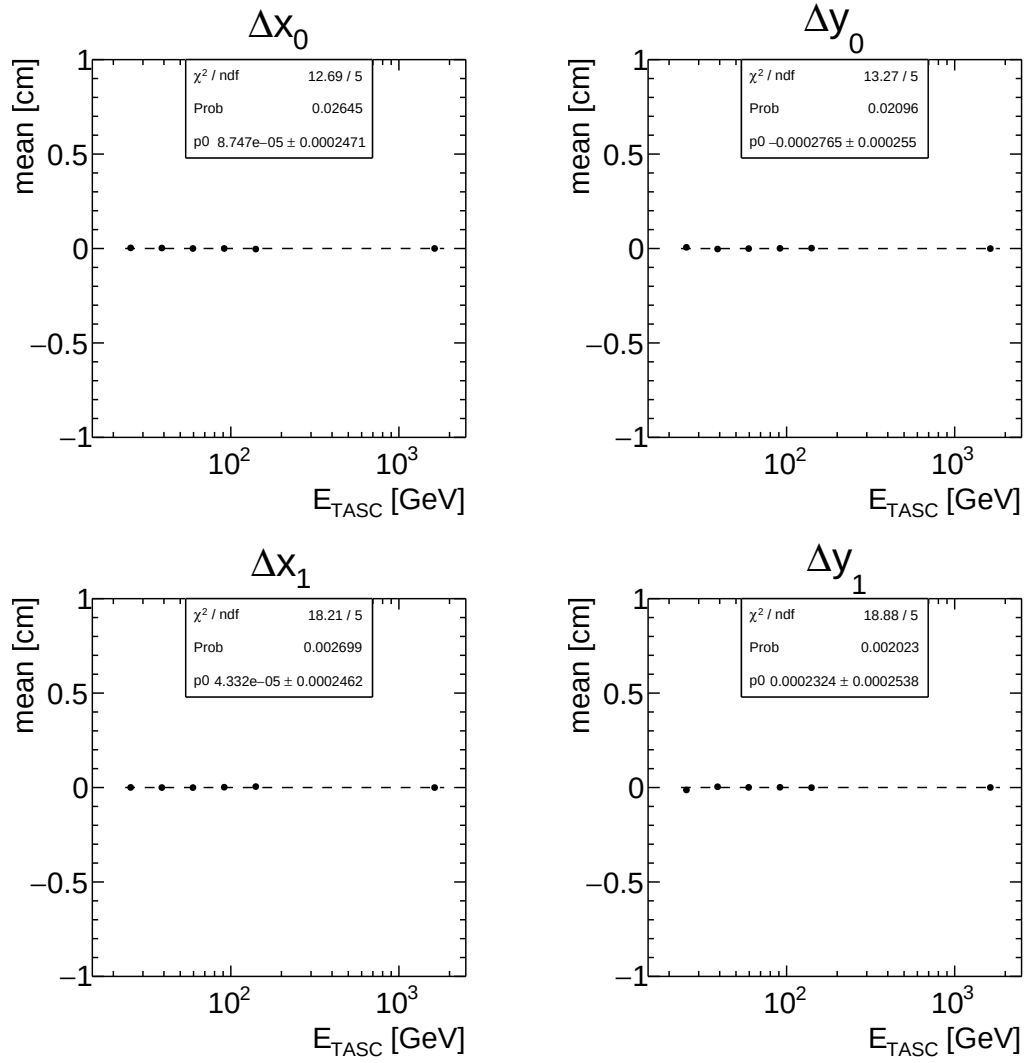


Figure B.7. mean value of the residual distribution of the impact points predicted by the Kalman filter (in *black*) on the MC simulated helium data, as a function of the mean energy deposited in each range.

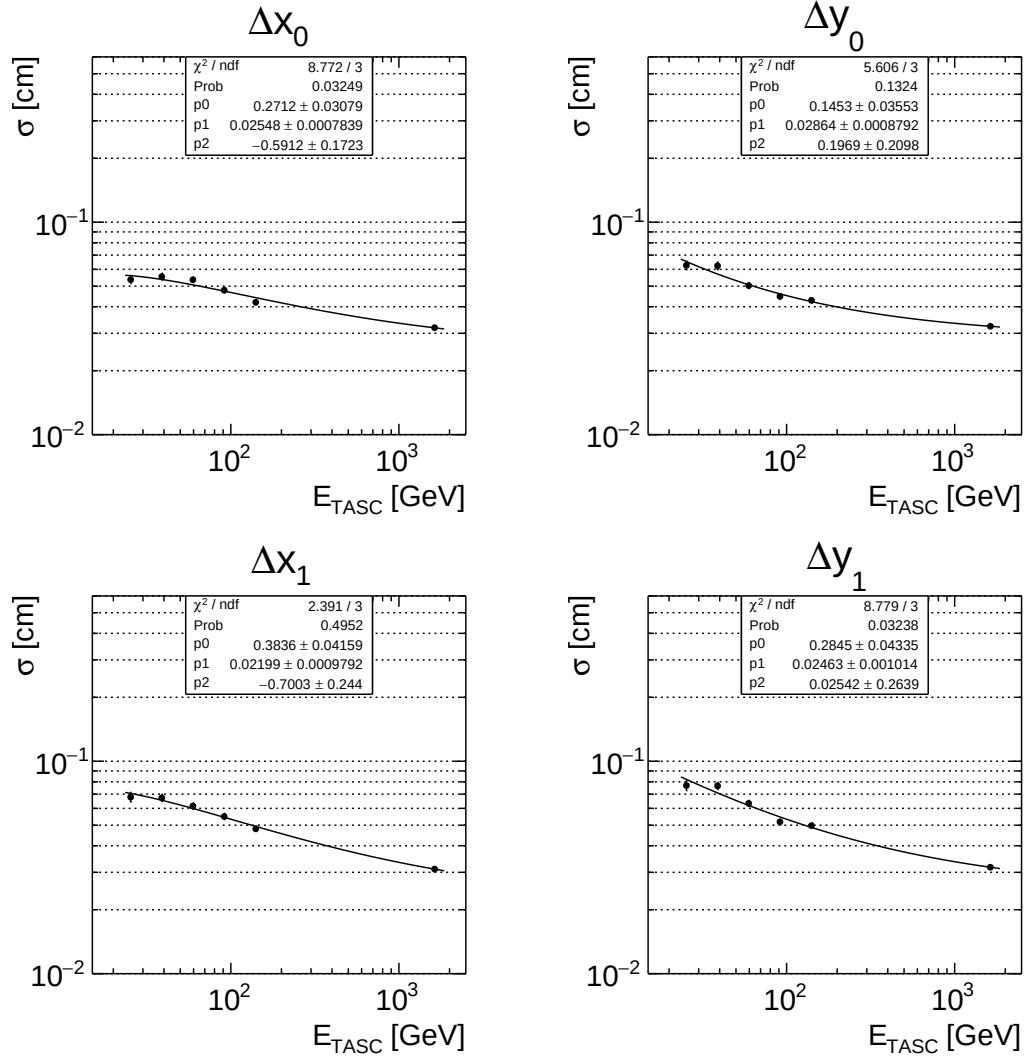


Figure B.8. spatial resolution of the Kalman filter (in *black*) on the MC simulated helium data, as a function of the mean energy deposited in each range.

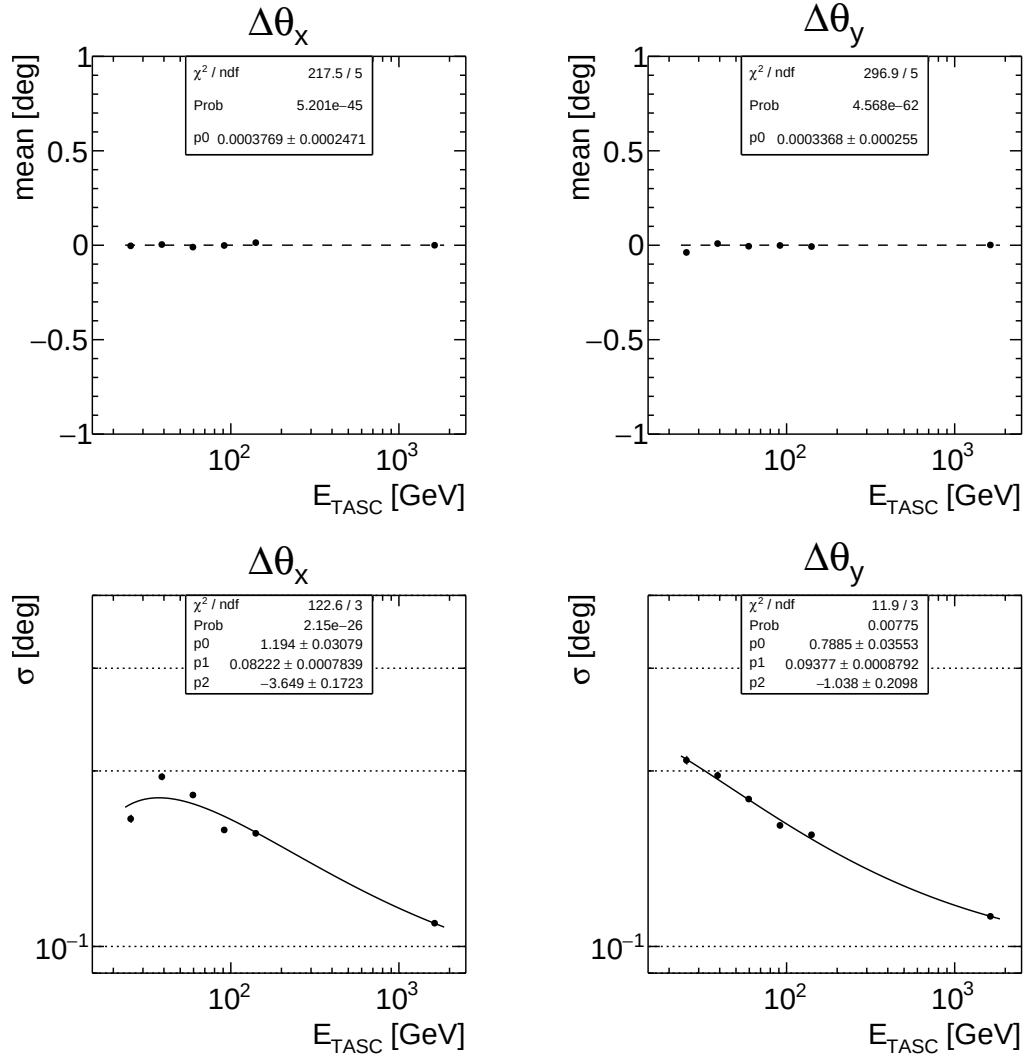


Figure B.9. in the graphs *above* the mean value of the residual distribution; *below* the angular resolution of the residual distribution of the angular variables of the Kalman filter (in *black*) on the MC simulated helium data, as a function of the mean energy deposited in each range.

B.2 Sample of MC simulated He nuclei

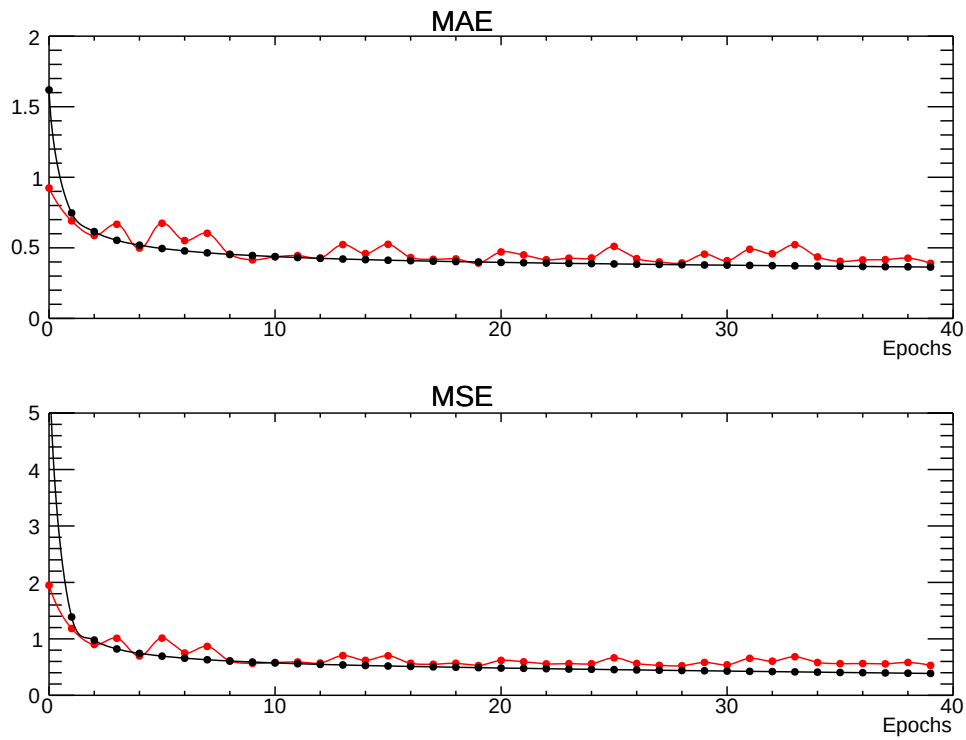


Figure B.10. graphs of the *mean absolute error* (mae) and *mean square error* (mse) related to the CNN trained on helium nuclei. In *black* the training curve, in *red* the validation curve.

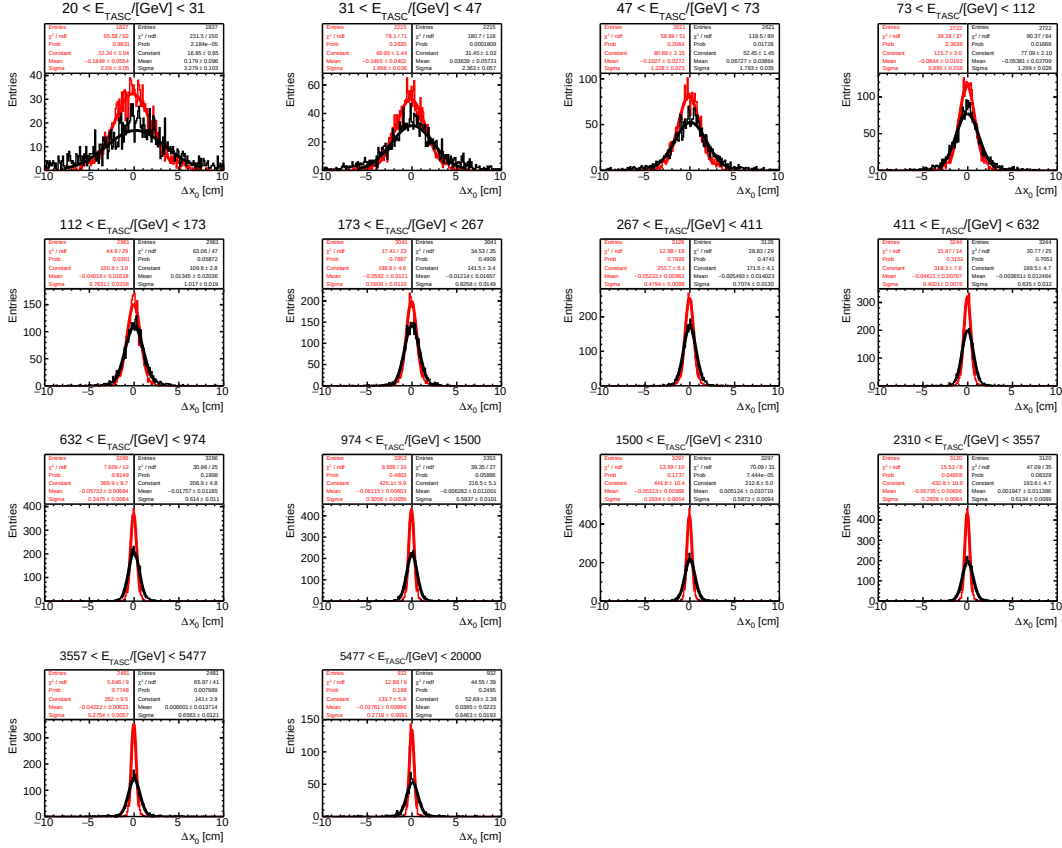


Figure B.11. distribution of the residuals of the impact point variable x_0 predicted by the CNN trained on helium nuclei (histogram in red) and by the shower axis method (histogram in black). The curves superimposed on the histograms constitute the Gaussian fits to the histograms of the same color.

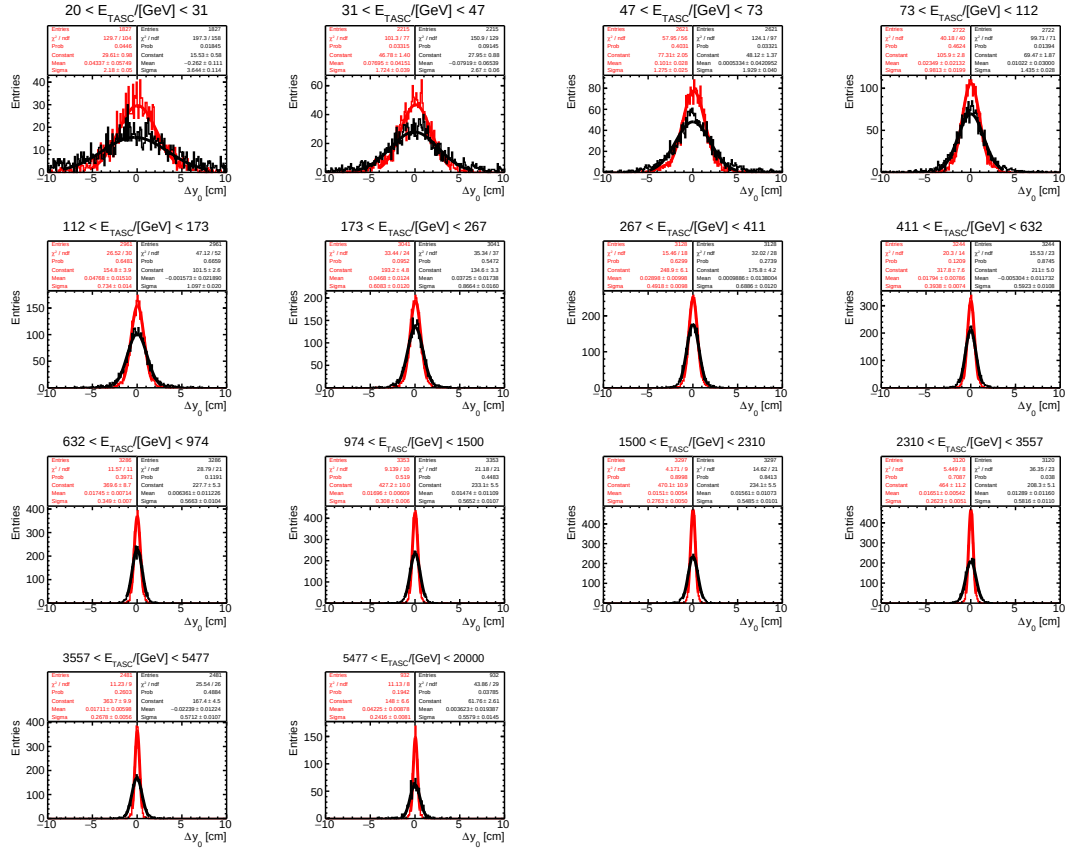


Figure B.12. distribution of the residuals of the impact point variable y_0 predicted by the CNN trained on helium nuclei (histogram in red) and by the shower axis method (histogram in black). The curves superimposed on the histograms constitute the Gaussian fits to the histograms of the same color.

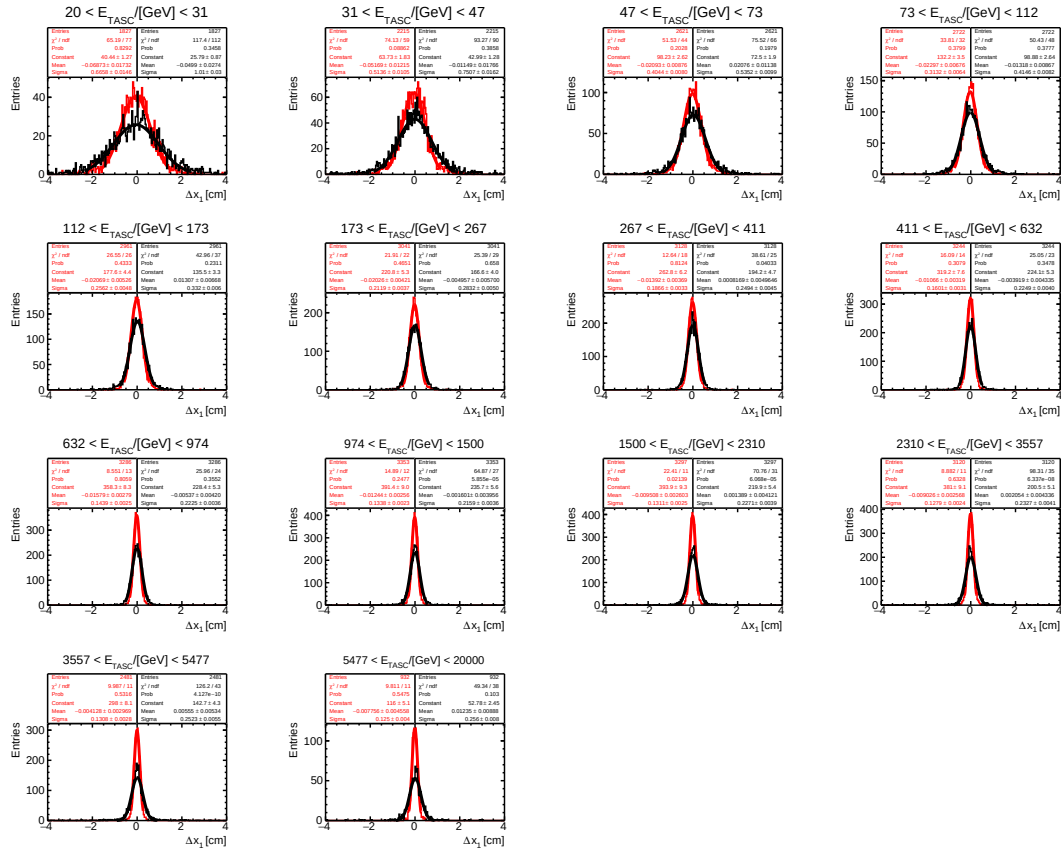


Figure B.13. distribution of the residuals of the impact point variable x_1 predicted by the CNN trained on helium nuclei (histogram in red) and by the shower axis method (histogram in black). The curves superimposed on the histograms constitute the Gaussian fits to the histograms of the same color.

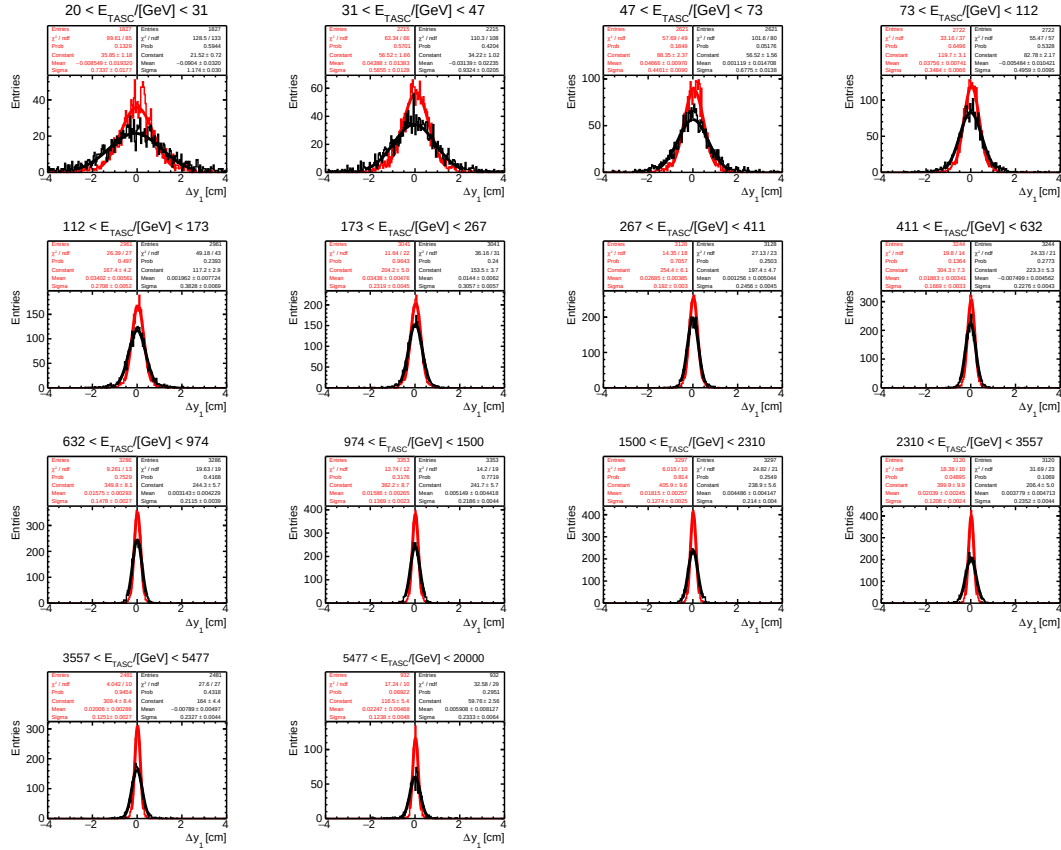


Figure B.14. distribution of the residuals of the impact point variable y_1 predicted by the CNN trained on helium nuclei (histogram in *red*) and by the shower axis method (histogram in *black*). The curves superimposed on the histograms constitute the Gaussian fits to the histograms of the same color.

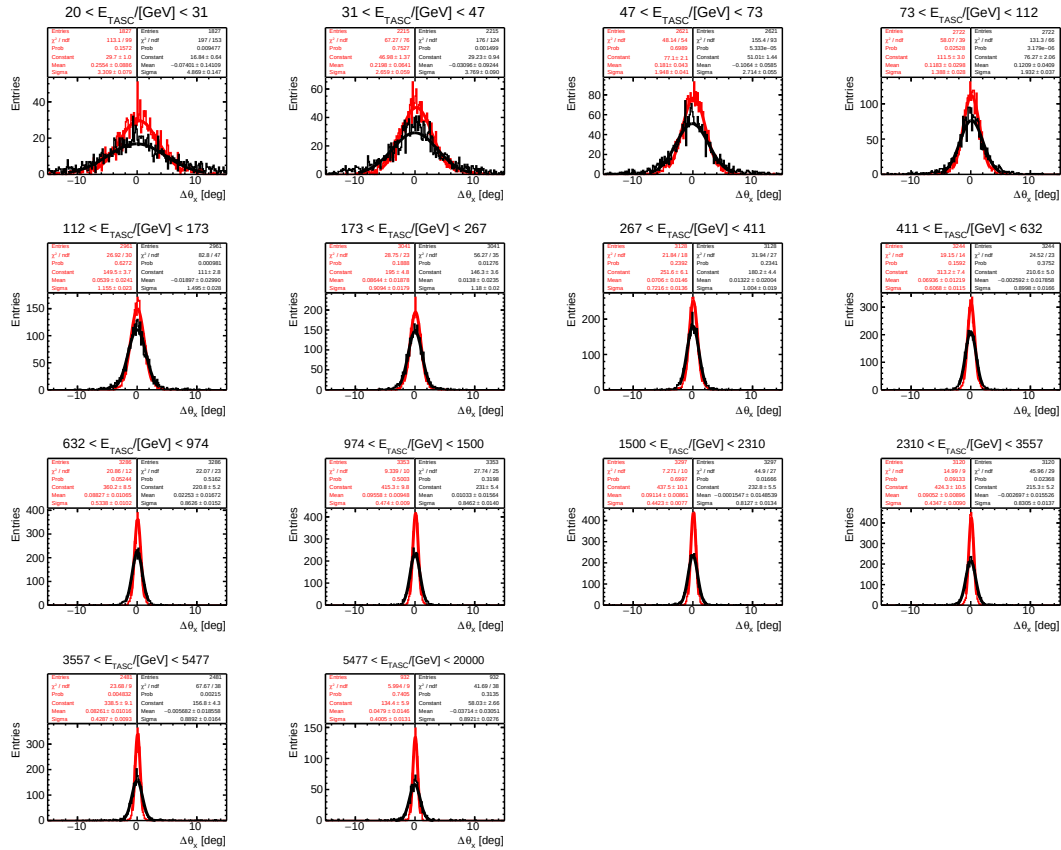


Figure B.15. distribution of the residuals of the angular variable θ_x predicted by the CNN trained on helium nuclei (histogram in *red*) and by the shower axis method (histogram in *black*). The curves superimposed on the histograms constitute the Gaussian fits to the histograms of the same color.

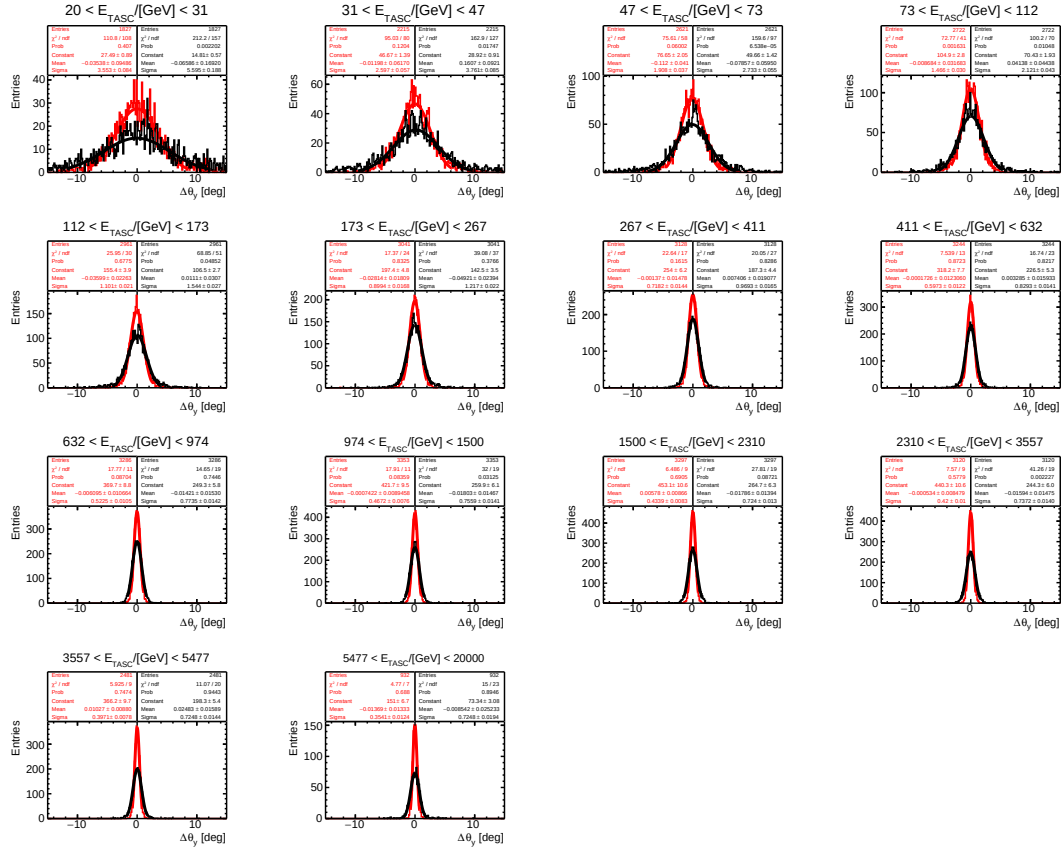


Figure B.16. distribution of the residuals of the angular variable θ_y predicted by the CNN trained on helium nuclei (histogram in *red*) and by the shower axis method (histogram in *black*). The curves superimposed on the histograms constitute the Gaussian fits to the histograms of the same color.

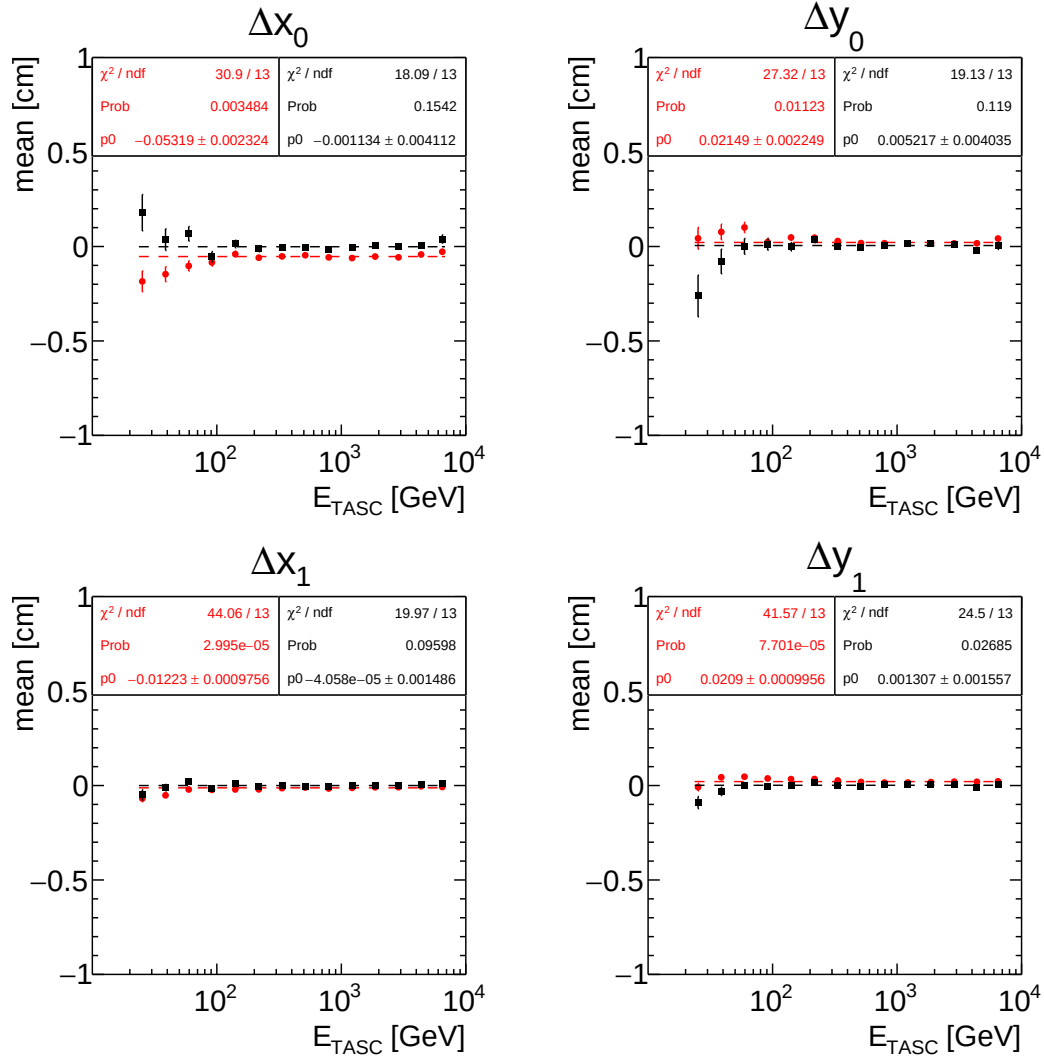


Figure B.17. mean value of the distribution of the impact point residuals of the CNN trained on helium nuclei (in *red*) and of the shower axis method (in *black*), as a function of the mean energy deposited in each range.

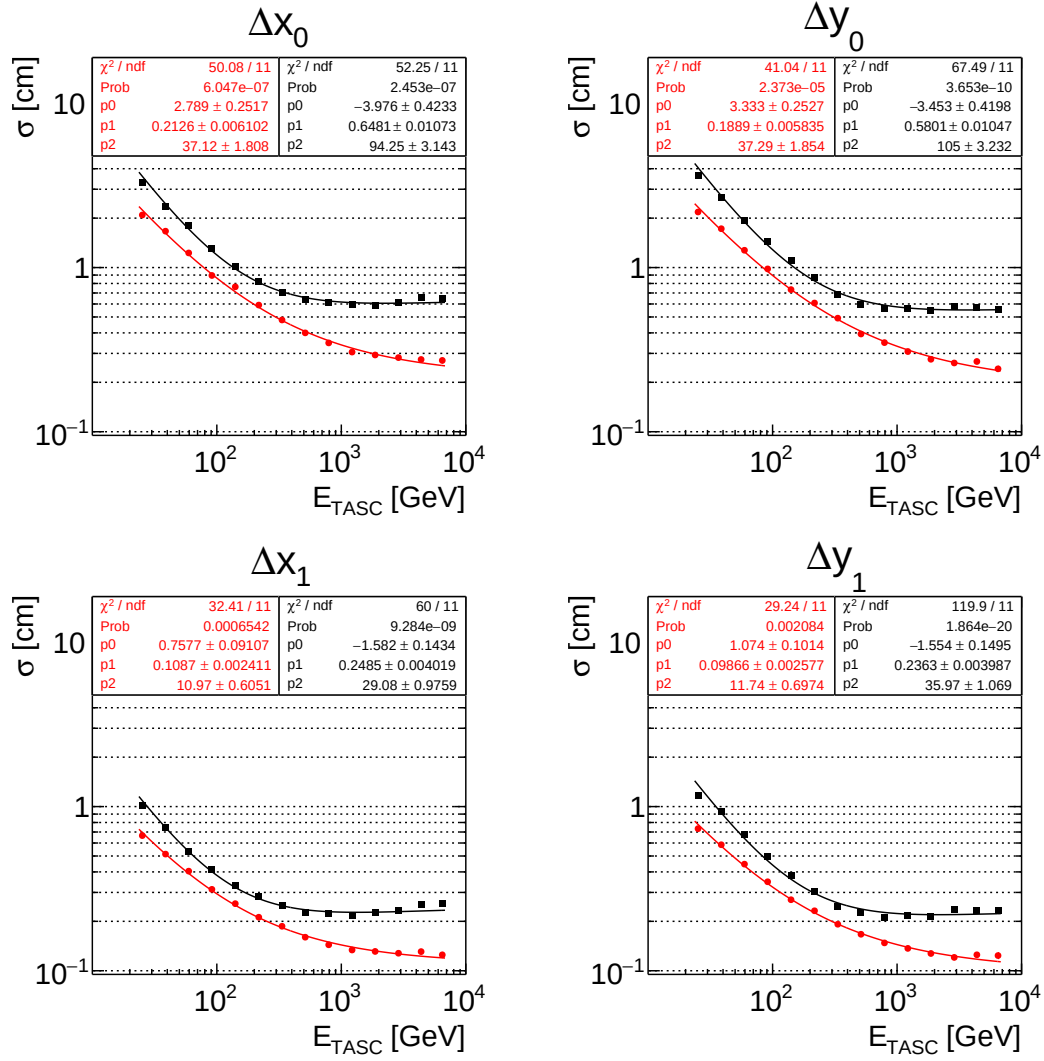


Figure B.18. spatial resolution of the CNN trained on helium nuclei (in red) and of the shower axis method (in black), as a function of the mean energy deposited in each range.

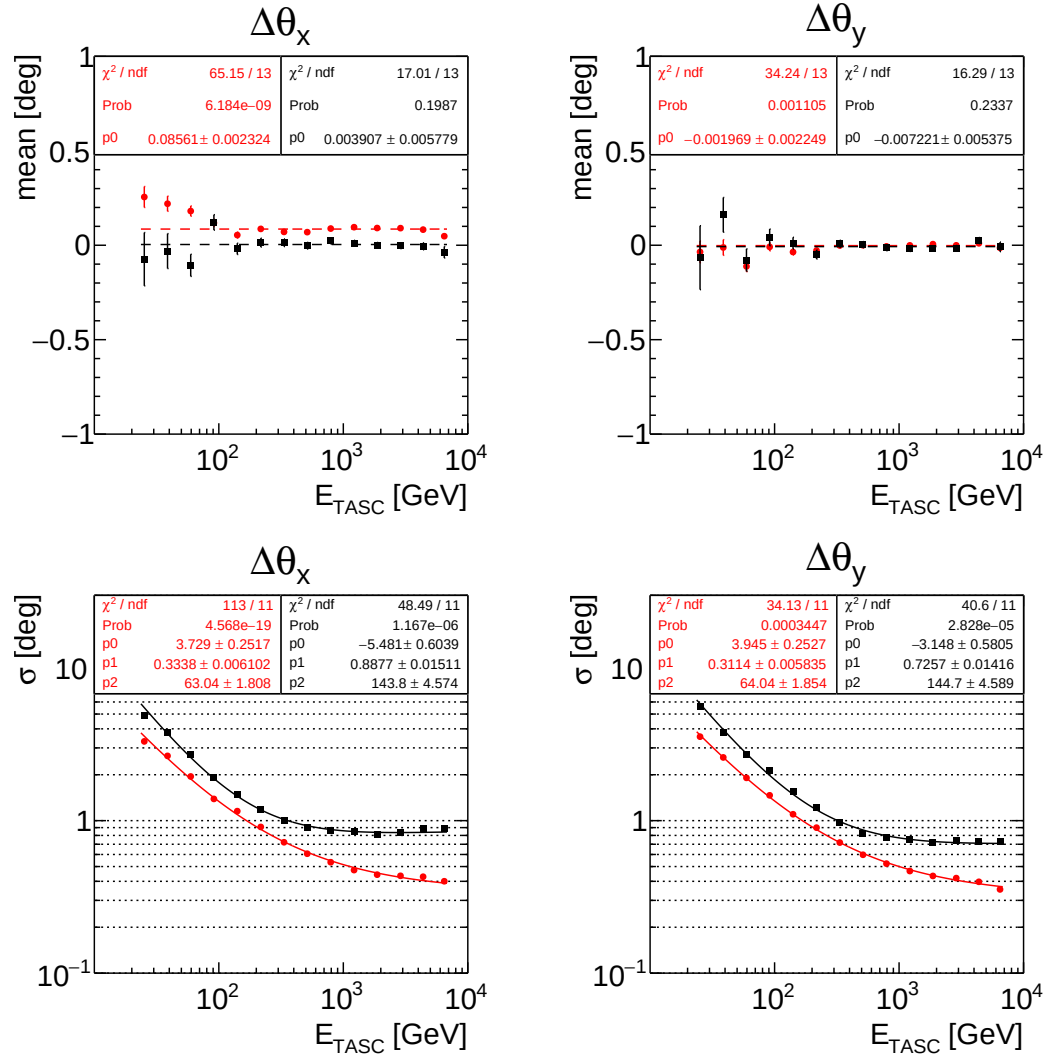


Figure B.19. in the *top* graphs the mean value of the residuals distribution; *below* the angular resolution of the residuals distribution of the angular variables of the CNN trained on helium nuclei (in *red*) and of the shower axis method (in *black*), as a function of the mean energy deposited in each range.

B.3 Sample of MC simulated p and He nuclei

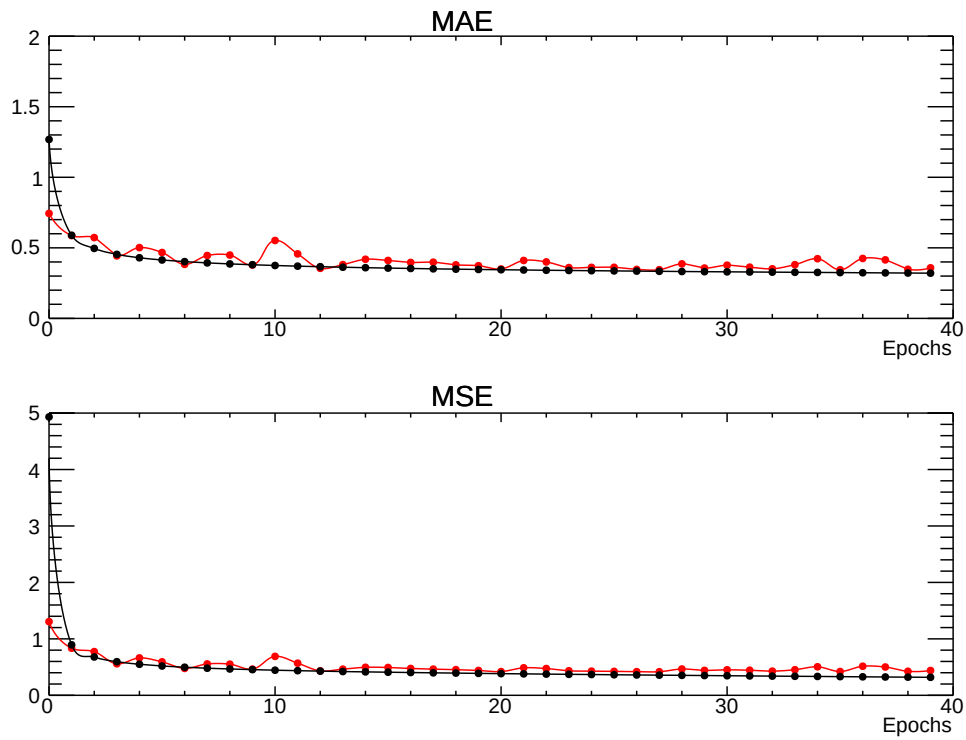


Figure B.20. graphs of the *mean absolute error* (mae) and *mean square error* (mse) relating to the CNN trained on combined p and He data. In *black* the training curve, in *red* the validation curve

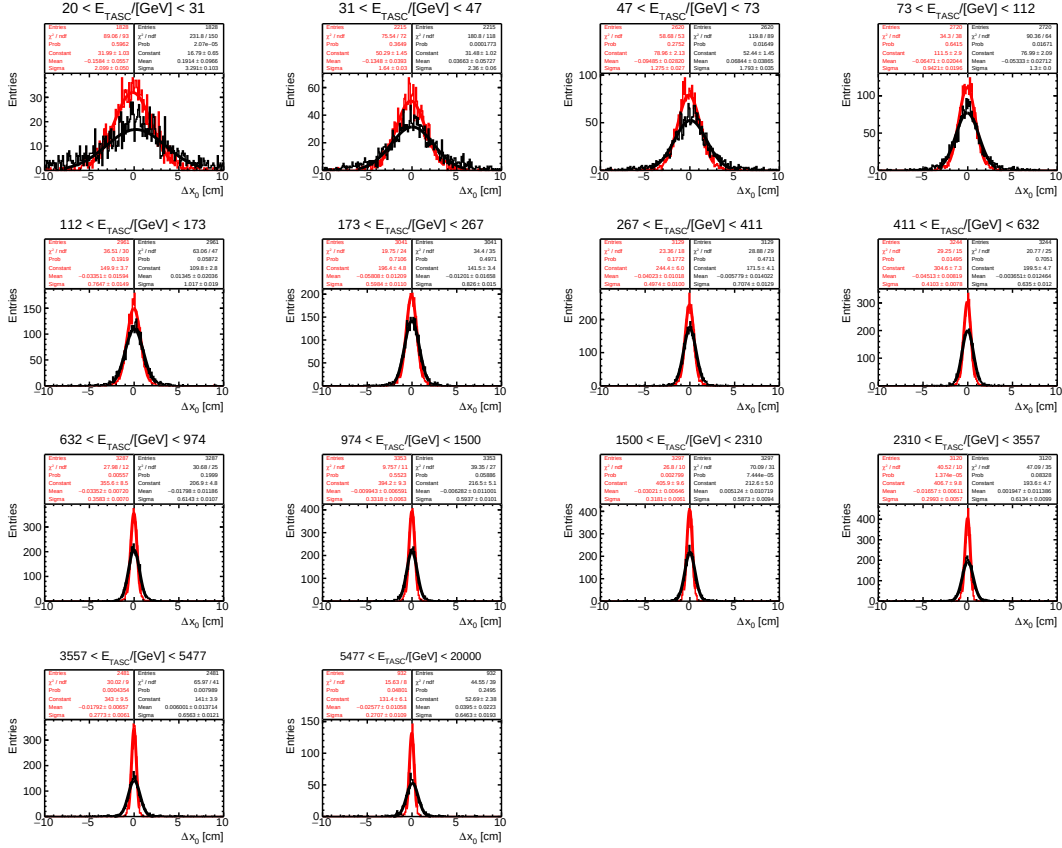


Figure B.21. residual distribution of the impact point variable x_0 predicted by the CNN trained on combined p and He data (red histogram) and by the shower axis method (black histogram). The curves superimposed on the histograms constitute the Gaussian fits to the histograms of the same color.

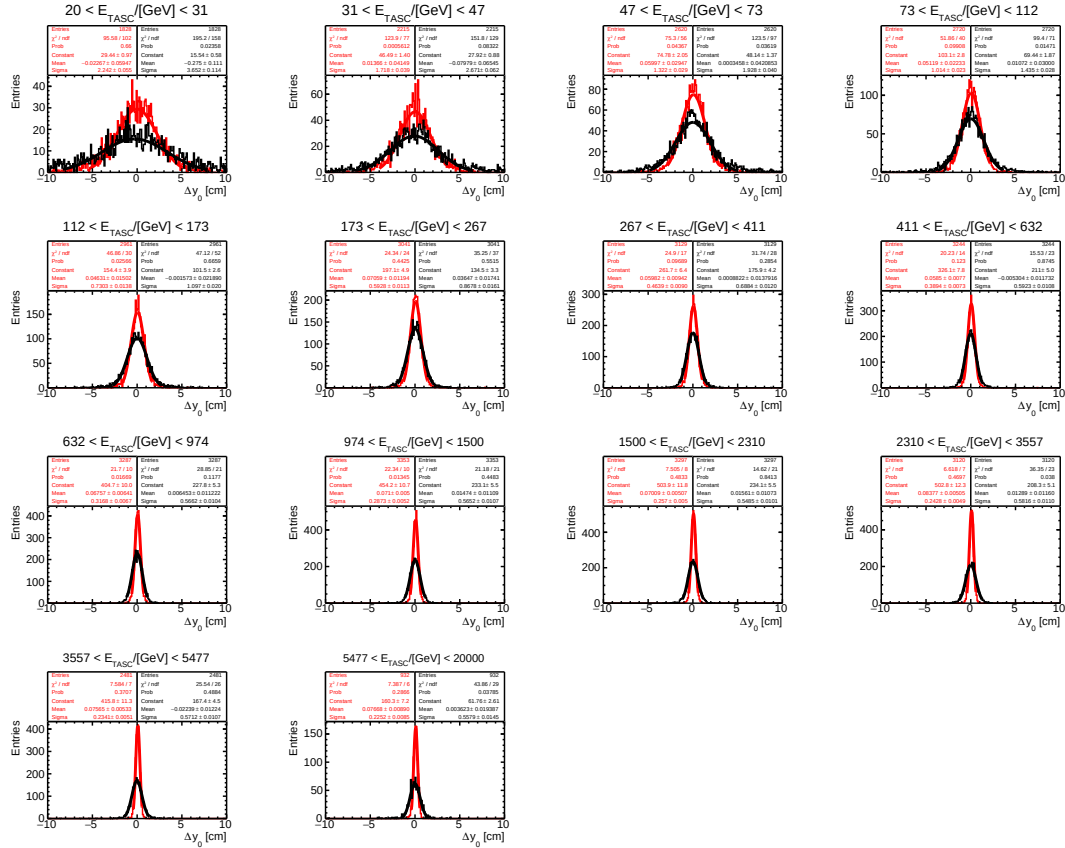


Figure B.22. residual distribution of the impact point variable y_0 predicted by the CNN trained on combined p and He data (*red* histogram) and by the shower axis method (*black* histogram). The curves superimposed on the histograms constitute the Gaussian fits to the histograms of the same color.

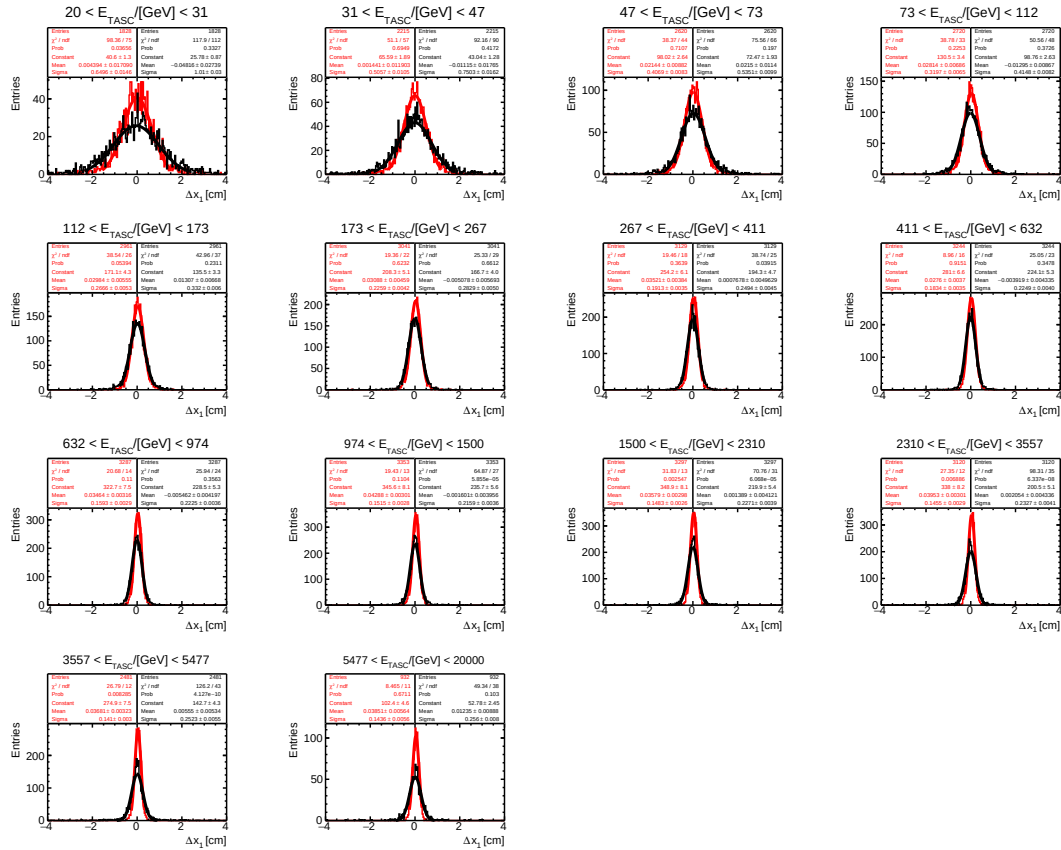


Figure B.23. residual distribution of the impact point variable x_1 predicted by the CNN trained on combined p and He data (red histogram) and by the shower axis method (black histogram). The curves superimposed on the histograms constitute the Gaussian fits to the histograms of the same color.

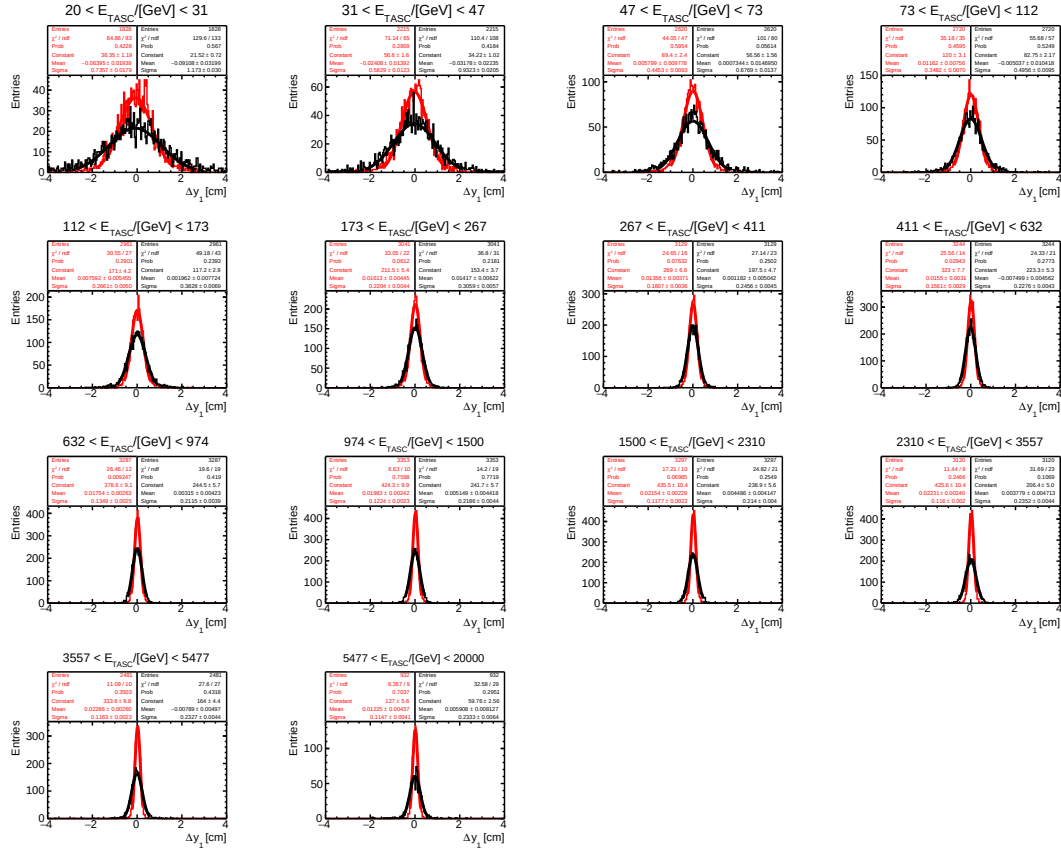


Figure B.24. residual distribution of the impact point variable y_1 predicted by the CNN trained on combined p and He data (*red* histogram) and by the shower axis method (*black* histogram). The curves superimposed on the histograms constitute the Gaussian fits to the histograms of the same color.

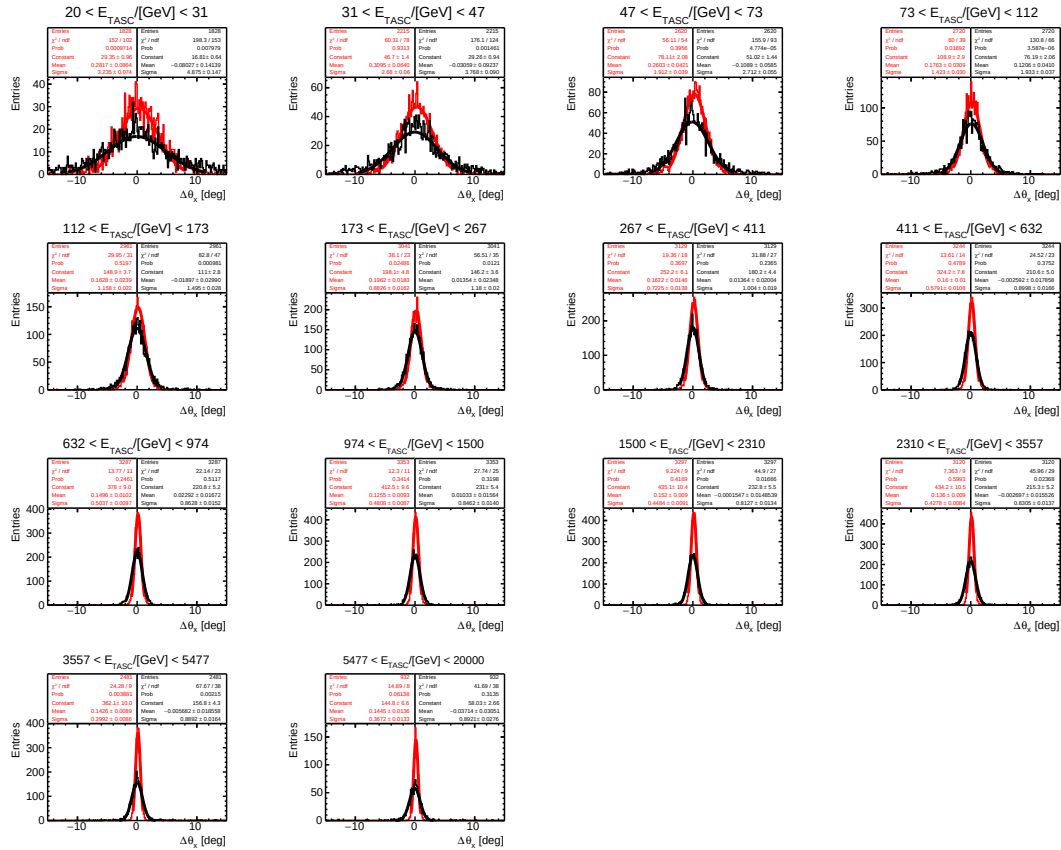


Figure B.25. residual distribution of the angular variable θ_x predicted by the CNN trained on combined p and He data (*red* histogram) and by the shower axis method (*black* histogram). The curves superimposed on the histograms constitute the Gaussian fits to the histograms of the same color.

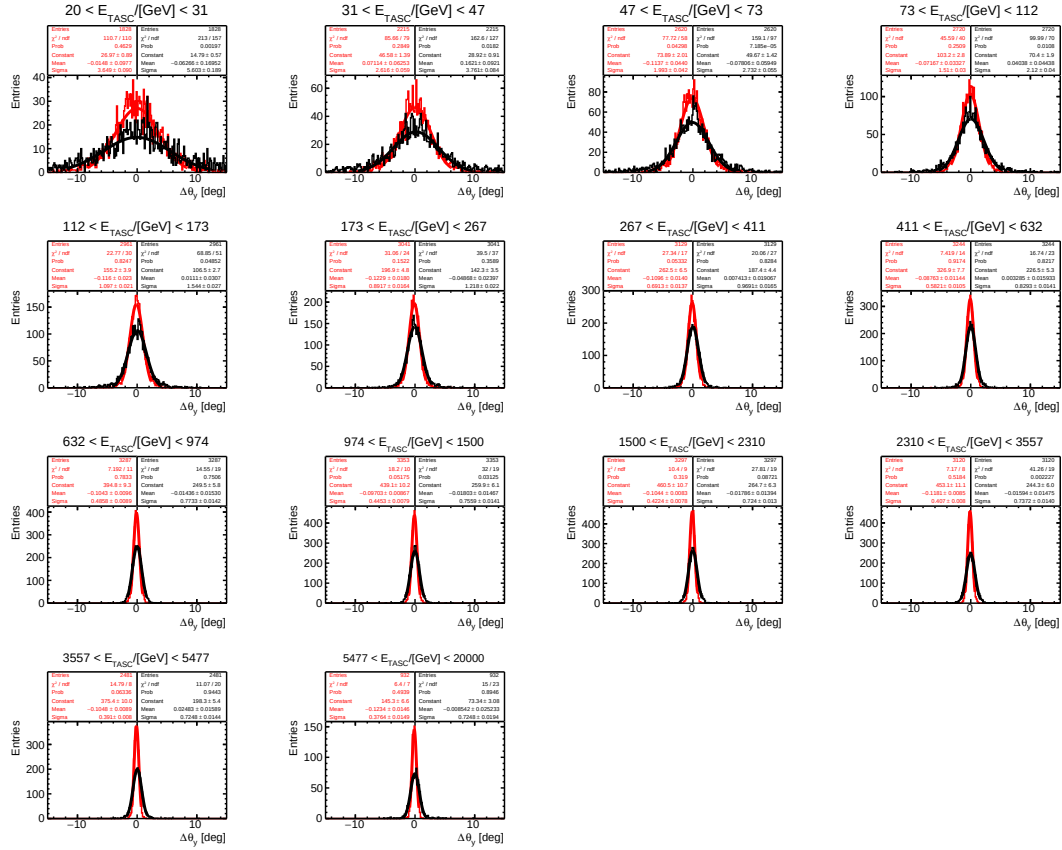


Figure B.26. residual distribution of the angular variable θ_y predicted by the CNN trained on combined p and He data (*red* histogram) and by the shower axis method (*black* histogram). The curves superimposed on the histograms constitute the Gaussian fits to the histograms of the same color.

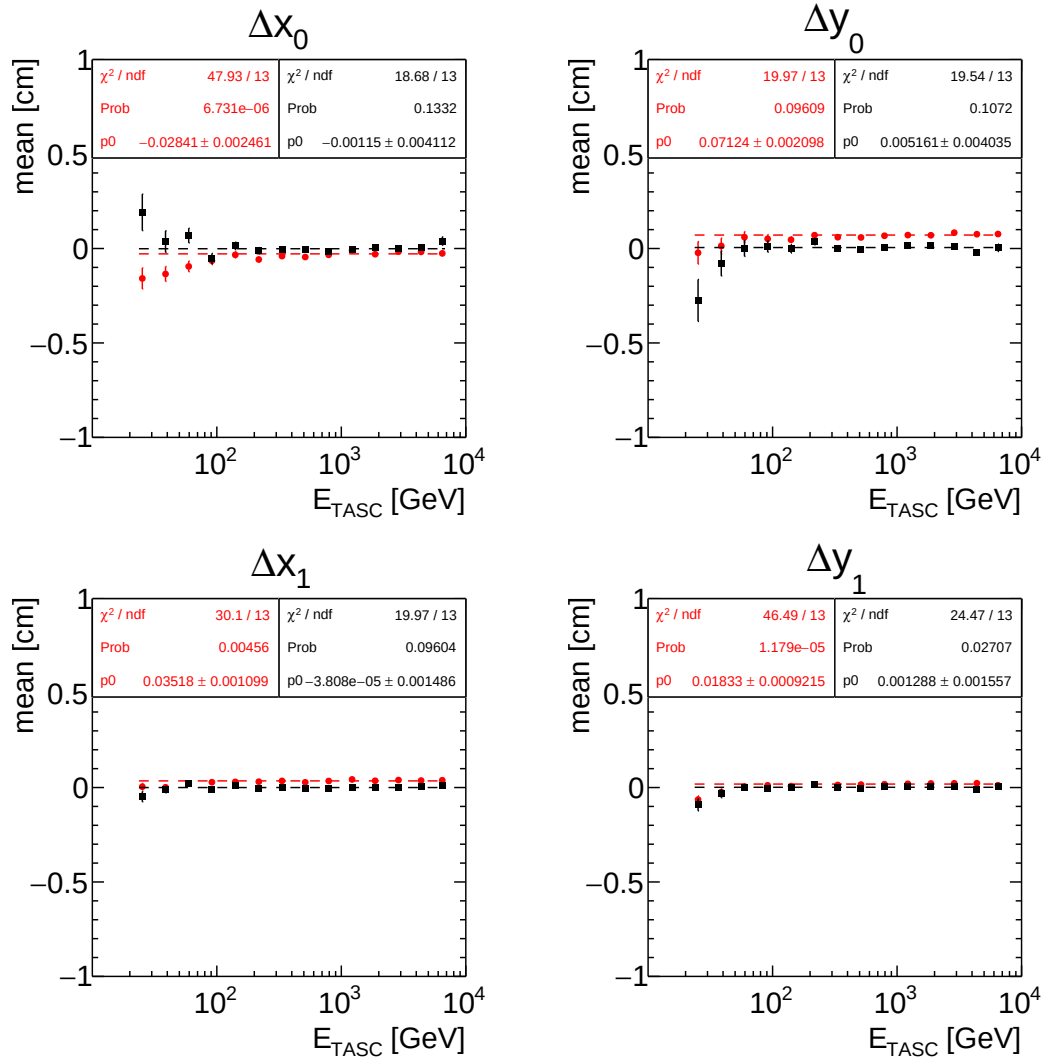


Figure B.27. spatial resolution of the residual distribution of the impact points of the CNN trained on combined p and He data (in red) and of the shower axis method (in black), as a function of the average energy deposited in each range.

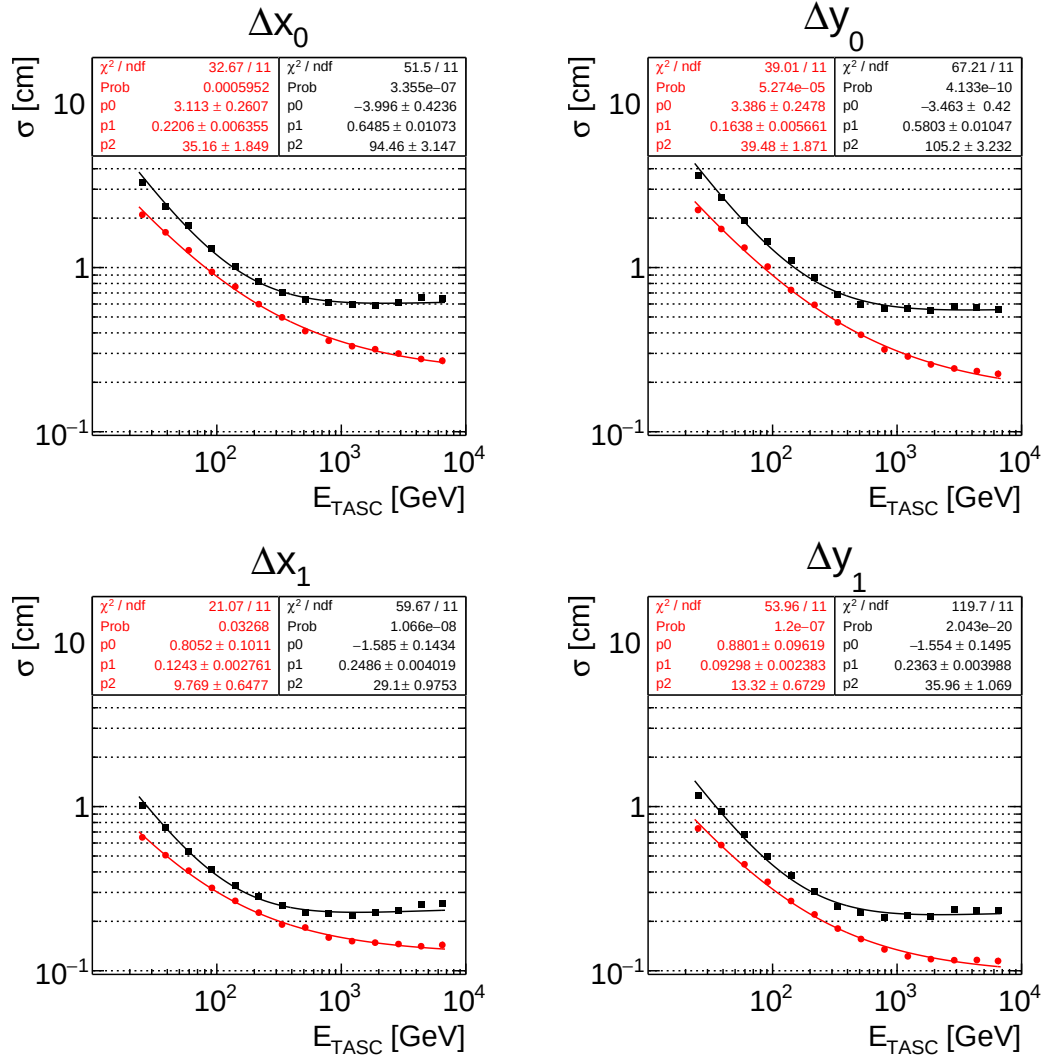


Figure B.28. spatial resolution of the CNN trained on combined p and He data (in *red*) and of the shower axis method (in *black*), as a function of the average energy deposited in each range.

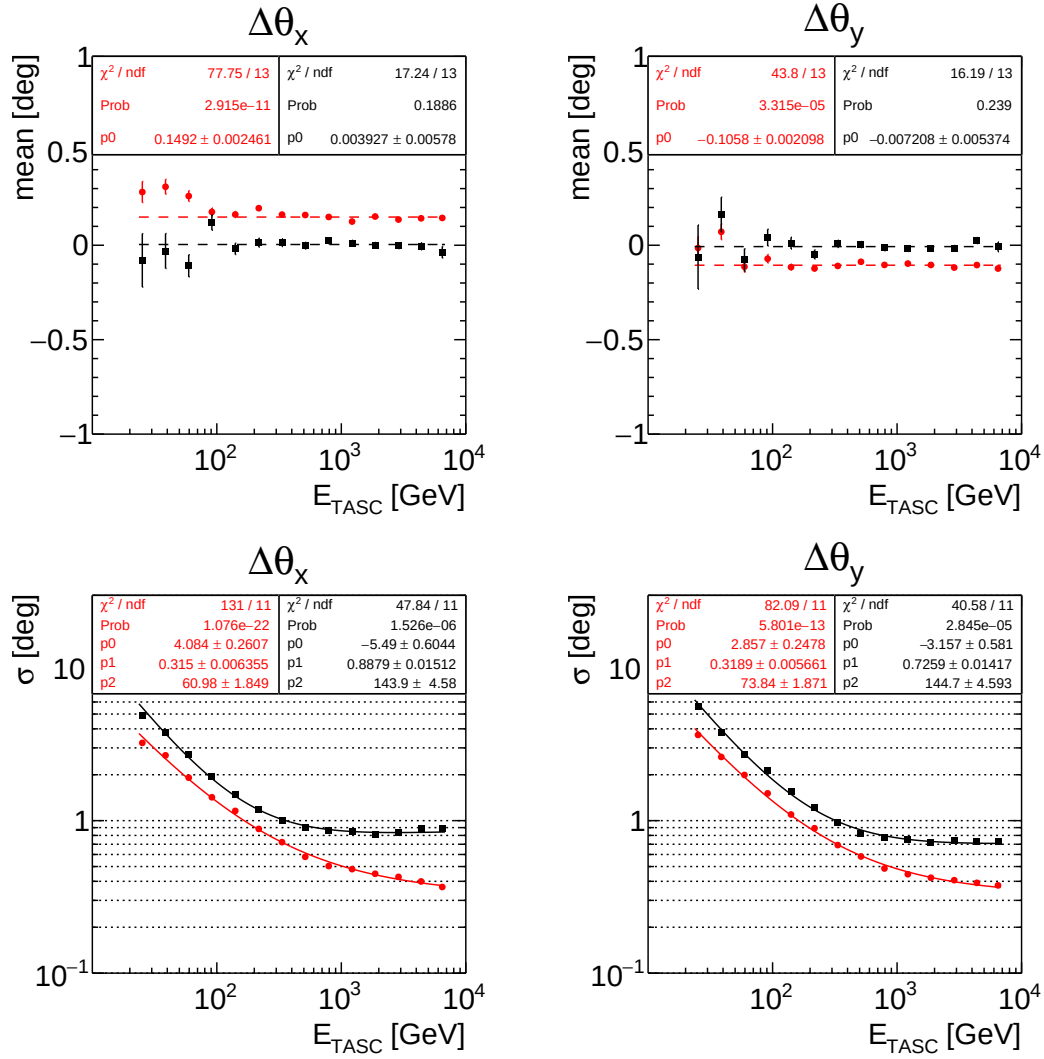


Figure B.29. in the graphs *above* the mean value of the residual distribution; *below* the angular resolution of the residual distribution of the angular variables of the CNN trained on combined p and He data (in *red*) and of the shower axis method (in *black*), as a function of the average energy deposited in each range.

B.4 Sample of He nuclei from flight data

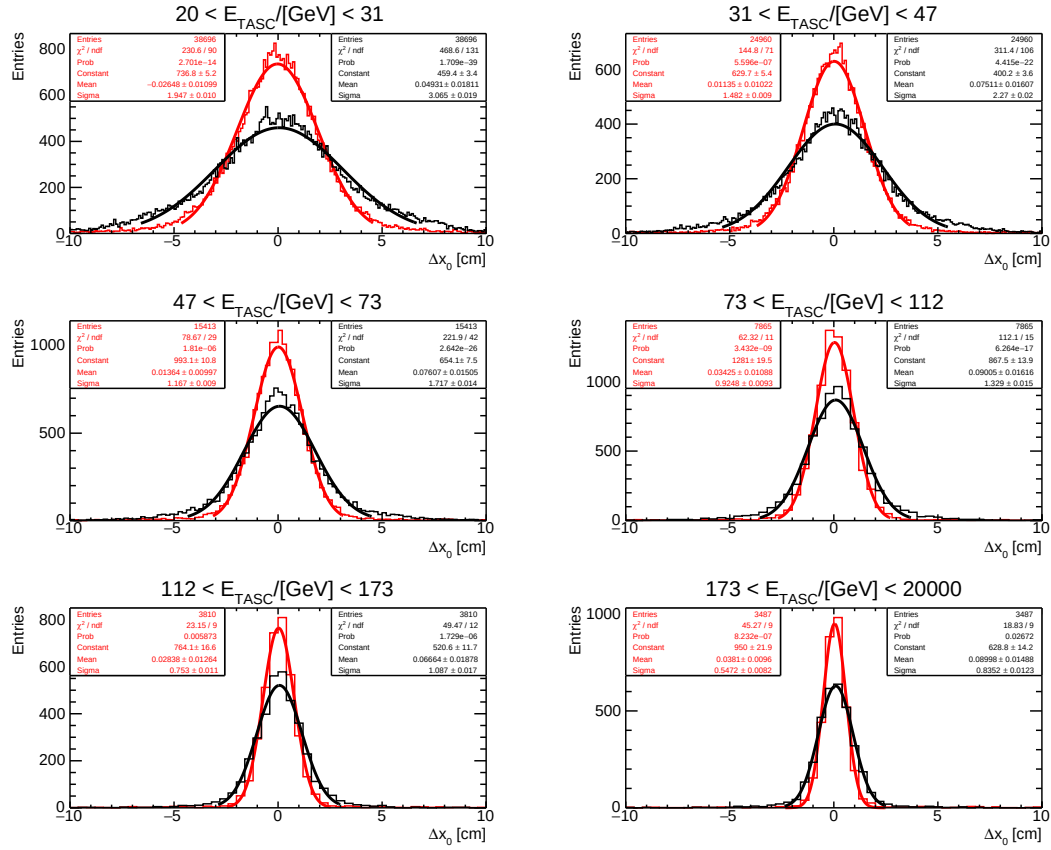


Figure B.30. residual distribution of the impact point variable x_0 predicted by the CNN trained on helium nuclei (*red* histogram) and by the shower axis method (*black* histogram). The curves superimposed on the histograms constitute the Gaussian fits to the histograms of the same color.

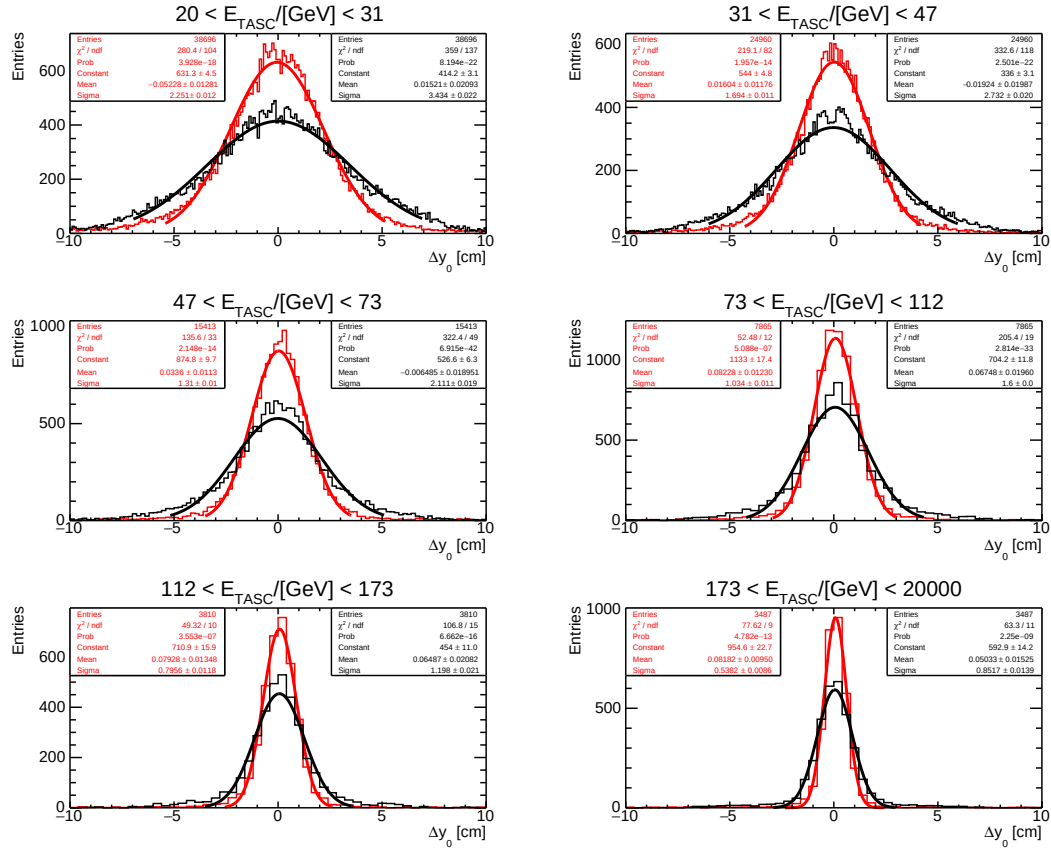


Figure B.31. residual distribution of the impact point variable y_0 predicted by the CNN trained on helium nuclei (*red* histogram) and by the shower axis method (*black* histogram). The curves superimposed on the histograms constitute the Gaussian fits to the histograms of the same color.

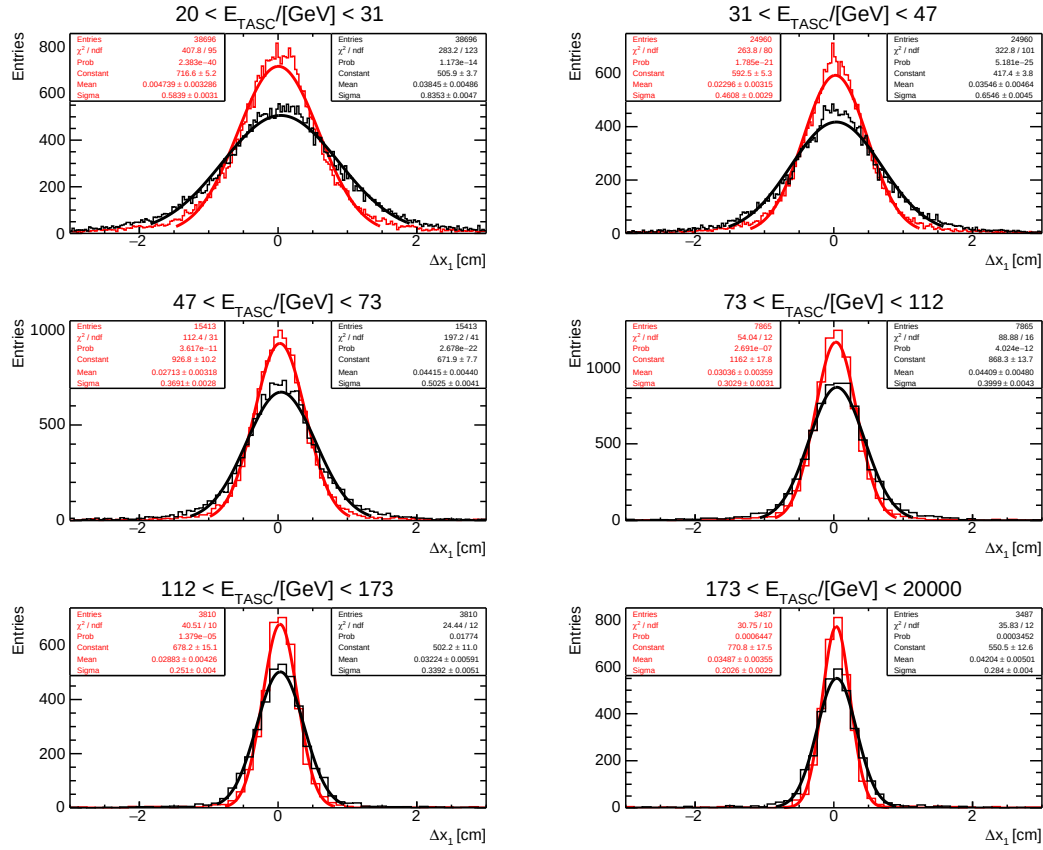


Figure B.32. residual distribution of the impact point variable x_1 predicted by the CNN trained on helium nuclei (*red* histogram) and by the shower axis method (*black* histogram). The curves superimposed on the histograms constitute the Gaussian fits to the histograms of the same color.

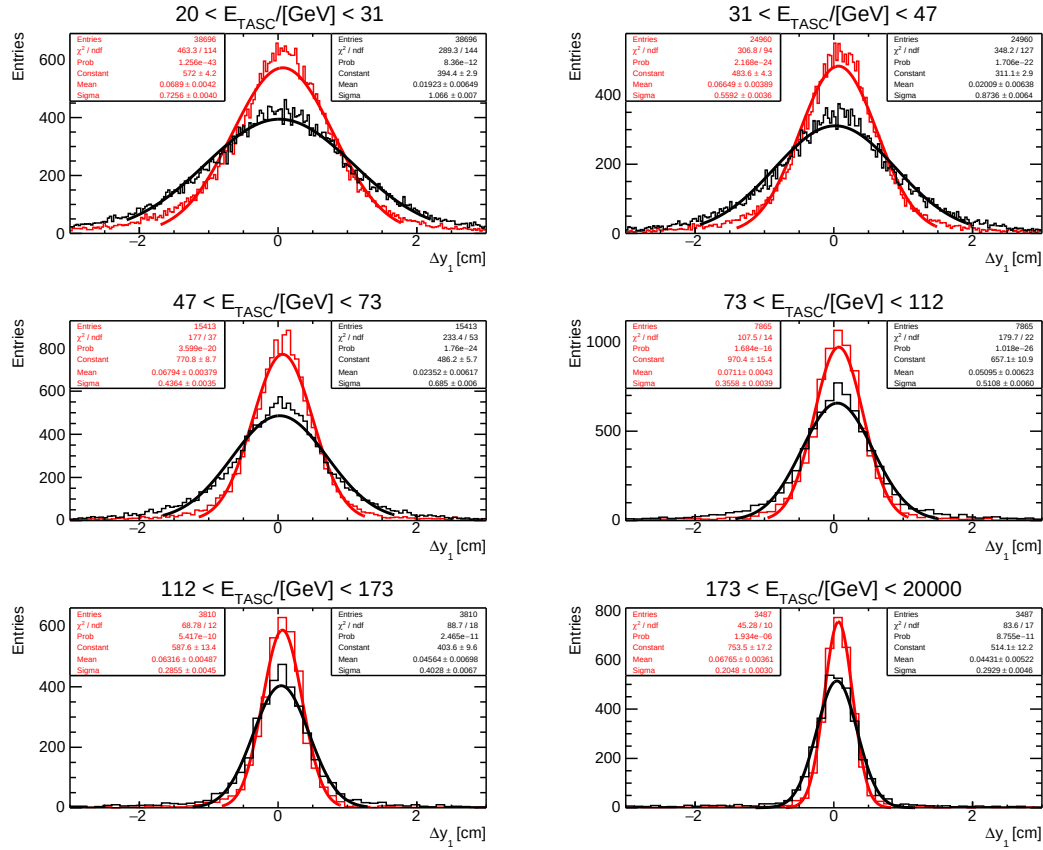


Figure B.33. residual distribution of the impact point variable y_1 predicted by the CNN trained on helium nuclei (*red* histogram) and by the shower axis method (*black* histogram). The curves superimposed on the histograms constitute the Gaussian fits to the histograms of the same color.

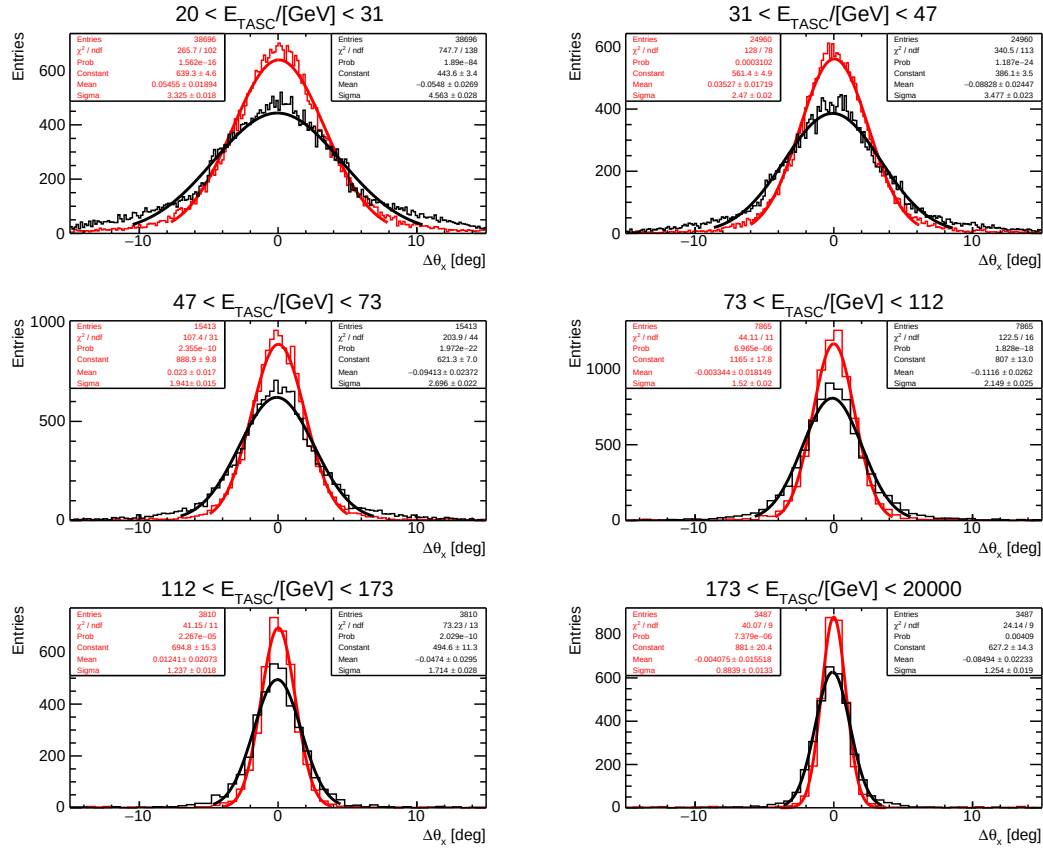


Figure B.34. residual distribution of the angular variable θ_x predicted by the CNN trained on helium nuclei (*red* histogram) and by the shower axis method (*black* histogram). The curves superimposed on the histograms constitute the Gaussian fits to the histograms of the same color.

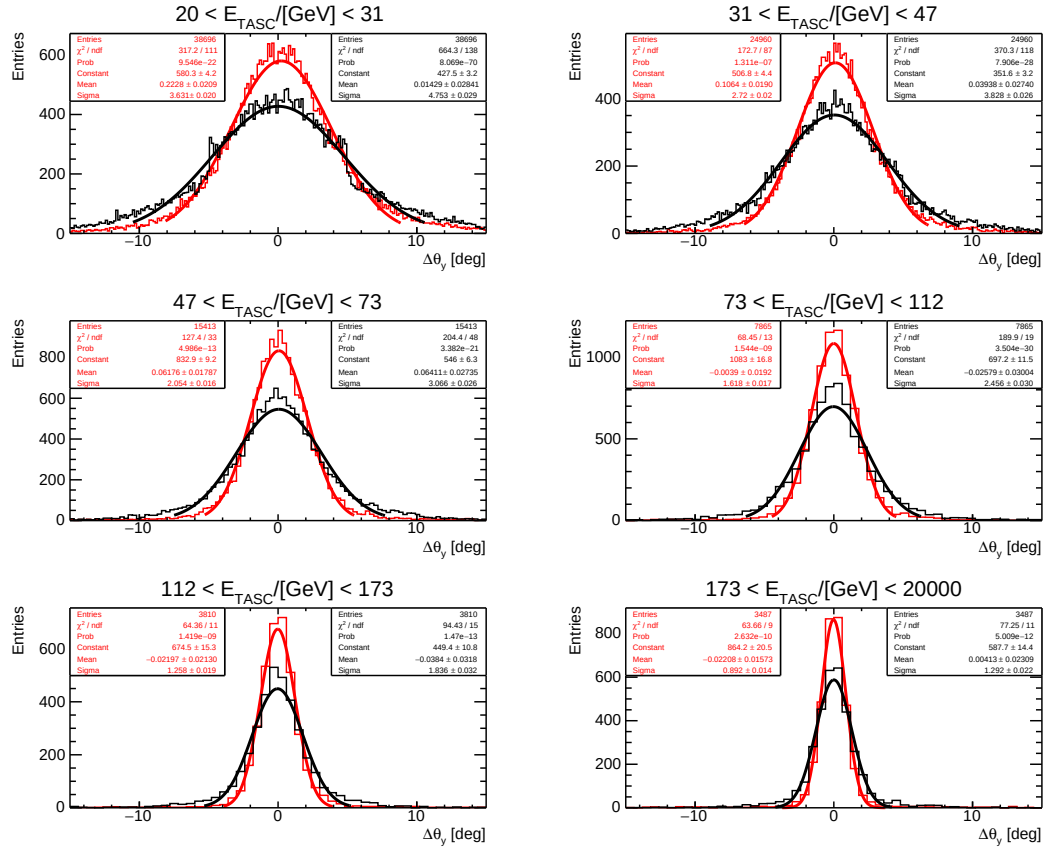


Figure B.35. residual distribution of the angular variable θ_y predicted by the CNN trained on helium nuclei (red histogram) and by the shower axis method (black histogram). The curves superimposed on the histograms constitute the Gaussian fits to the histograms of the same color.

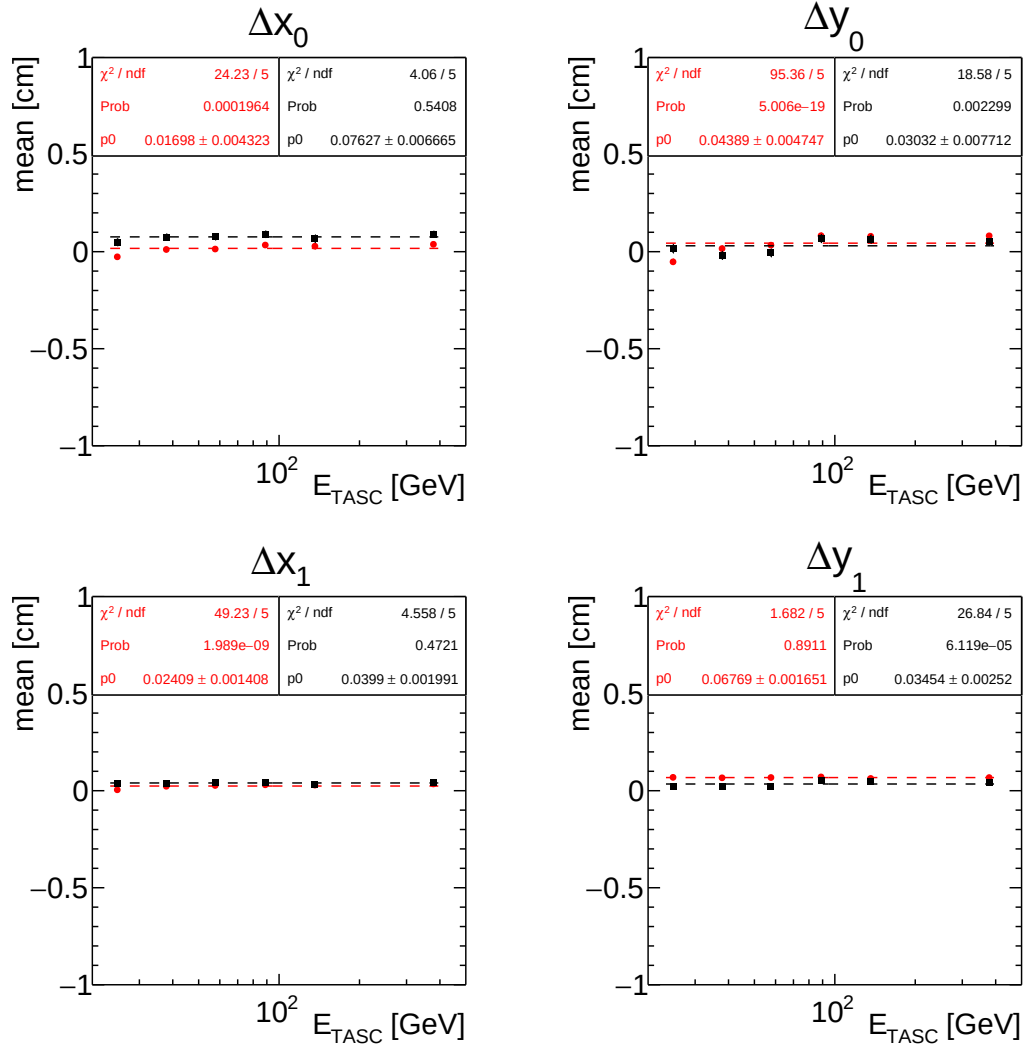


Figure B.36. mean value of the impact point residual distribution of the CNN trained on helium nuclei (in *red*) and of the shower axis method (in *black*), as a function of the mean deposited energy in each range.

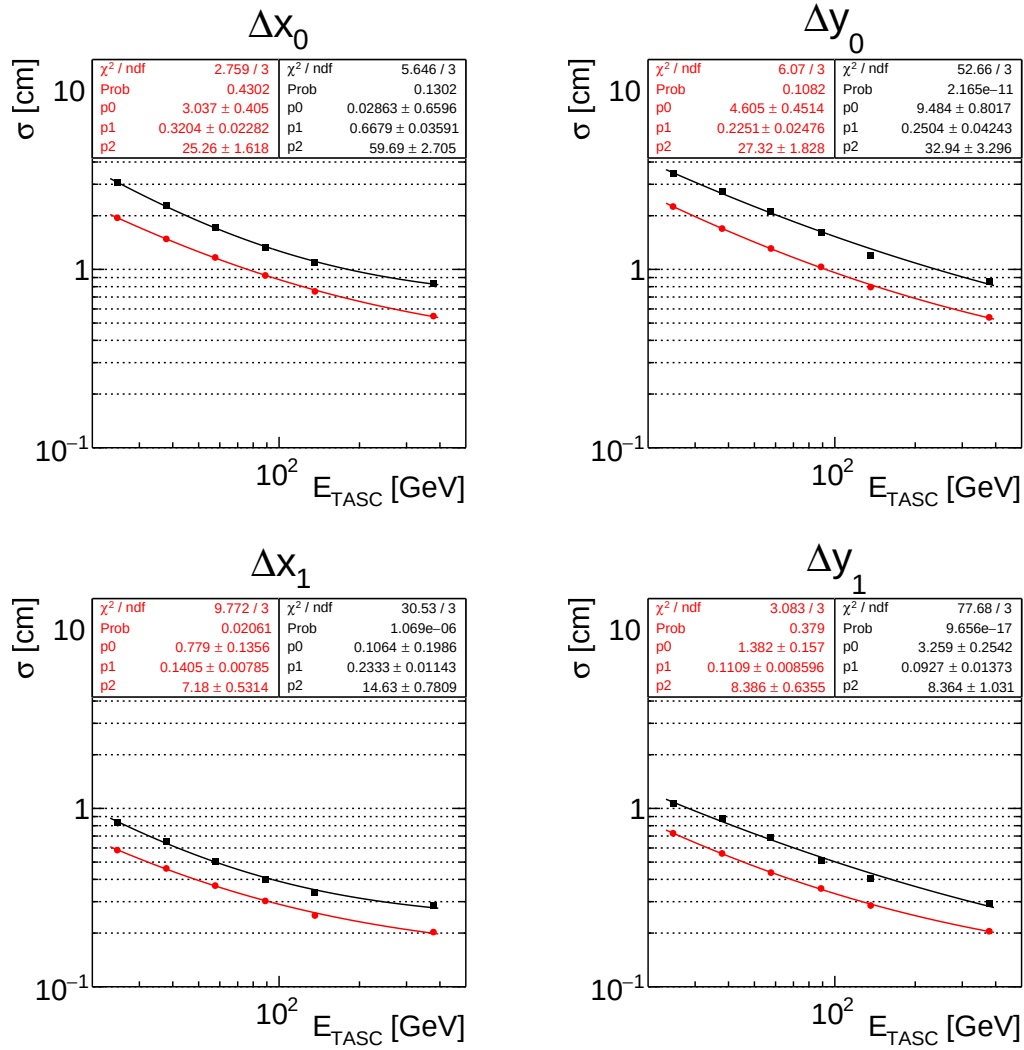


Figure B.37. spatial resolution of the CNN trained on helium nuclei (in *red*) and of the shower axis method (in *black*), as a function of the mean deposited energy in each range.

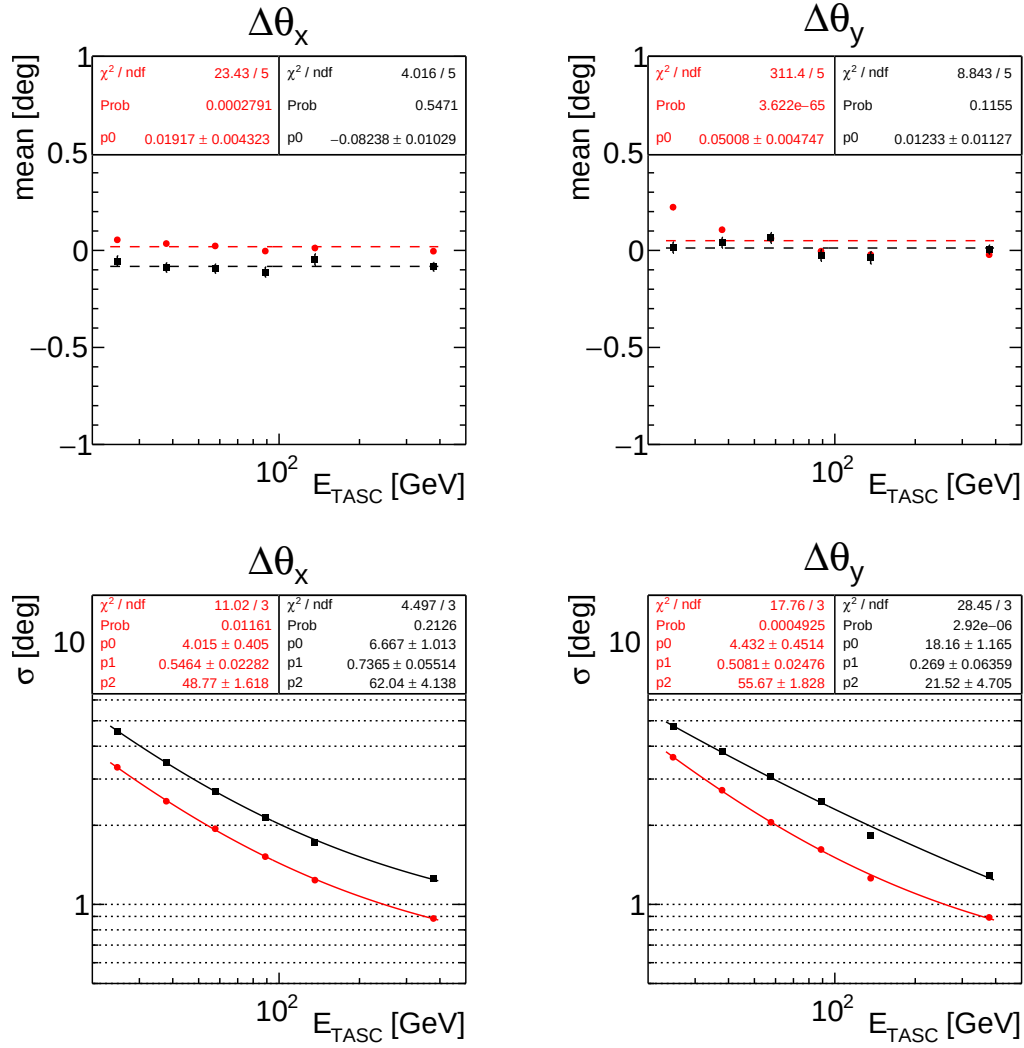


Figure B.38. in the *top* graphs, the mean value of the residual distribution; *below*, the angular resolution of the residual distribution of the angular variables of the CNN trained on helium nuclei (in *red*) and of the shower axis method (in *black*), as a function of the mean deposited energy in each range.

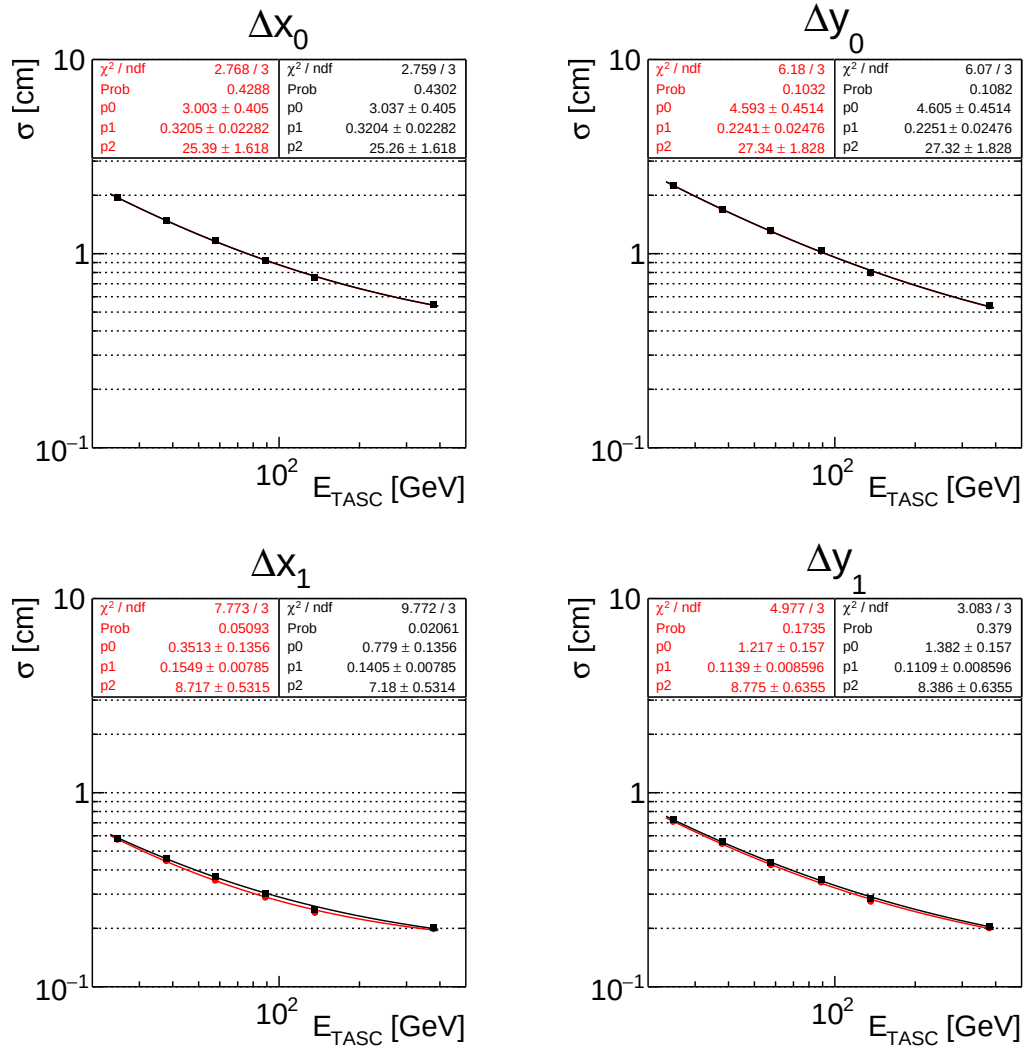


Figure B.39. *effective* spatial resolution of the CNN trained on helium nuclei (in red) and of the Kalman filter method (in black), as a function of the mean deposited energy in each range.

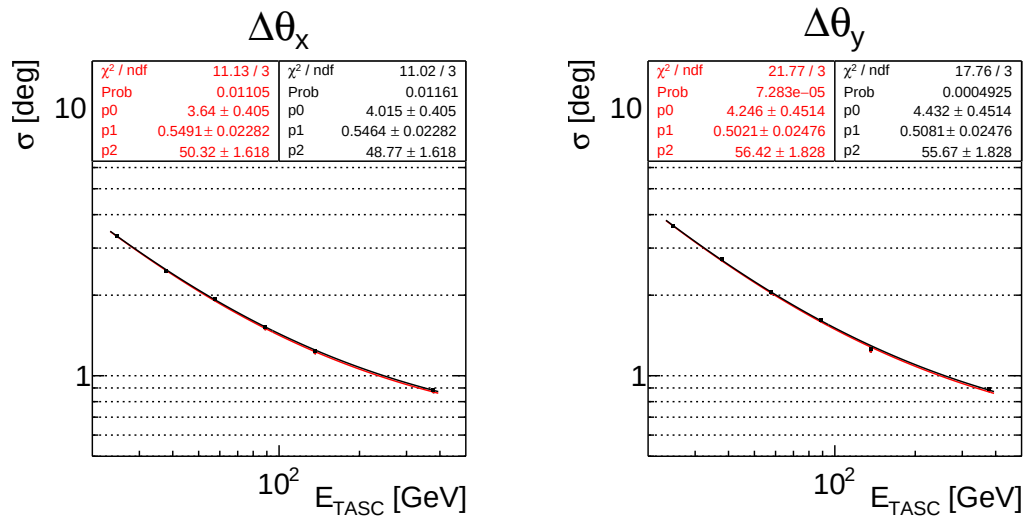


Figure B.40. *effective* angular resolution of the CNN trained on helium nuclei (in *red*) and of the Kalman filter method (in *black*), as a function of the mean deposited energy in each range.

B.5 Sample of p and He nuclei from flight data

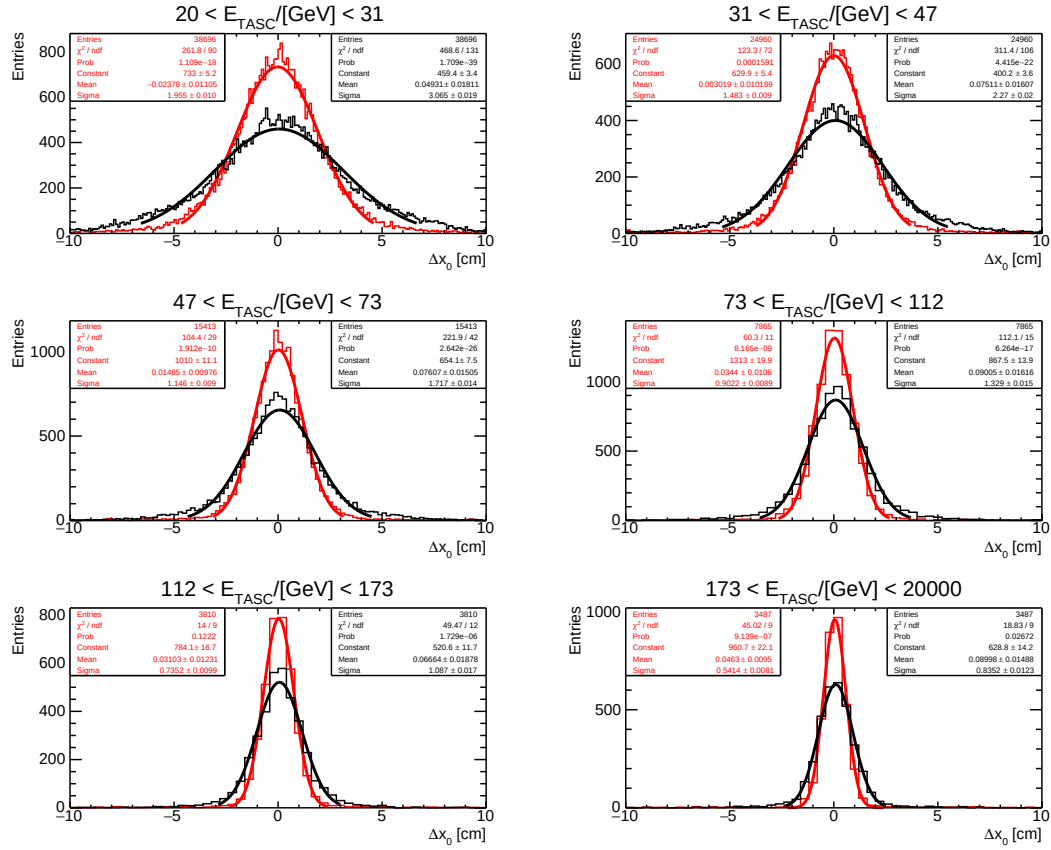


Figure B.41. residual distribution of the impact point variable x_0 predicted by the CNN trained on combined p and He data (red histogram) and by the shower axis method (black histogram). The curves superimposed on the histograms are the Gaussian fits to the histograms of the same color.

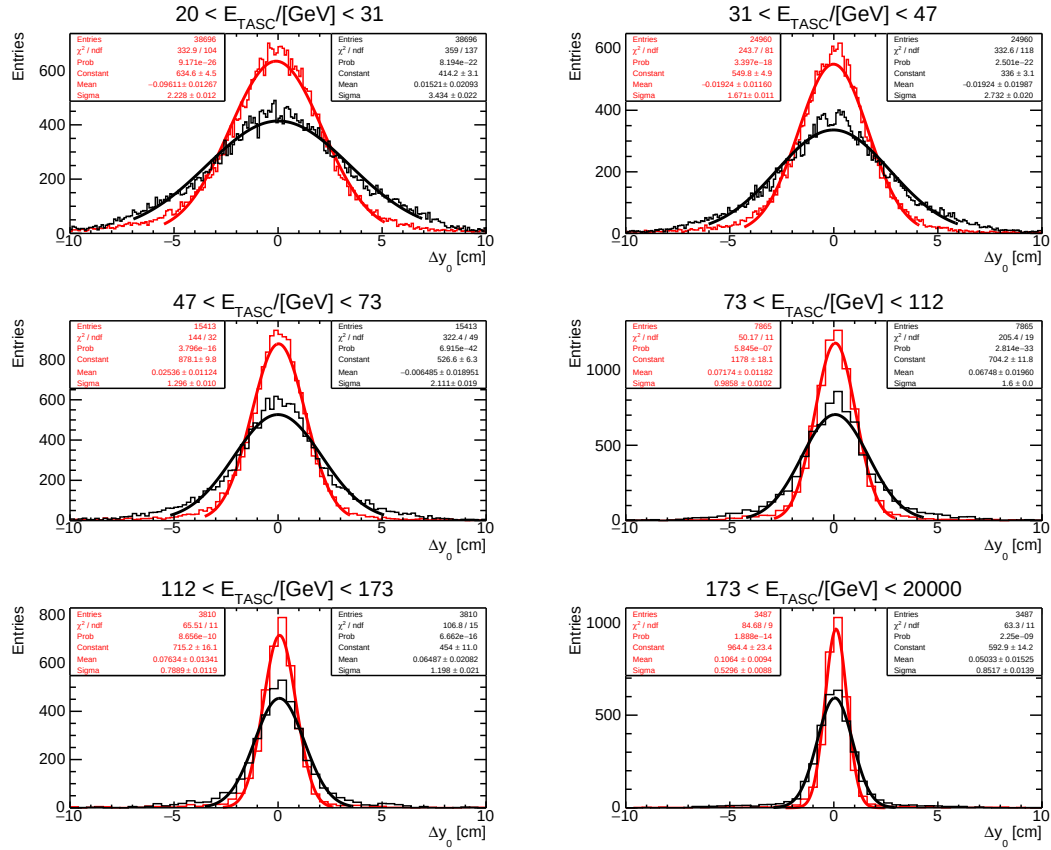


Figure B.42. residual distribution of the impact point variable y_0 predicted by the CNN trained on combined p and He data (*red* histogram) and by the shower axis method (*black* histogram). The curves superimposed on the histograms are the Gaussian fits to the histograms of the same color.

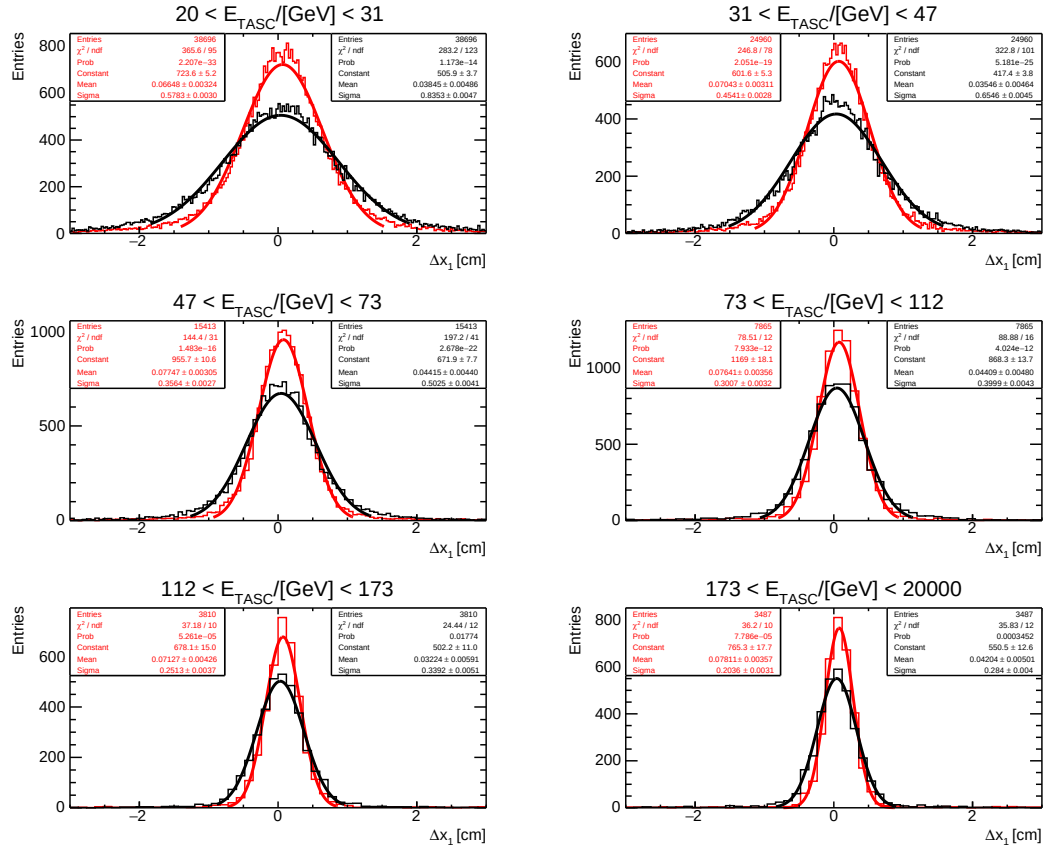


Figure B.43. residual distribution of the impact point variable x_1 predicted by the CNN trained on combined p and He data (*red* histogram) and by the shower axis method (*black* histogram). The curves superimposed on the histograms are the Gaussian fits to the histograms of the same color.

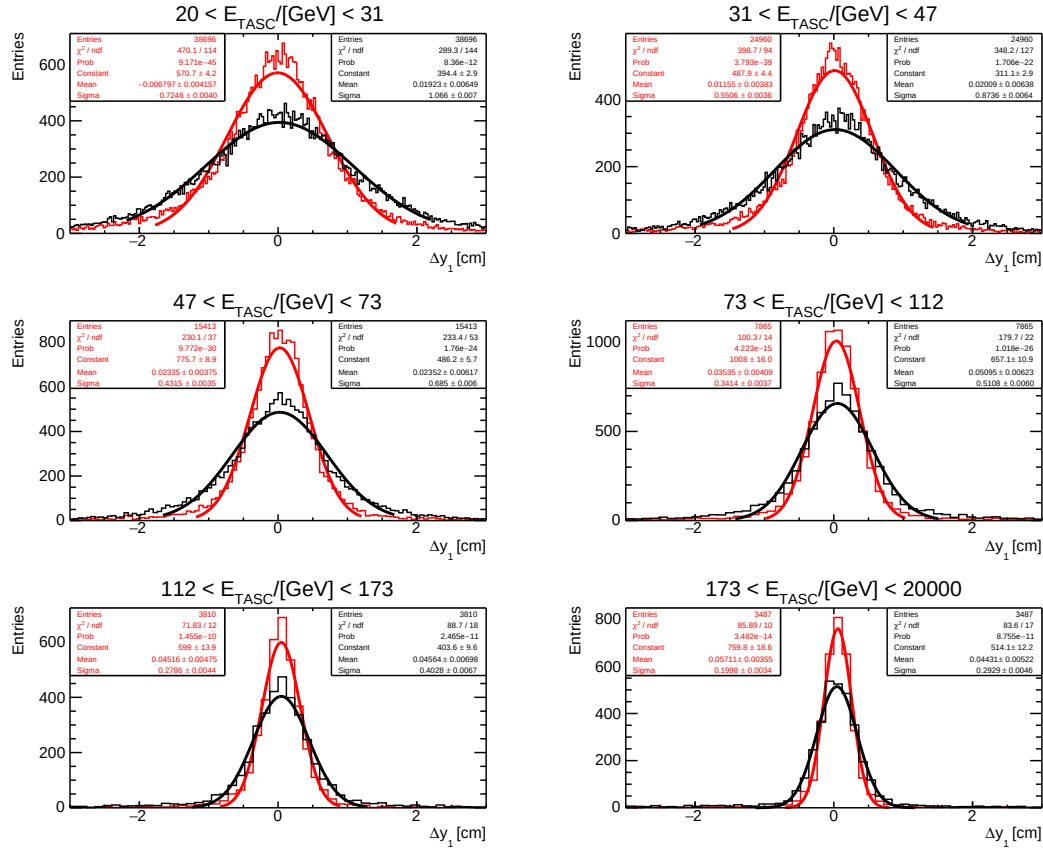


Figure B.44. residual distribution of the impact point variable y_1 predicted by the CNN trained on combined p and He data (*red* histogram) and by the shower axis method (*black* histogram). The curves superimposed on the histograms are the Gaussian fits to the histograms of the same color.

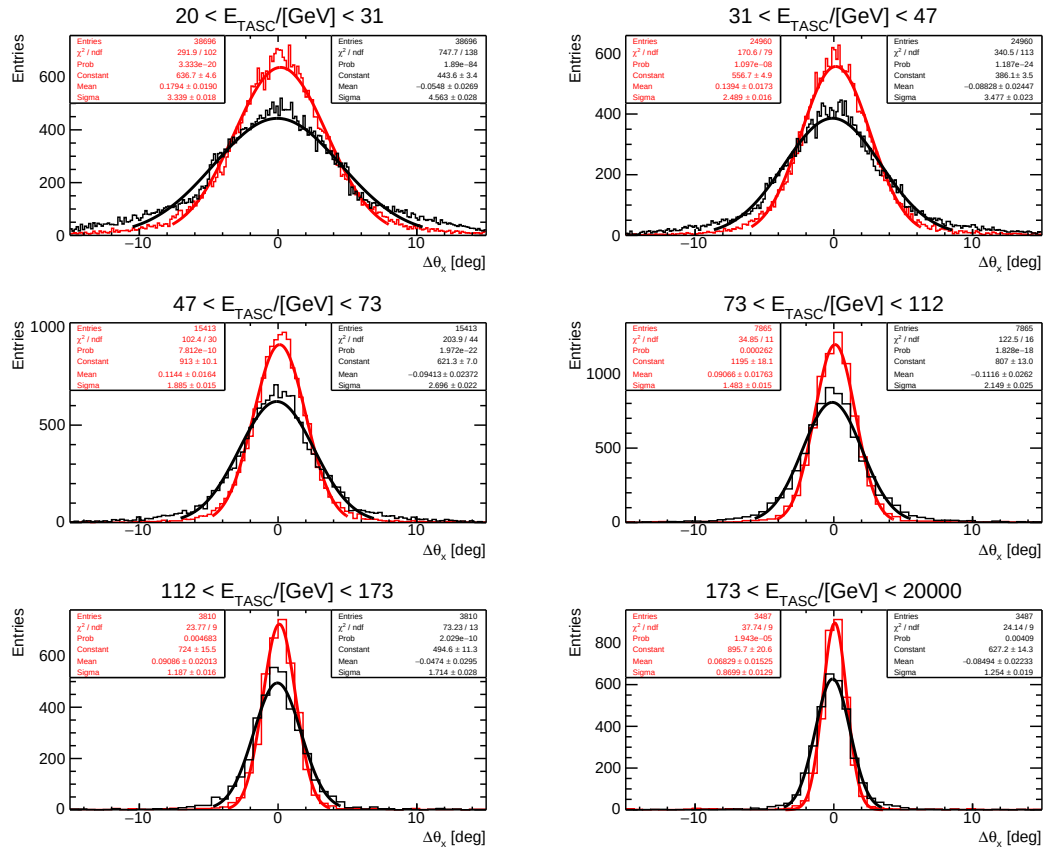


Figure B.45. residual distribution of the angular variable θ_x predicted by the CNN trained on combined p and He data (red histogram) and by the shower axis method (black histogram). The curves superimposed on the histograms are the Gaussian fits to the histograms of the same color.

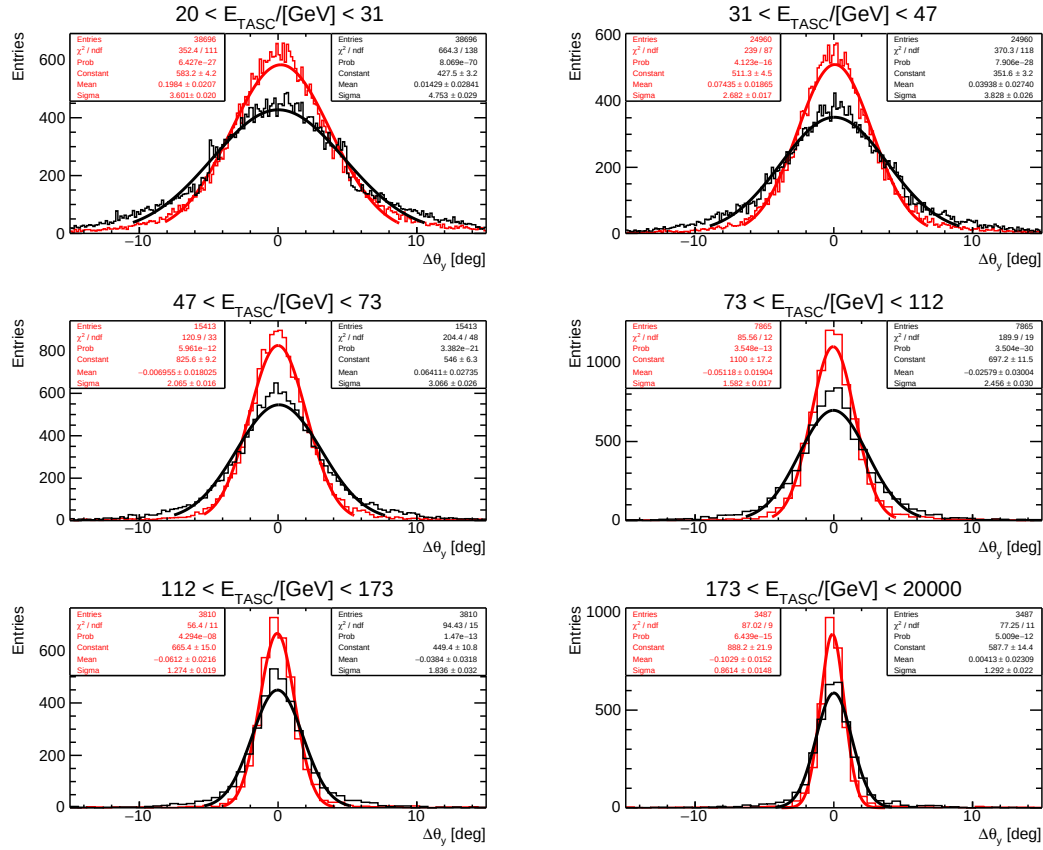


Figure B.46. residual distribution of the angular variable θ_y predicted by the CNN trained on combined p and He data (*red* histogram) and by the shower axis method (*black* histogram). The curves superimposed on the histograms are the Gaussian fits to the histograms of the same color.

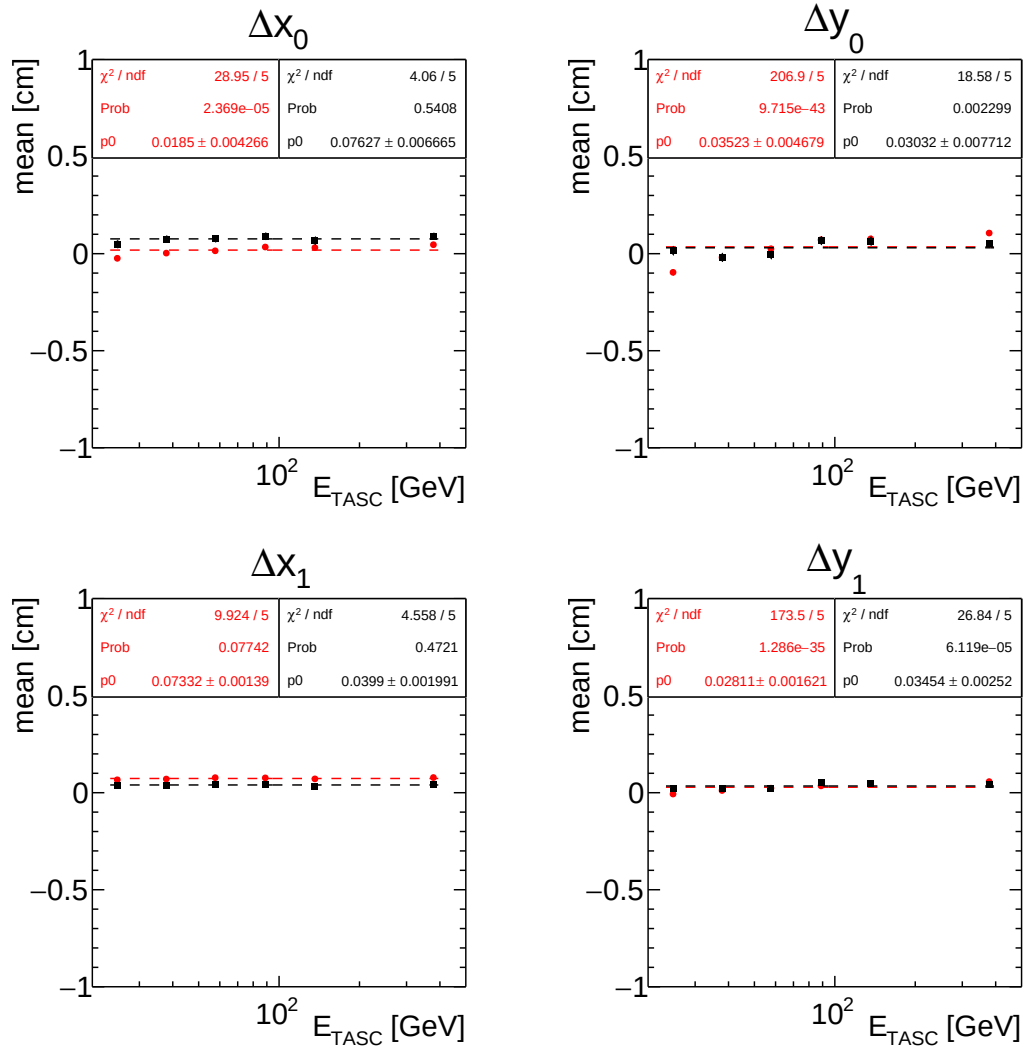


Figure B.47. mean value of the residual distribution of the impact points of the CNN trained on combined p and He data (in *red*) and of the shower axis method (in *black*), as a function of the mean energy deposited in each range.

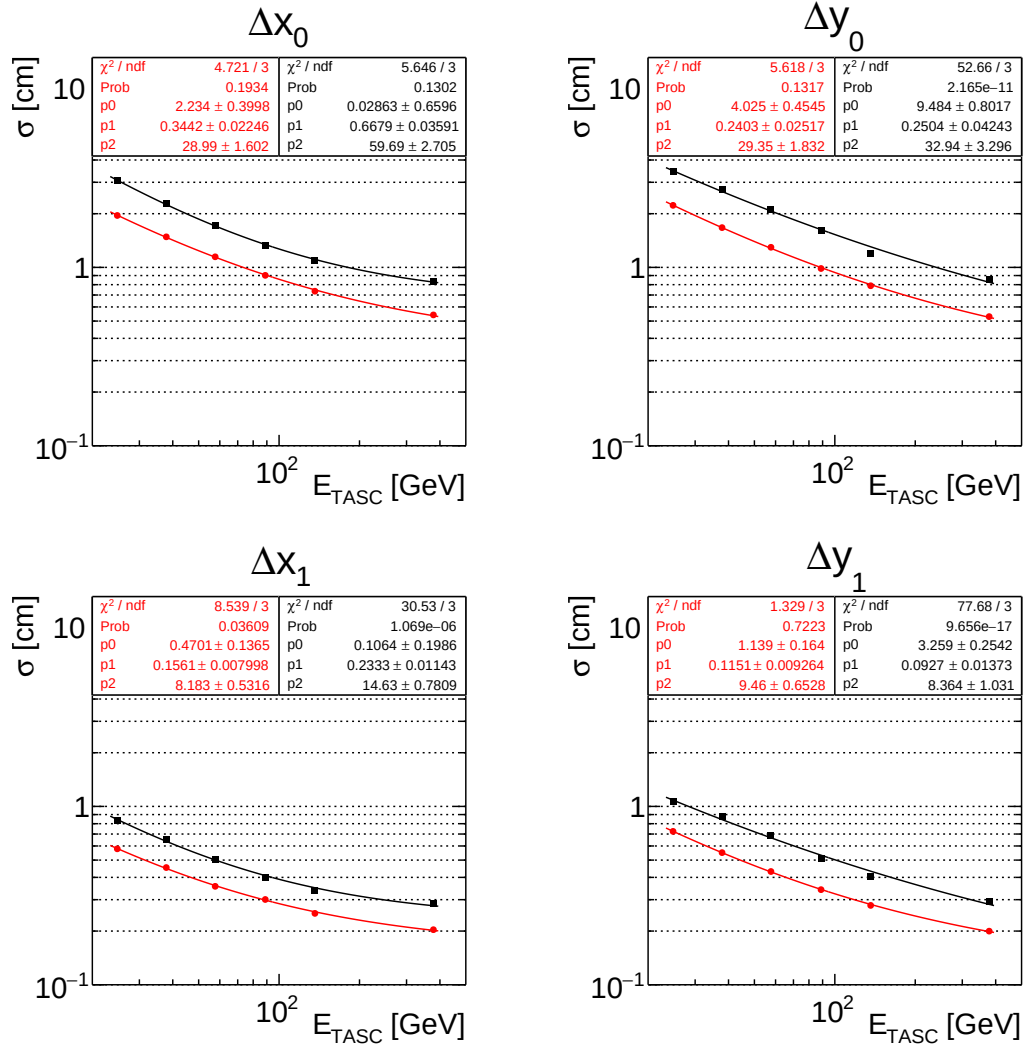


Figure B.48. spatial resolution of the CNN trained on combined p and He data (in red) and of the shower axis method (in black), as a function of the mean energy deposited in each range.

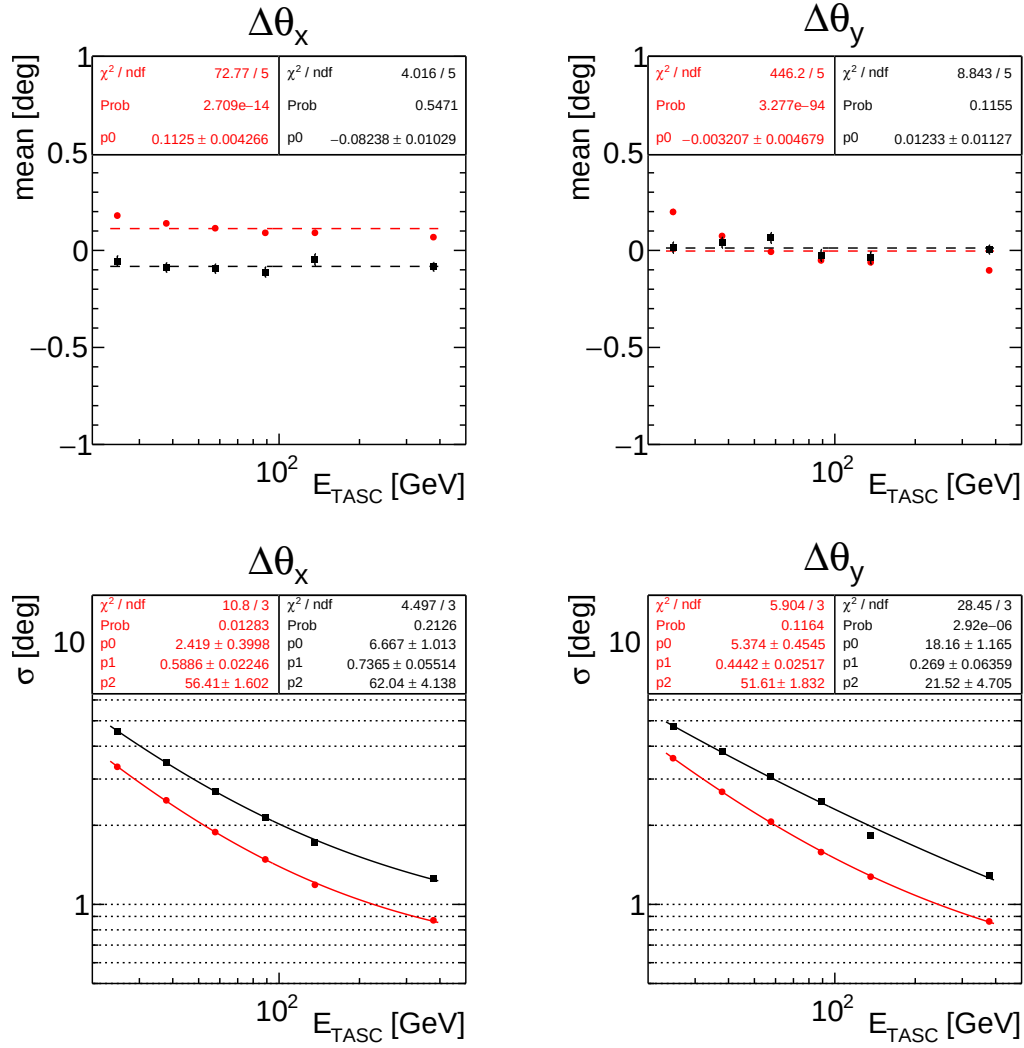


Figure B.49. in the graphs *above* the mean value of the residual distribution; *below* the angular resolution of the residual distribution of the angular variables of the CNN trained on combined p and He data (in *red*) and of the shower axis method (in *black*), as a function of the mean energy deposited in each range.

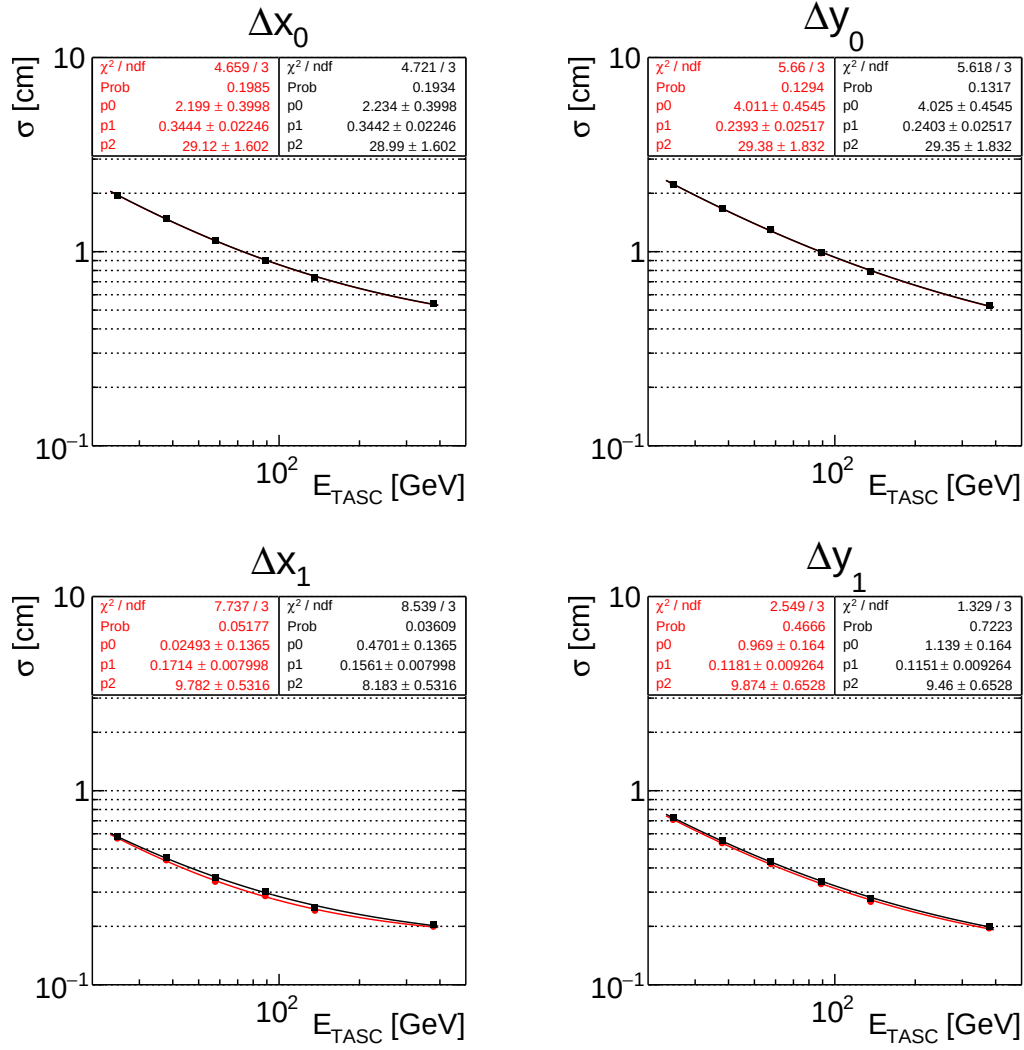


Figure B.50. *effective* spatial resolution of the CNN trained on combined p and He data (in *red*) and of the Kalman filter method (in *black*), as a function of the mean energy deposited in each range.

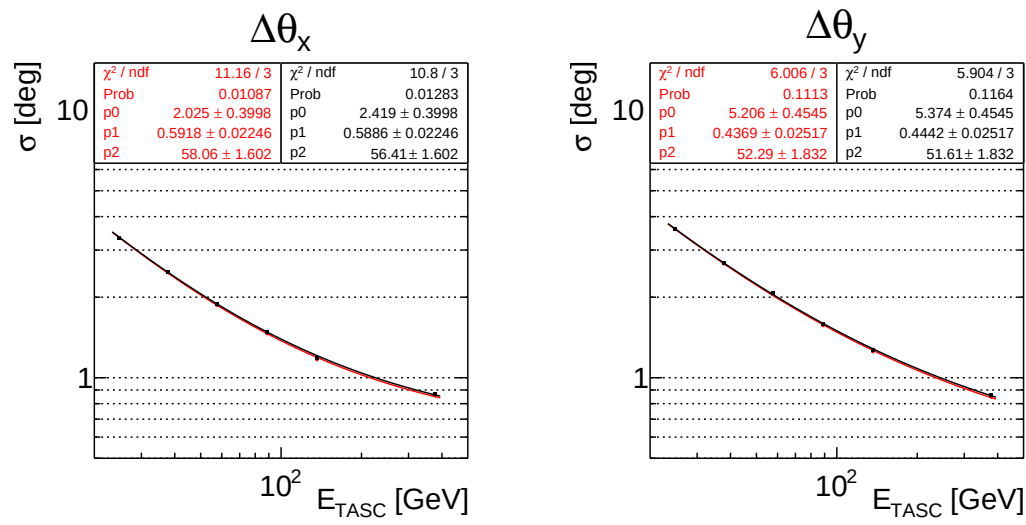


Figure B.51. *effective* angular resolution of the CNN trained on combined p and He data (in *red*) and of the Kalman filter method (in *black*), as a function of the mean energy deposited in each range.

B.6 Nuclei charge selection

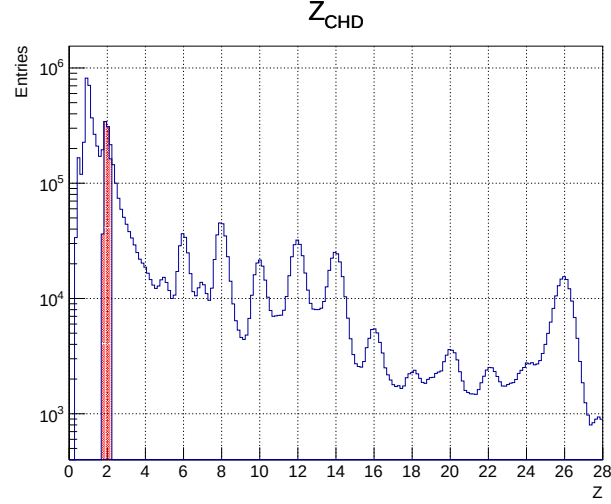


Figure B.52. in *blue* the reconstructed charge in the two planes of the CHD in the range (0, 28); in *red* the selected helium nuclei in the range (1.8, 2.2).

Peaks corresponding to the nuclei (p, He, C, O, Ne, Mg, Si, S, Ar, Ca, Ti, Cr, Fe) are observed.

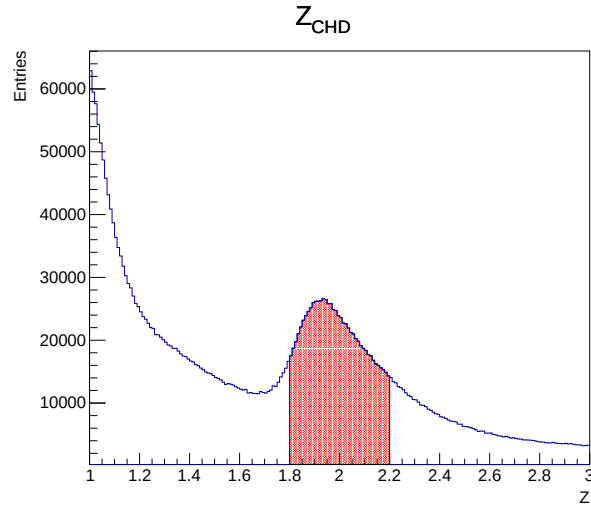


Figure B.53. in *blue* the zoom of the reconstructed charge in the two planes of the CHD in the range (1, 3); in *red* the selected helium nuclei in the range (1.8, 2.2).

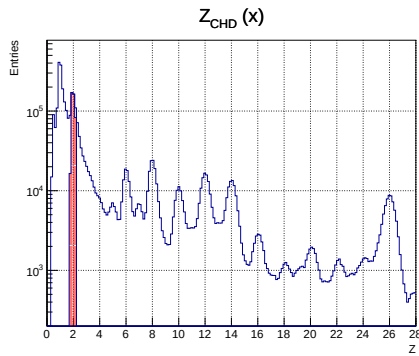


Figure B.54. in *blue* the reconstructed charge in the xz plane of the CHD in the range (0, 28); in *red* the selected helium nuclei in the range (1.8, 2.2).

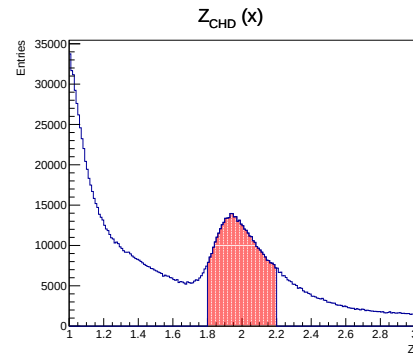


Figure B.55. in *blue* the zoom of the reconstructed charge in the xz plane of the CHD in the range (1, 3); in *red* the selected helium nuclei in the range (1.8, 2.2).

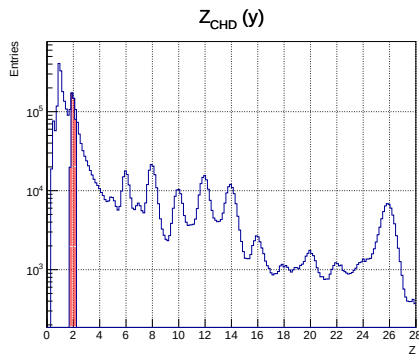


Figure B.56. in *blue* the reconstructed charge in the yz plane of the CHD in the range (0, 28); in *red* the selected helium nuclei in the range (1.8, 2.2).

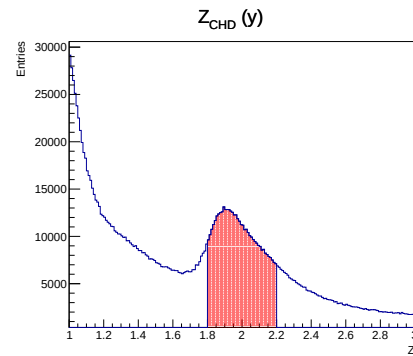


Figure B.57. in *blue* the zoom of the reconstructed charge in the yz plane of the CHD in the range (1, 3); in *red* the selected helium nuclei in the range (1.8, 2.2).

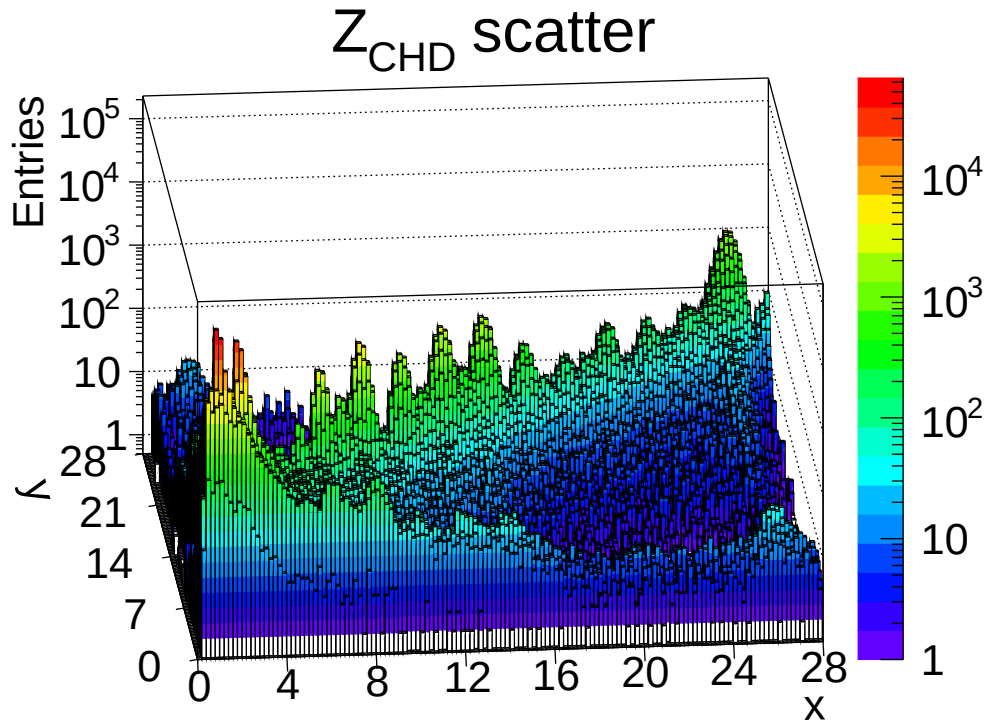


Figure B.58. three-dimensional scatter plot of the reconstructed charge in the xz and yz planes of the CHD in the range $(0, 28)$; the scale on the z -axis is logarithmic; it is possible to observe the peaks relative to the nuclei measured in the calorimeter, as reported in B.52.

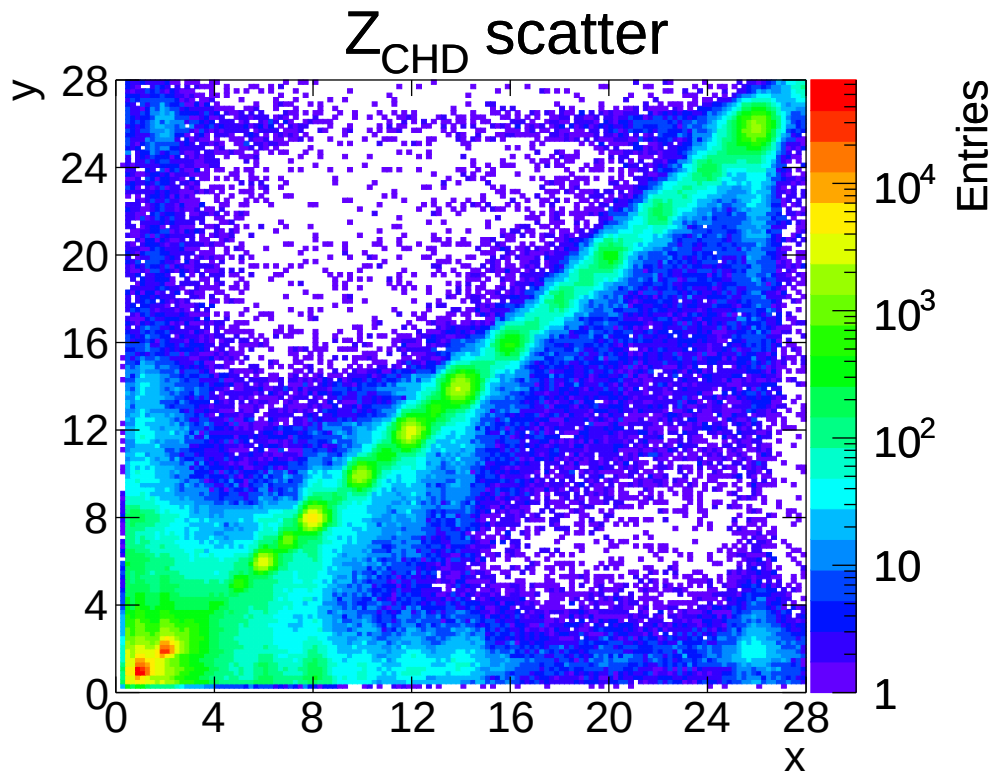


Figure B.59. two-dimensional scatter plot (with *top* view) of the reconstructed charge in the xz and yz planes of the CHD in the range $(0, 28)$; the scale on the z -axis is logarithmic; it is possible to observe the peaks relative to the nuclei measured in the calorimeter, as reported in B.52.

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